Chapter 3
Transport Layer

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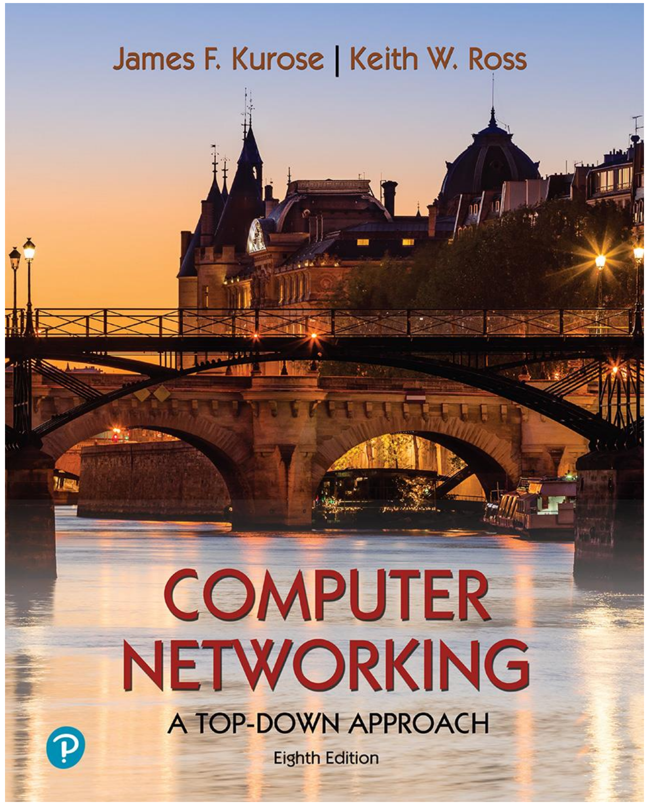


Computer Networking: A Top-Down Approach
8th edition
Jim Kurose, Keith Ross
Pearson, 2020

Transport layer: overview

Our goal:

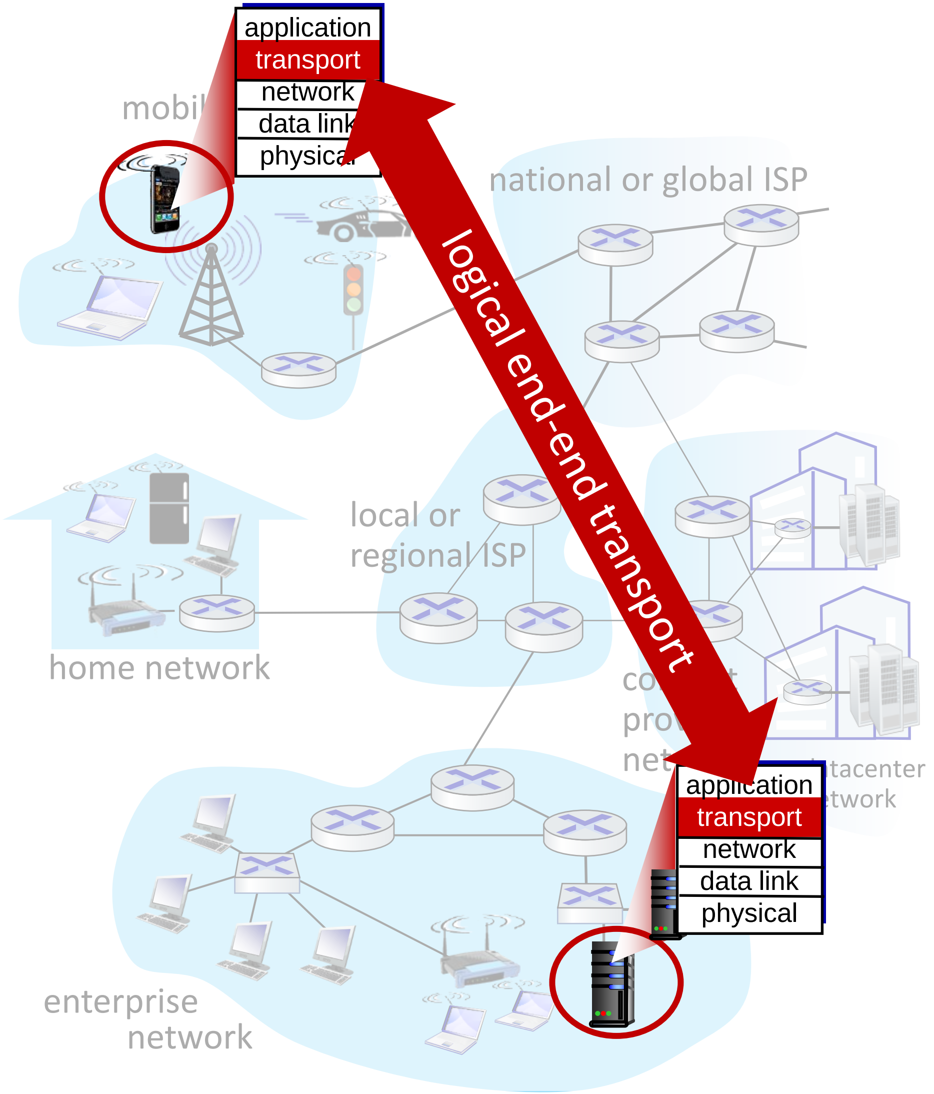
- understand principles behind transport layer services:
  - multiplexing, demultiplexing
  - reliable data transfer
  - flow control
  - congestion control
- learn about Internet transport layer protocols:
  - UDP: connectionless transport
  - TCP: connection-oriented reliable transport
  - TCP congestion control

Transport layer: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- Principles of reliable data transfer
- Connection-oriented transport: TCP
- Principles of congestion control
- TCP congestion control
- Evolution of transport-layer functionality

Transport services and protocols

- provide *logical communication* between application processes running on different hosts
- transport protocols actions in end systems:
  - sender: breaks application messages into *segments*, passes to network layer
  - receiver: reassembles segments into messages, passes to application layer
- two transport protocols available to Internet applications
  - TCP, UDP


application
transport
network
data link
physical
mobil
national or global ISP
logical end-end transport
local or
regional ISP
home network
co
pro
net
tacenter
twork
application
transport
network
data link
physical
enterprise
network

Transport vs. network layer services and protocols

© 1922. WM. EVANS.

THERE was an old woman who lived in a shoe,
She had so many children, she didn't know what to do.
She gave them some milk and nice butter bread,
She kissed them all round and put them to bed.

household analogy:

12 kids in Ann's house sending letters to 12 kids in Bill's house:

- hosts = houses
- processes = kids
- app messages = letters in envelopes

Transport vs. network layer services and protocols

- network layer: logical communication between *hosts*
- transport layer*:* logical communication between *processes*
    - relies on, enhances, network layer services

household analogy:

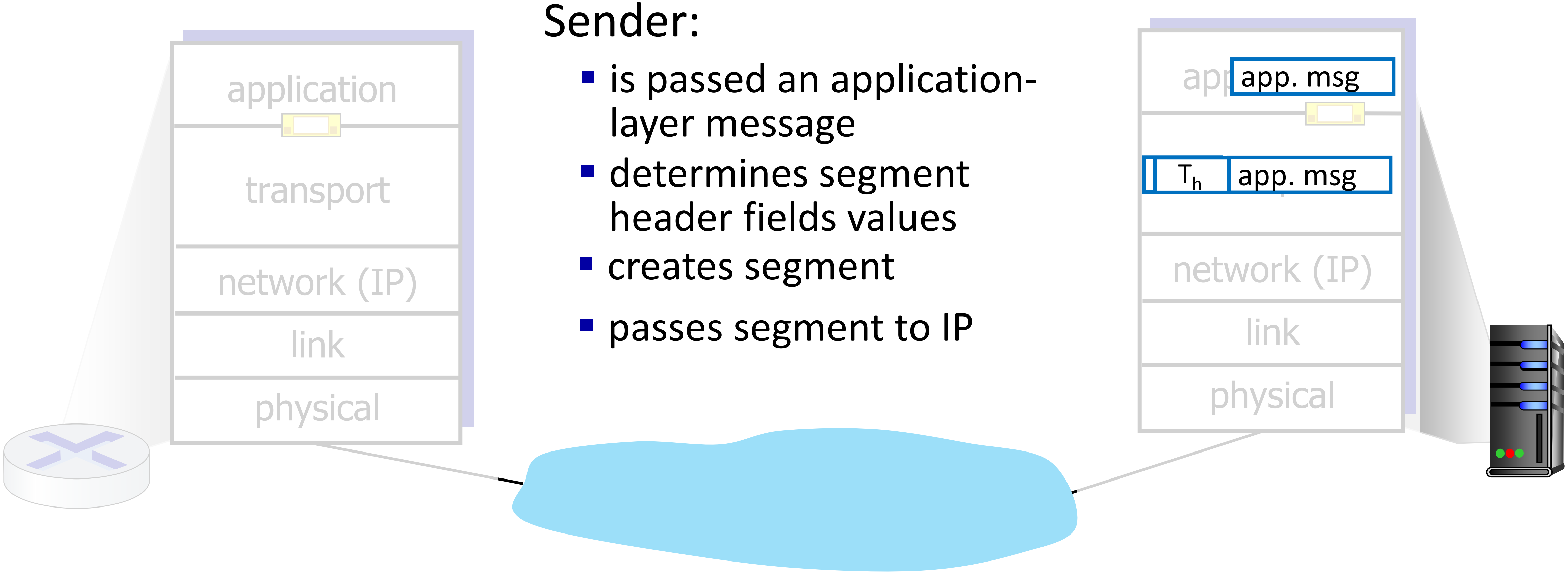
12 kids in Ann's house sending letters to 12 kids in Bill's house:

- hosts = houses
- processes = kids
- app messages = letters in envelopes

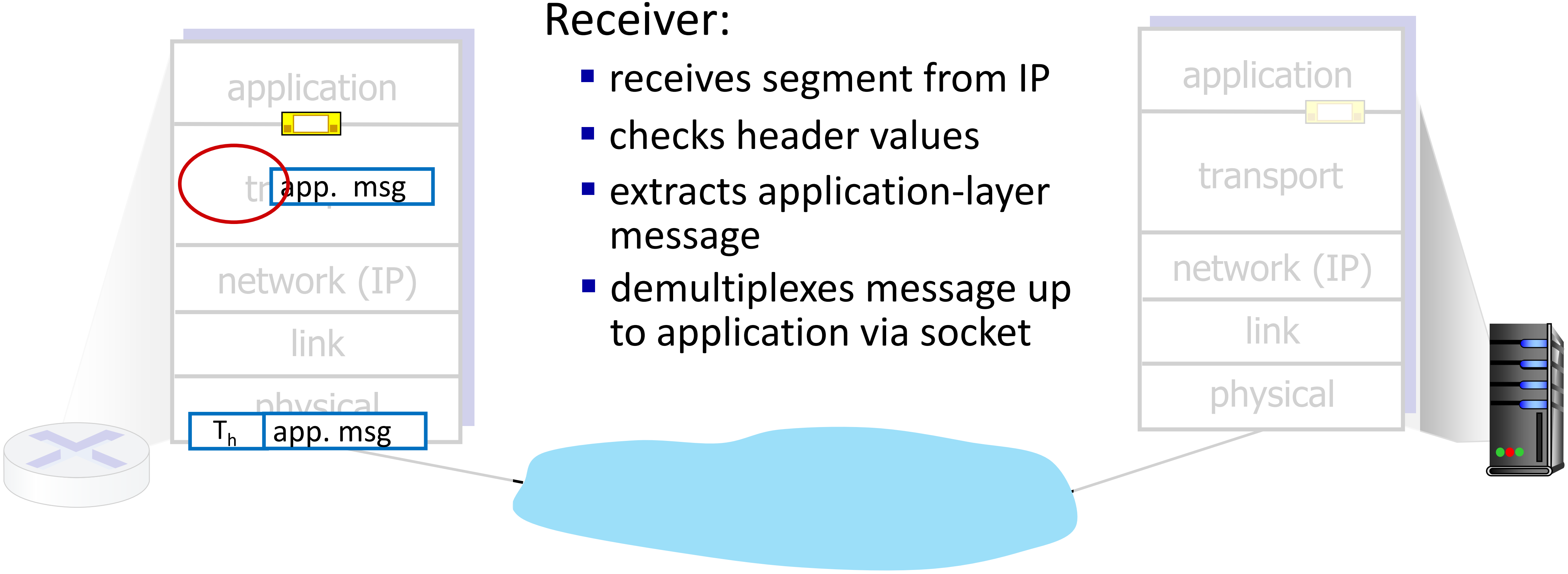
Transport Layer Actions

Sender:

- is passed an application-layer message
- determines segment header fields values
- creates segment
- passes segment to IP


application
transport
network (IP)
link
physical
app. msg
T_h
app. msg
network (IP)
link
physical

Transport Layer Actions



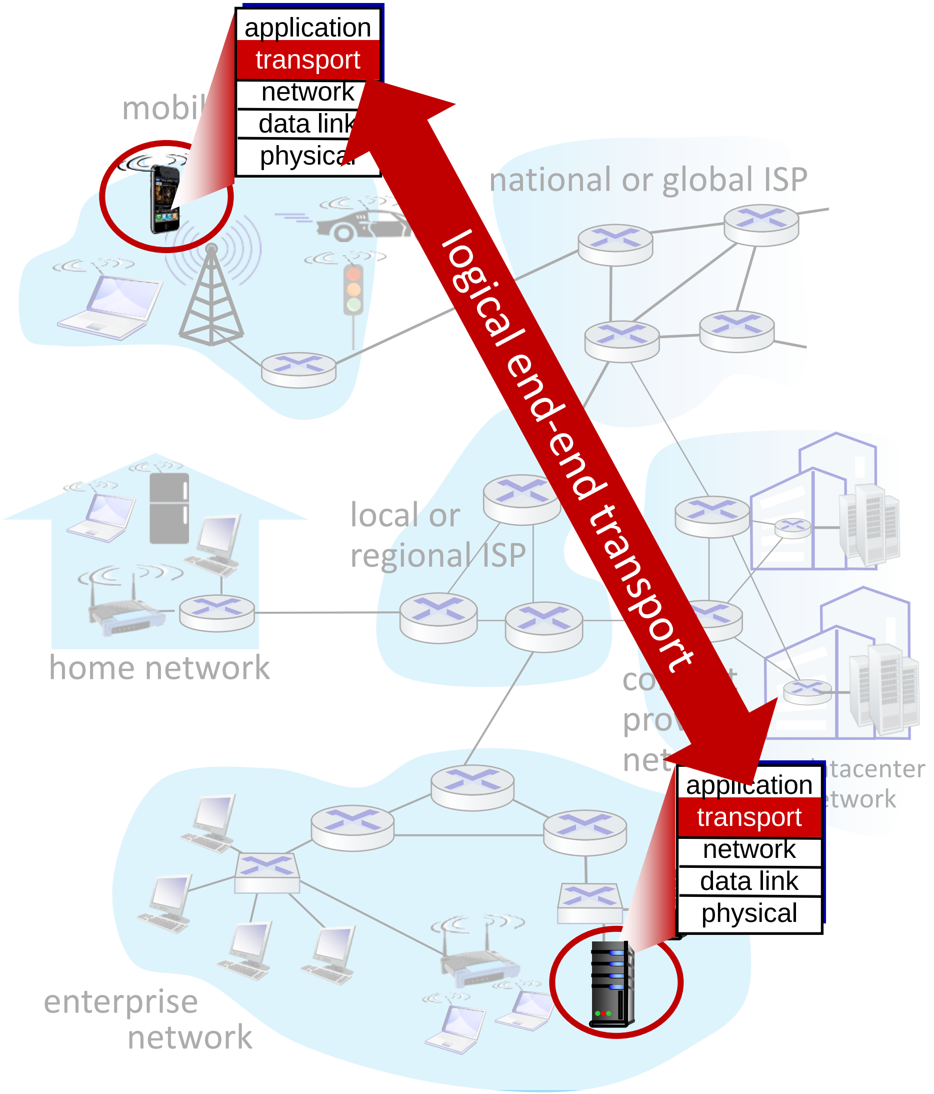
application
transport
app. msg
network (IP)
link
physical
T_h
app. msg
application
transport
network (IP)
link
physical

Receiver:

- receives segment from IP
- checks header values
- extracts application-layer message
- demultiplexes message up to application via socket

Two principal Internet transport protocols

- **TCP:** Transmission Control Protocol
  - reliable, in-order delivery
  - congestion control
  - flow control
  - connection setup
- **UDP:** User Datagram Protocol
  - unreliable, unordered delivery
  - no-frills extension of “best-effort” IP
- services not available:
  - delay guarantees
  - bandwidth guarantees


application
transport
network
data link
physical
mobil
national or global ISP
logical end-end transport
local or
regional ISP
home network
tacenter
twork
application
transport
network
data link
physical
enterprise
network

Chapter 3: roadmap

- Transport-layer services
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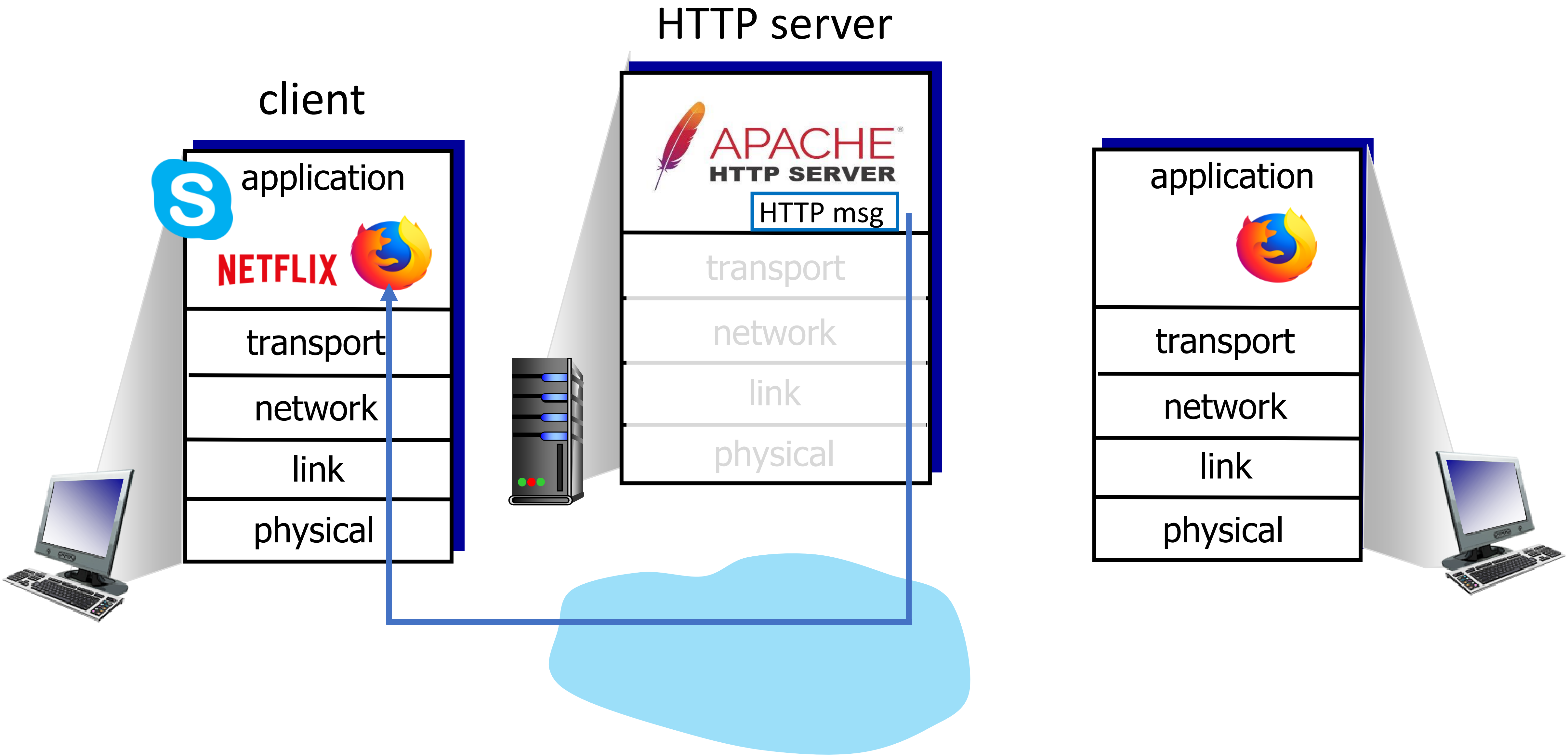
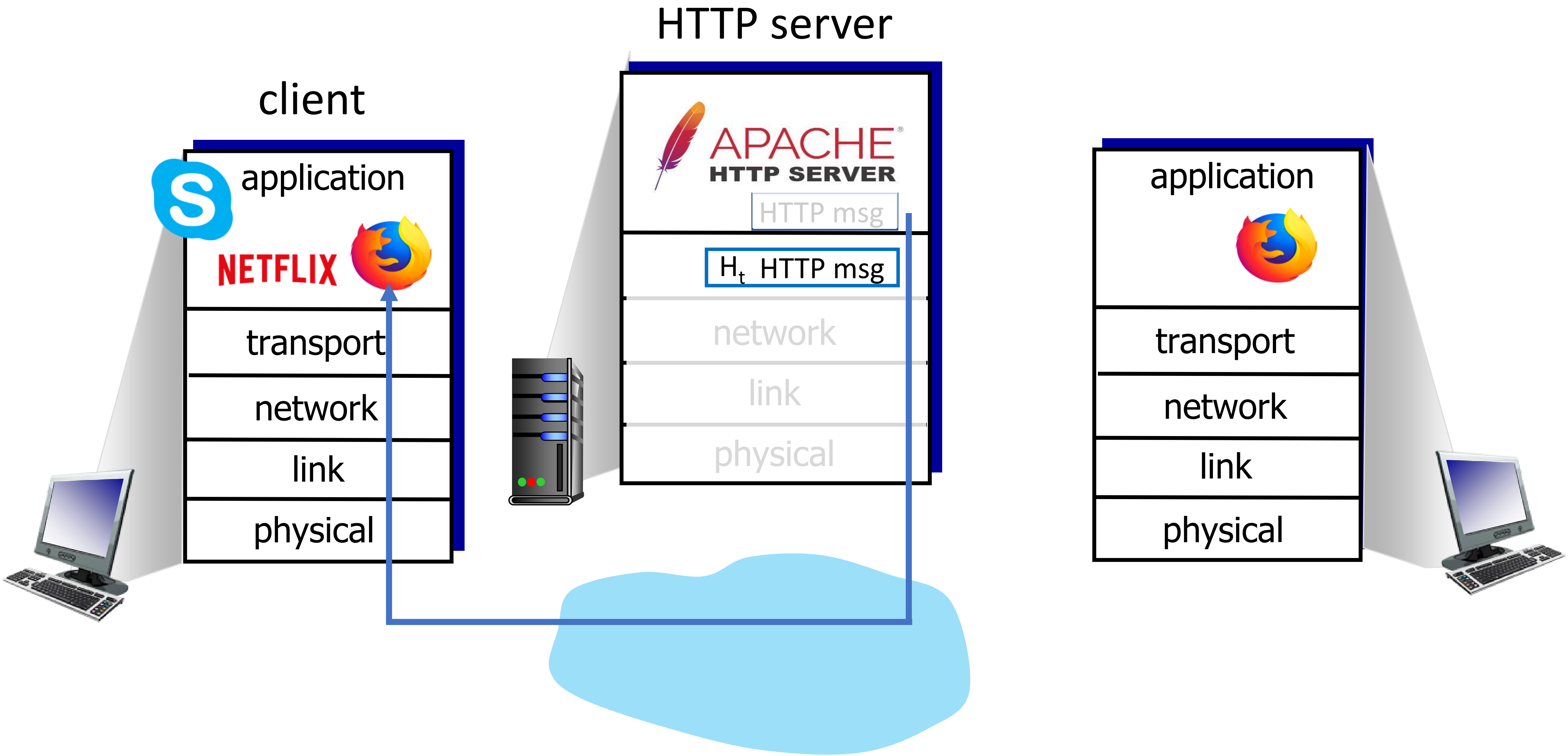
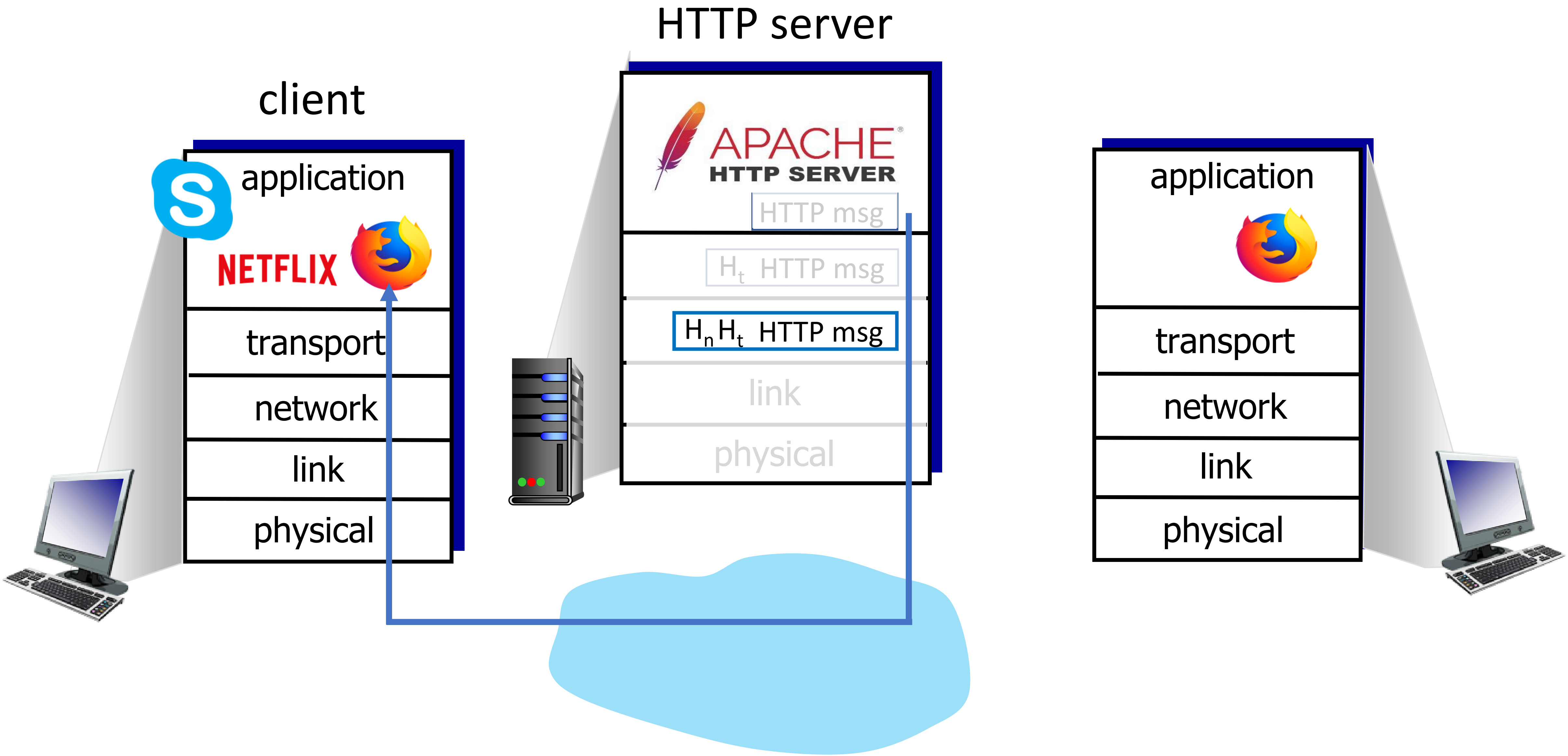
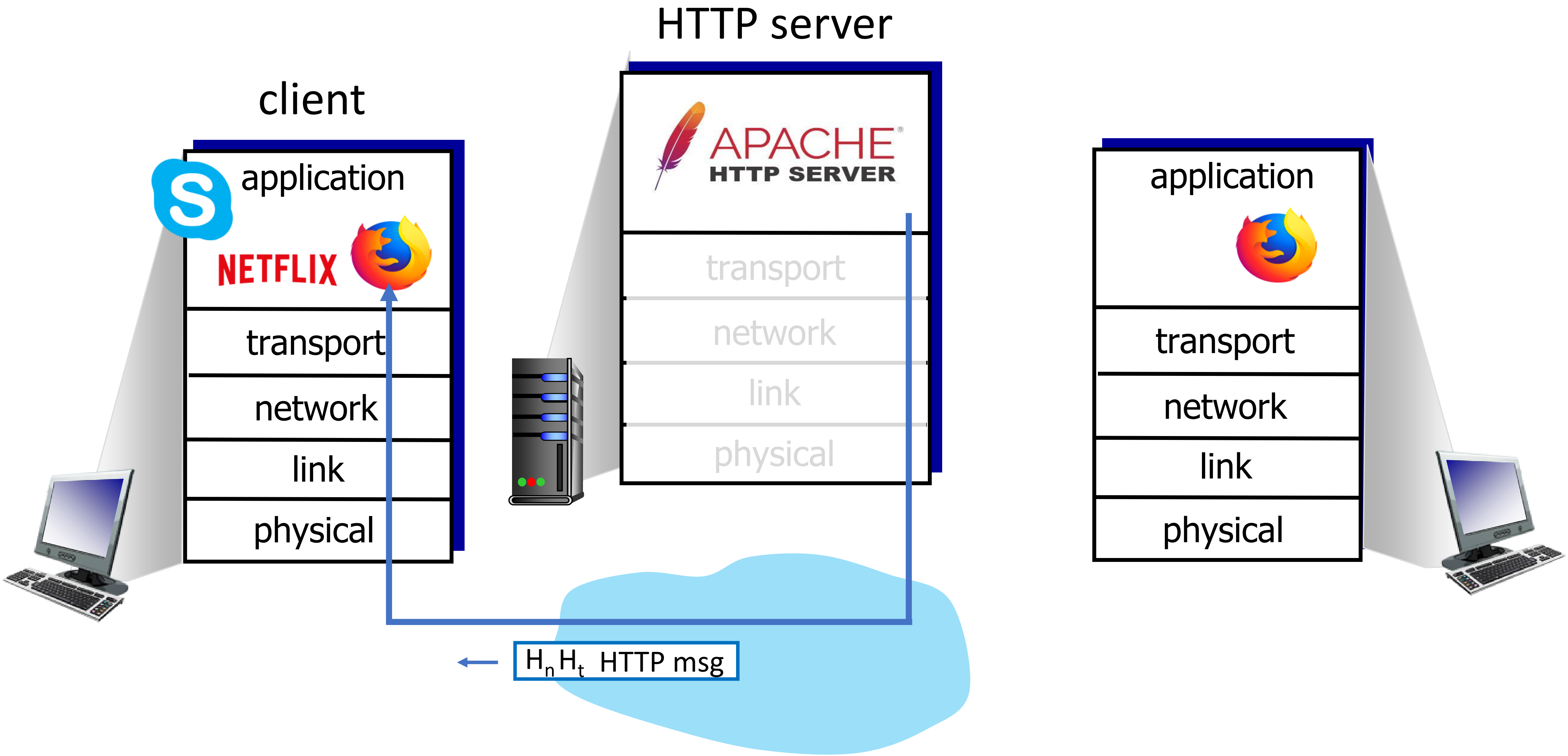
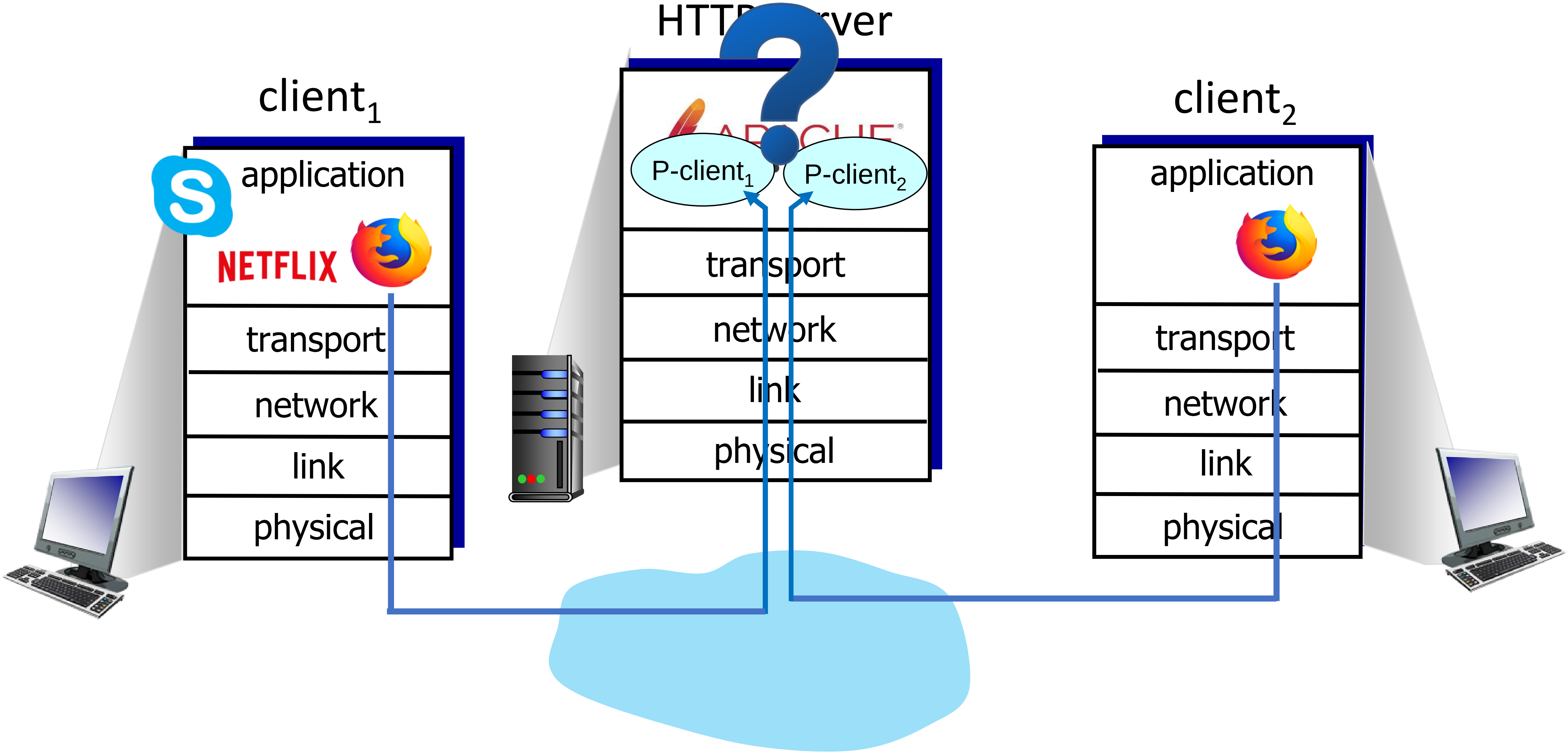

HTTP server
client
application
NETFLIX
transport
network
link
physical
APACHE
HTTP SERVER
HTTP msg
transport
network
link
physical
application
transport
network
link
physical


HTTP server
client
application
NETFLIX
transport
network
link
physical
APACHE
HTTP SERVER
HTTP msg
H$_t$ HTTP msg
network
link
physical
application
transport
network
link
physical


HTTP server
client
application
NETFLIX
transport
network
link
physical
APACHE
HTTP SERVER
HTTP msg
H_t HTTP msg
$H_n H_t$ HTTP msg
link
physical
application
transport
network
link
physical


HTTP server
client
application
NETFLIX
transport
network
link
physical
APACHE
HTTP SERVER
transport
network
link
physical
application
transport
network
link
physical
$H_n H_t$ HTTP msg



client$_1$

application

transport

network

link

physical

HTTP rver

P-client$_1$

P-client$_2$

transport

network

link

physical

client$_2$

application

transport

network

link

physical

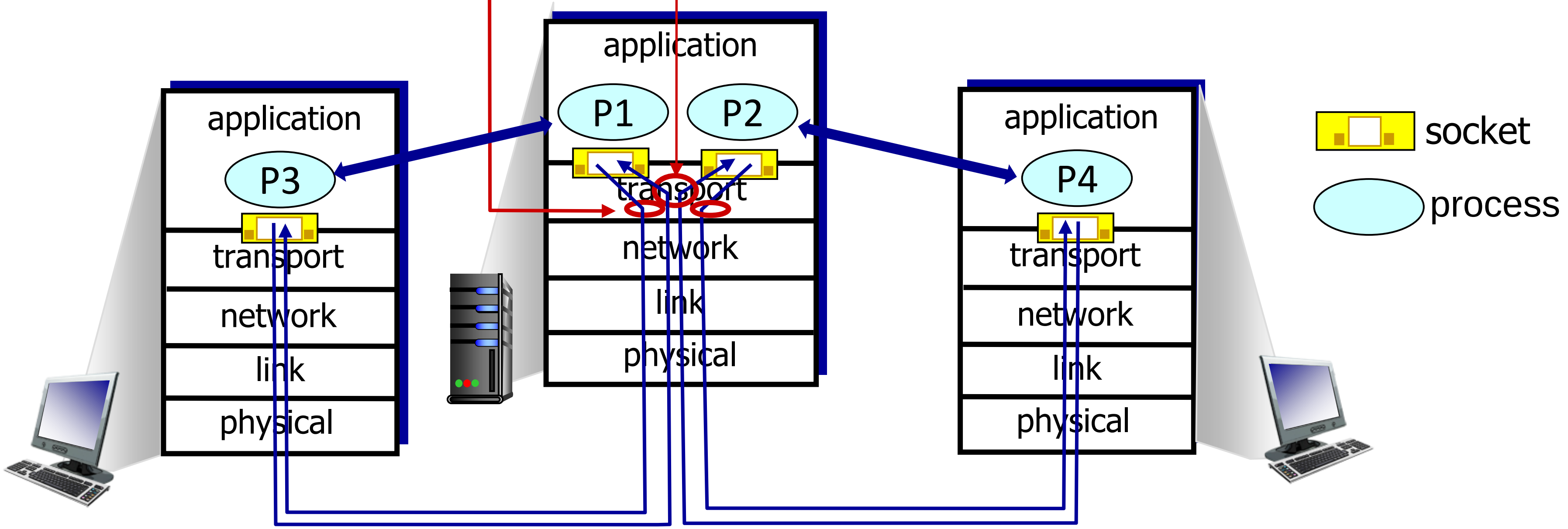
Multiplexing/demultiplexing

multiplexing at sender:

handle data from multiple sockets, add transport header (later used for demultiplexing)

demultiplexing at receiver:

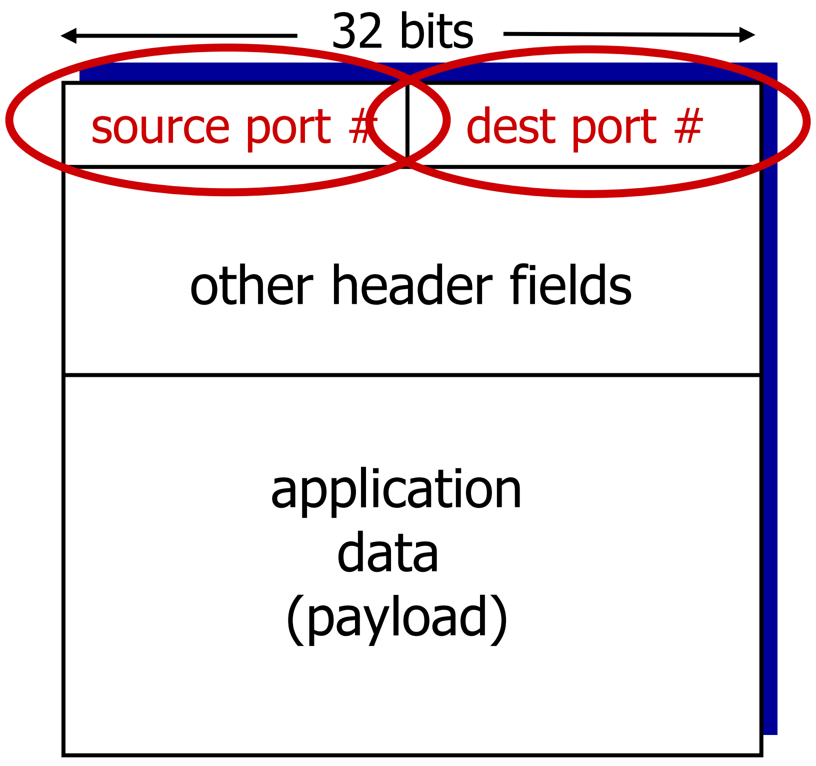
use header info to deliver received segments to correct socket



application
P1
P2
transport
network
link
physical
application
P3
transport
network
link
physical
application
P4
transport
network
link
physical
socket
process

How demultiplexing works

- host receives IP datagrams
  - each datagram has source IP address, destination IP address
  - each datagram carries one transport-layer segment
  - each segment has source, destination port number
- host uses *IP addresses & port numbers* to direct segment to appropriate socket


32 bits
source port #
dest port #
other header fields
application
data
(payload)

TCP/UDP segment format

Connectionless demultiplexing

Recall:

- when creating socket, must specify *host-local* port #:

```
DatagramSocket mySocket1
 = new DatagramSocket(12534);
```

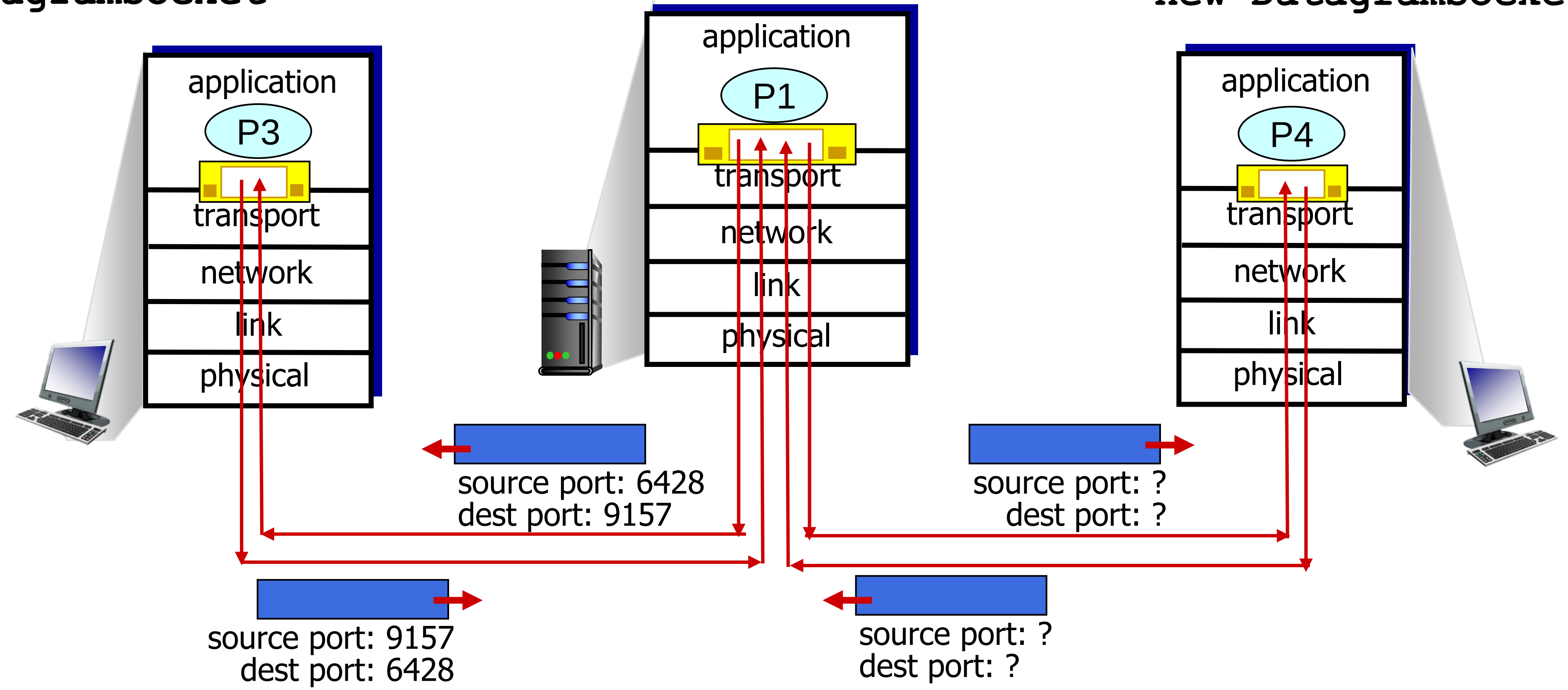
- when creating datagram to send into UDP socket, must specify
    - destination IP address
    - destination port #

when receiving host receives *UDP* segment:

- checks destination port # in segment
- directs UDP segment to socket with that port #

IP/UDP datagrams with *same dest. port #,* but different source IP addresses and/or source port numbers will be directed to *same socket* at receiving host

Connectionless demultiplexing: an example



```
DatagramSocket mySocket2 =
 new DatagramSocket
 (9157);
```

```
DatagramSocket
 serverSocket = new
 DatagramSocket
 (6428);
```

```
DatagramSocket mySocket1 =
 new DatagramSocket (5775);
```

application
P3
transport
network
link
physical
application
P1
transport
network
link
physical
application
P4
transport
network
link
physical

source port: 6428
dest port: 9157

source port: ?
dest port: ?

source port: 9157
dest port: 6428

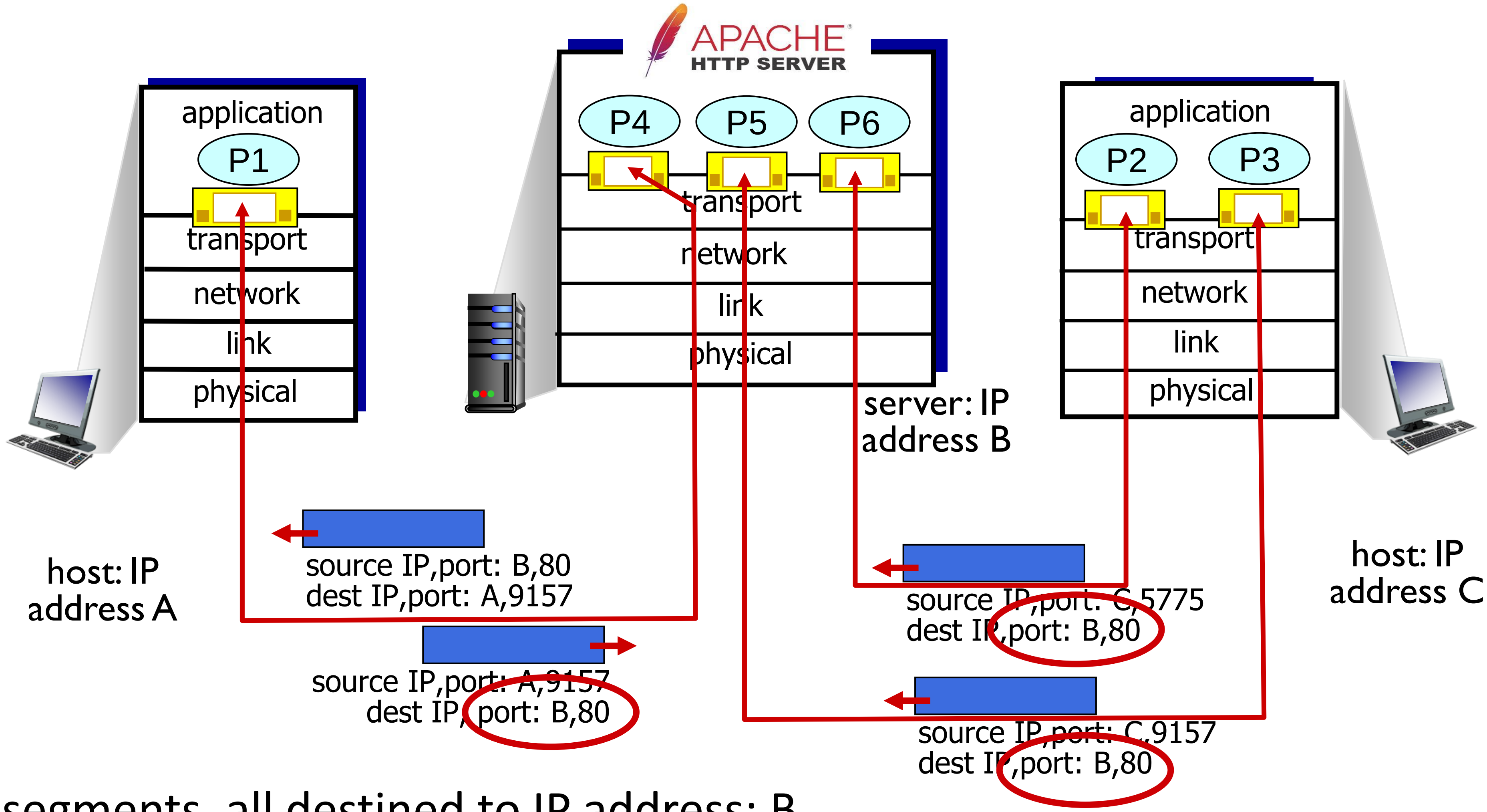
source port: ?
dest port: ?

Connection-oriented demultiplexing

- TCP socket identified by 4-tuple:
  - source IP address
  - source port number
  - dest IP address
  - dest port number
- demux: receiver uses *all four values (4-tuple)* to direct segment to appropriate socket

- server may support many simultaneous TCP sockets:
  - each socket identified by its own 4-tuple
  - each socket associated with a different connecting client

Connection-oriented demultiplexing: example


APACHE
HTTP SERVER
application
P1
transport
network
link
physical
P4
P5
P6
transport
network
link
physical
server: IP address B
application
P2
P3
transport
network
link
physical
host: IP address A
host: IP address C
source IP,port: B,80
dest IP,port: A,9157
source IP,port: A,9157
dest IP, port: B,80
source IP,port: C,5775
dest IP,port: B,80
source IP,port: C,9157
dest IP,port: B,80

Three segments, all destined to IP address: B,
dest port: 80 are demultiplexed to *different* sockets

Summary

- Multiplexing, demultiplexing: based on segment, datagram header field values
- **UDP:** demultiplexing using destination port number (only)
- **TCP:** demultiplexing using 4-tuple: source and destination IP addresses, and port numbers
- Multiplexing/demultiplexing happen at *all* layers

Chapter 3: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- **Connectionless transport: UDP**
- Principles of reliable data transfer
- Connection-oriented transport: TCP
- Principles of congestion control
- TCP congestion control
- Evolution of transport-layer functionality

UDP: User Datagram Protocol

- “no frills,” “bare bones” Internet transport protocol
- “best effort” service, UDP segments may be:
  - lost
  - delivered out-of-order to app
- *connectionless:*
  - no handshaking between UDP sender, receiver
  - each UDP segment handled independently of others

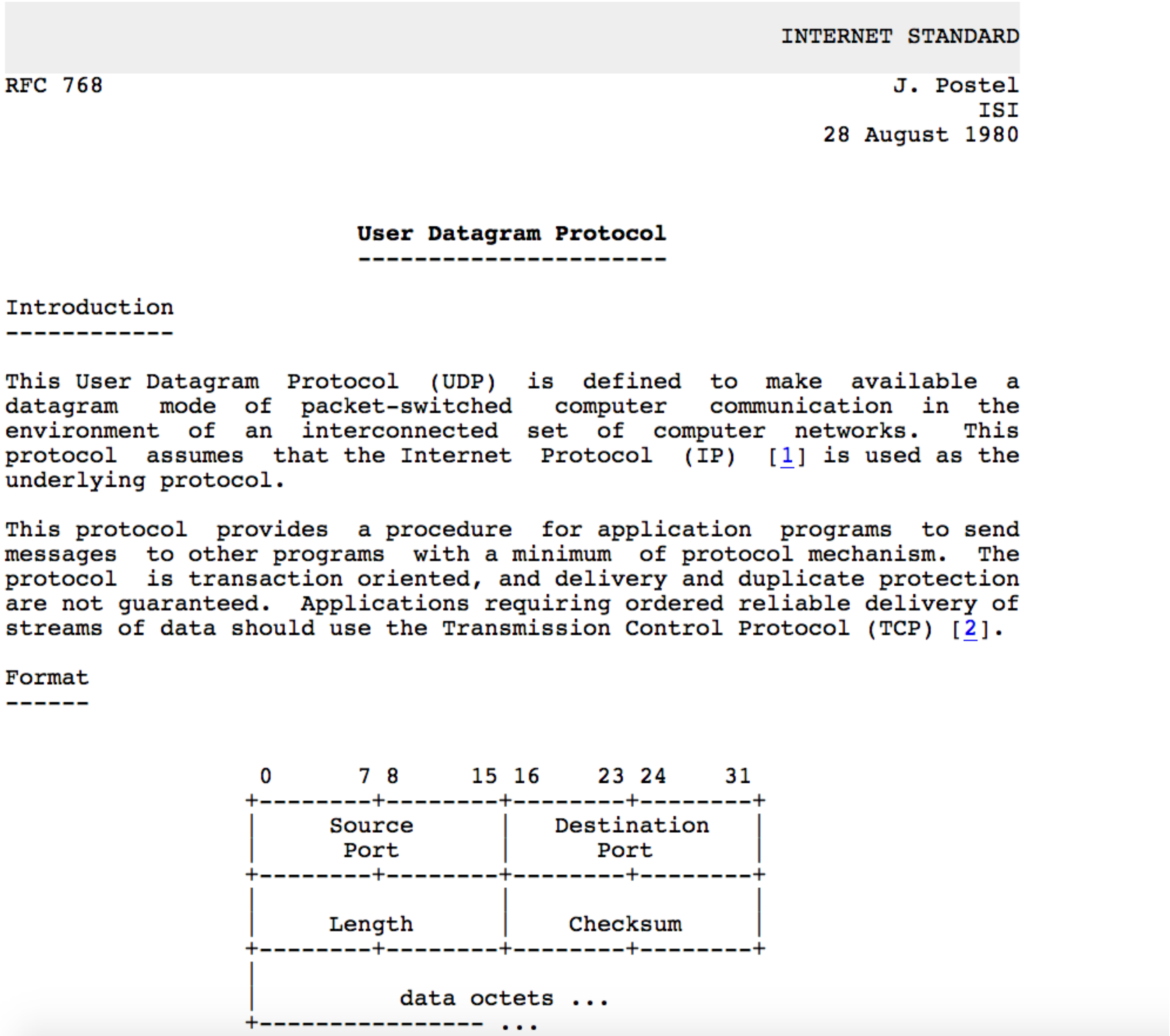
Why is there a UDP?

- no connection establishment (which can add RTT delay)
- simple: no connection state at sender, receiver
- small header size
- no congestion control
  - UDP can blast away as fast as desired!
  - can function in the face of congestion

UDP: User Datagram Protocol

- UDP use:
  - streaming multimedia apps (loss tolerant, rate sensitive)
  - DNS
  - SNMP
  - HTTP/3
- if reliable transfer needed over UDP (e.g., HTTP/3):
  - add needed reliability at application layer
  - add congestion control at application layer

UDP: User Datagram Protocol [RFC 768]



INTERNET STANDARD

RFC 768 J. Postel
ISI
28 August 1980

User Datagram Protocol

Introduction

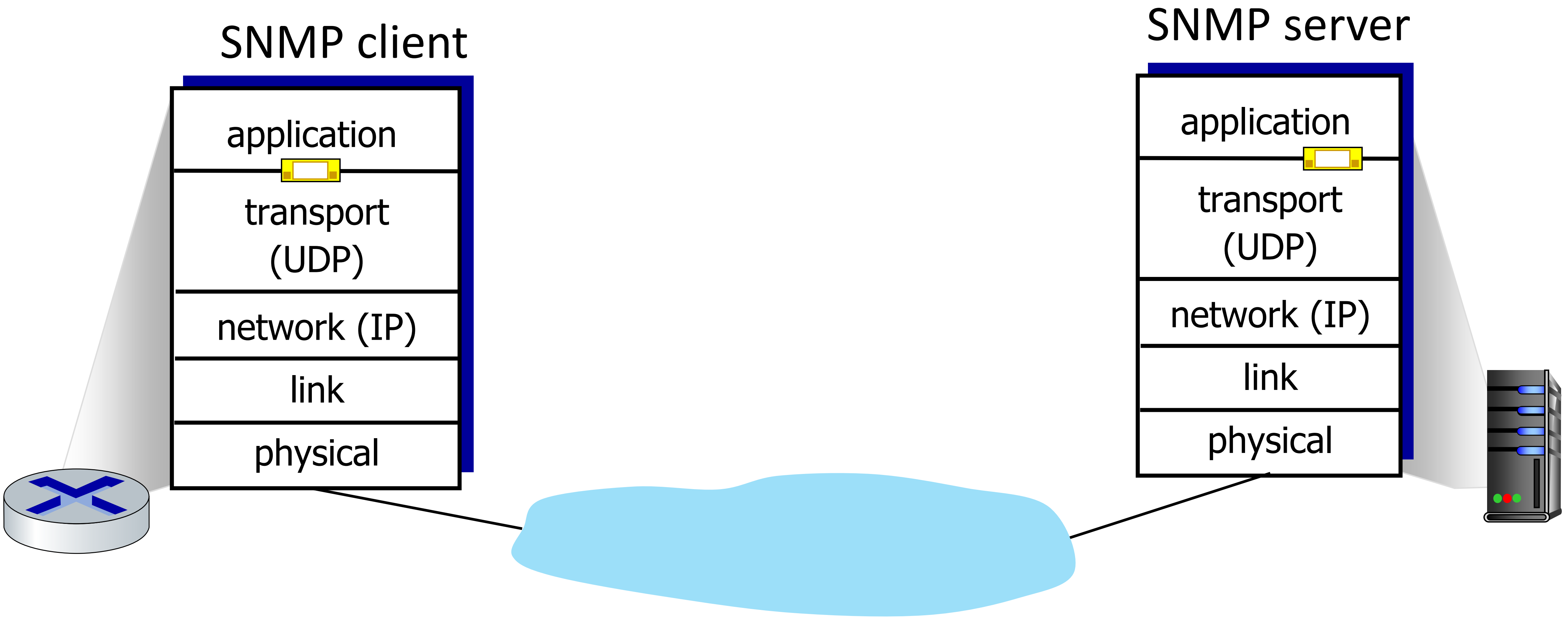
This User Datagram Protocol (UDP) is defined to make available a datagram mode of packet-switched computer communication in the environment of an interconnected set of computer networks. This protocol assumes that the Internet Protocol (IP) [1] is used as the underlying protocol.

This protocol provides a procedure for application programs to send messages to other programs with a minimum of protocol mechanism. The protocol is transaction oriented, and delivery and duplicate protection are not guaranteed. Applications requiring ordered reliable delivery of streams of data should use the Transmission Control Protocol (TCP) [2].

Format

```
 0      7 8     15 16    23 24    31
+--------+--------+--------+--------+
|     Source      |   Destination   |
|      Port       |      Port       |
+--------+--------+--------+--------+
|                 |                 |
|     Length      |    Checksum     |
+--------+--------+--------+--------+
|
|          data octets ...
+---------------- ...
```

UDP: Transport Layer Actions



SNMP client

application

transport (UDP)

network (IP)

link

physical

SNMP server

application

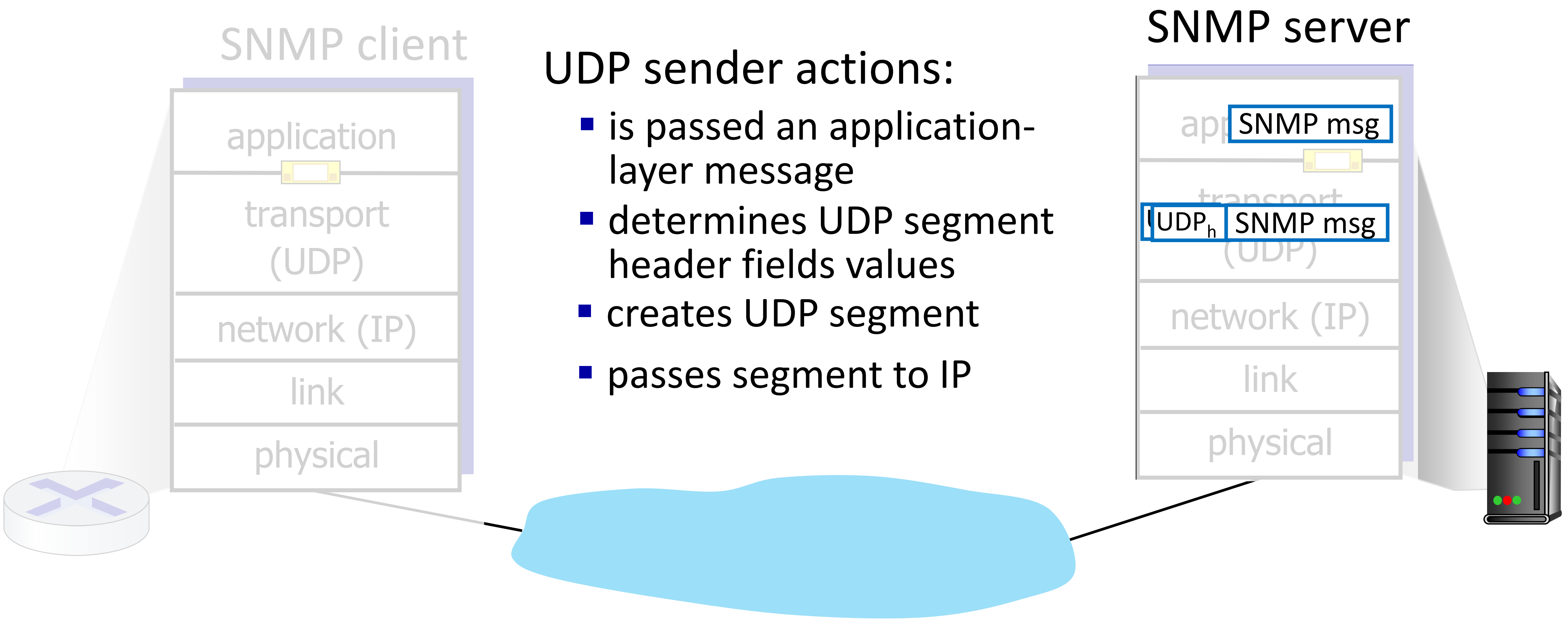
transport (UDP)

network (IP)

link

physical

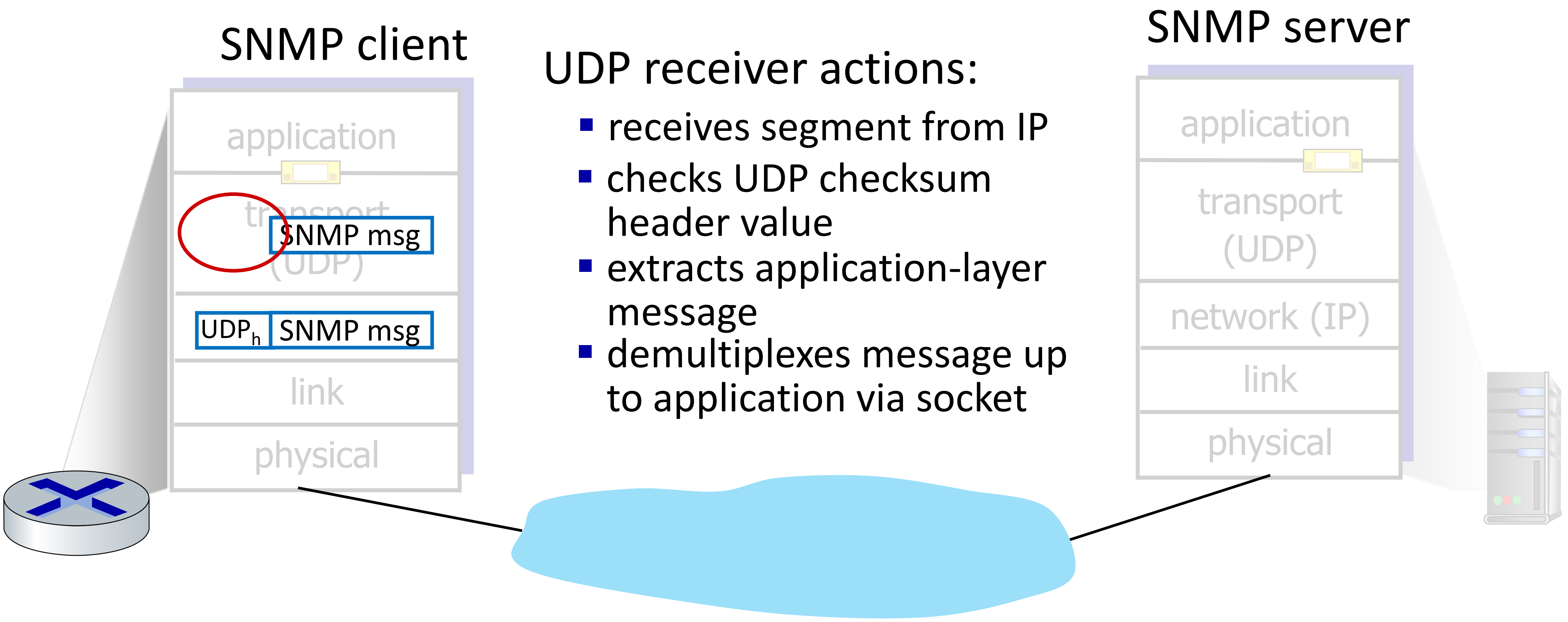
UDP: Transport Layer Actions


SNMP client
application
transport
(UDP)
network (IP)
link
physical
SNMP server
application
SNMP msg
transport
(UDP)
UDP$_h$
SNMP msg
network (IP)
link
physical

UDP sender actions:

- is passed an application-layer message
- determines UDP segment header fields values
- creates UDP segment
- passes segment to IP

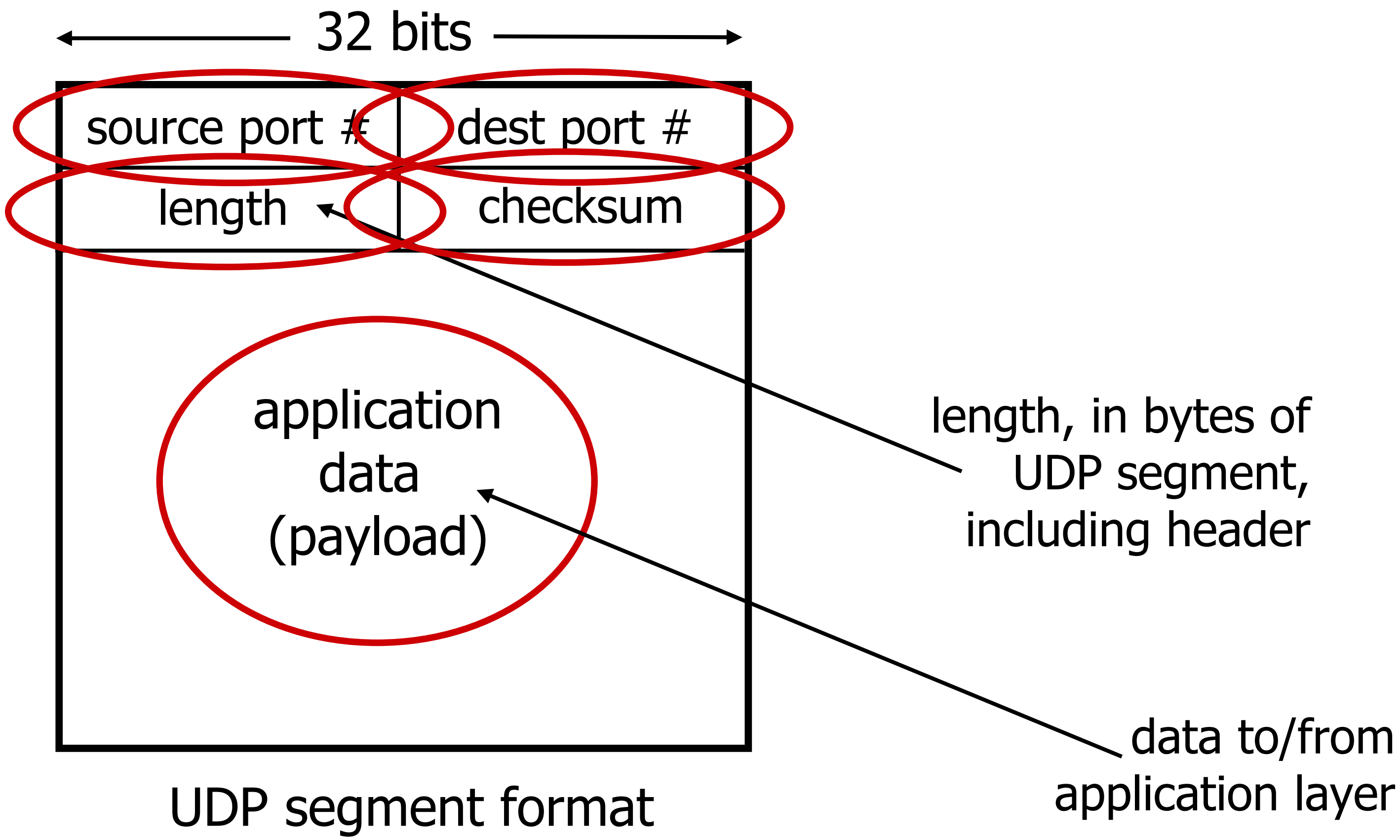
UDP: Transport Layer Actions


SNMP client
application
transport
(UDP)
SNMP msg
UDP$_h$ SNMP msg
link
physical
SNMP server
application
transport
(UDP)
network (IP)
link
physical

UDP receiver actions:

- receives segment from IP
- checks UDP checksum header value
- extracts application-layer message
- demultiplexes message up to application via socket

UDP segment header


32 bits
source port #
dest port #
length
checksum
application
data
(payload)

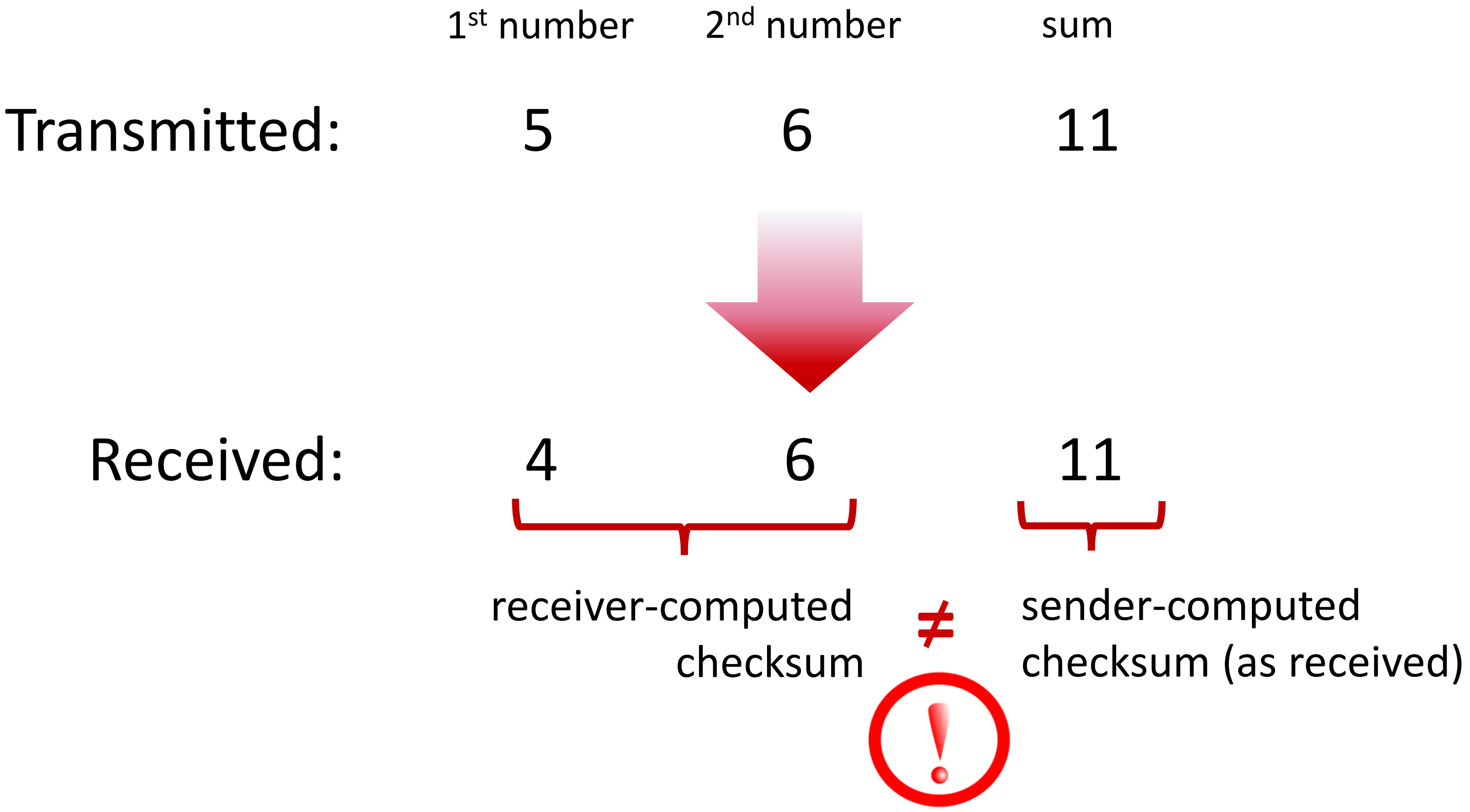
UDP segment format

length, in bytes of UDP segment, including header

data to/from application layer

UDP checksum

Goal: detect errors (*i.e.,* flipped bits) in transmitted segment



	1st number	2nd number	sum
Transmitted:	5	6	11
Received:	4	6	11

$\underbrace{4 \quad 6}_{\text{receiver-computed checksum}} \neq \underbrace{11}_{\text{sender-computed checksum (as received)}}$

UDP checksum

Goal: detect errors (*i.e.*, flipped bits) in transmitted segment

sender:

- treat contents of UDP segment (including UDP header fields and IP addresses) as sequence of 16-bit integers
- checksum: addition (one's complement sum) of segment content
- checksum value put into UDP checksum field

receiver:

- compute checksum of received segment
- check if computed checksum equals checksum field value:
    - Not equal - error detected
    - Equal - no error detected. *But maybe errors nonetheless?* More later ….

Internet checksum: an example

example: add two 16-bit integers

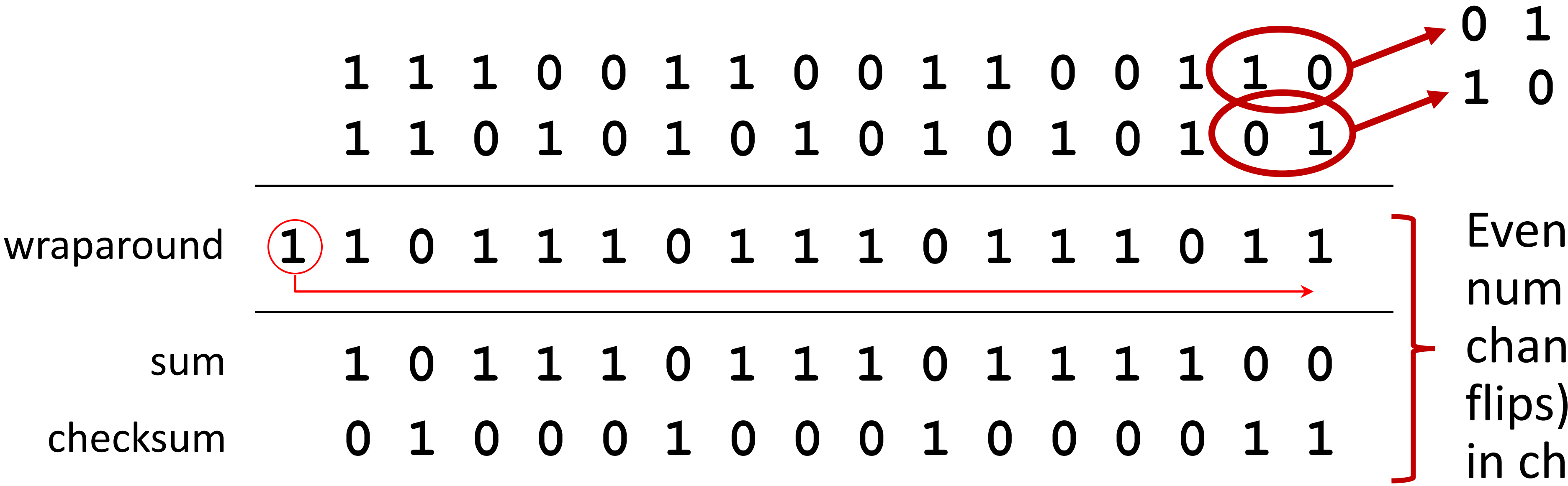
		1	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0
		1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
wraparound	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1	1
sum		1	0	1	1	1	0	1	1	1	0	1	1	1	1	0	0
checksum		0	1	0	0	0	1	0	0	0	1	0	0	0	0	1	1

Note: when adding numbers, a carryout from the most significant bit needs to be added to the result

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

Internet checksum: weak protection!

example: add two 16-bit integers



1 1 1 0 0 1 1 0 0 1 1 0 0 1 1 0 → 0 1

1 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 → 1 0

wraparound 1 1 0 1 1 1 0 1 1 1 0 1 1 1 0 1 1

sum 1 0 1 1 1 0 1 1 1 0 1 1 1 1 0 0

checksum 0 1 0 0 0 1 0 0 0 1 0 0 0 0 1 1

Even though numbers have changed (bit flips), *no* change in checksum!

Summary: UDP

- “no frills” protocol:
    - segments may be lost, delivered out of order
    - best effort service: “send and hope for the best”
- UDP has its plusses:
    - no setup/handshaking needed (no RTT incurred)
    - can function when network service is compromised
    - helps with reliability (checksum)
- build additional functionality on top of UDP in application layer (e.g., HTTP/3)

Chapter 3: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- **Principles of reliable data transfer**
- Connection-oriented transport: TCP
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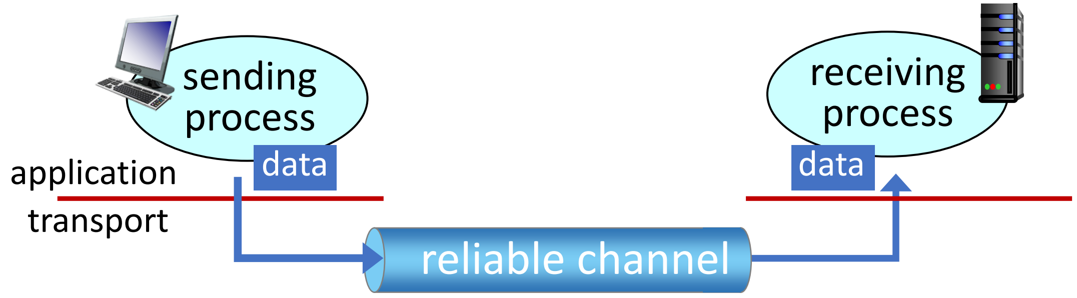
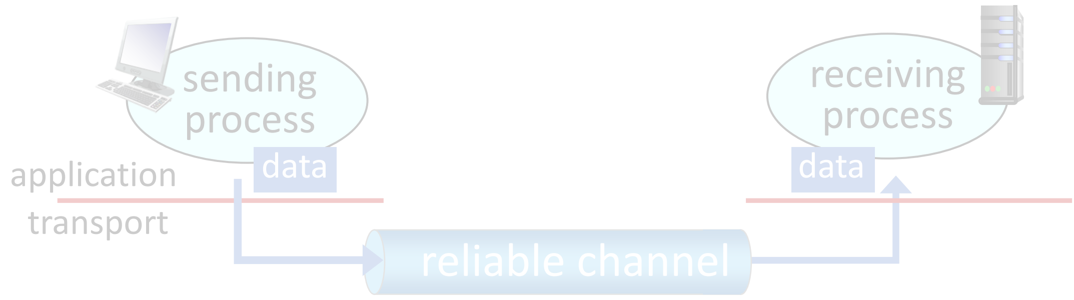
- Evolution of transport-layer functionality

Principles of reliable data transfer

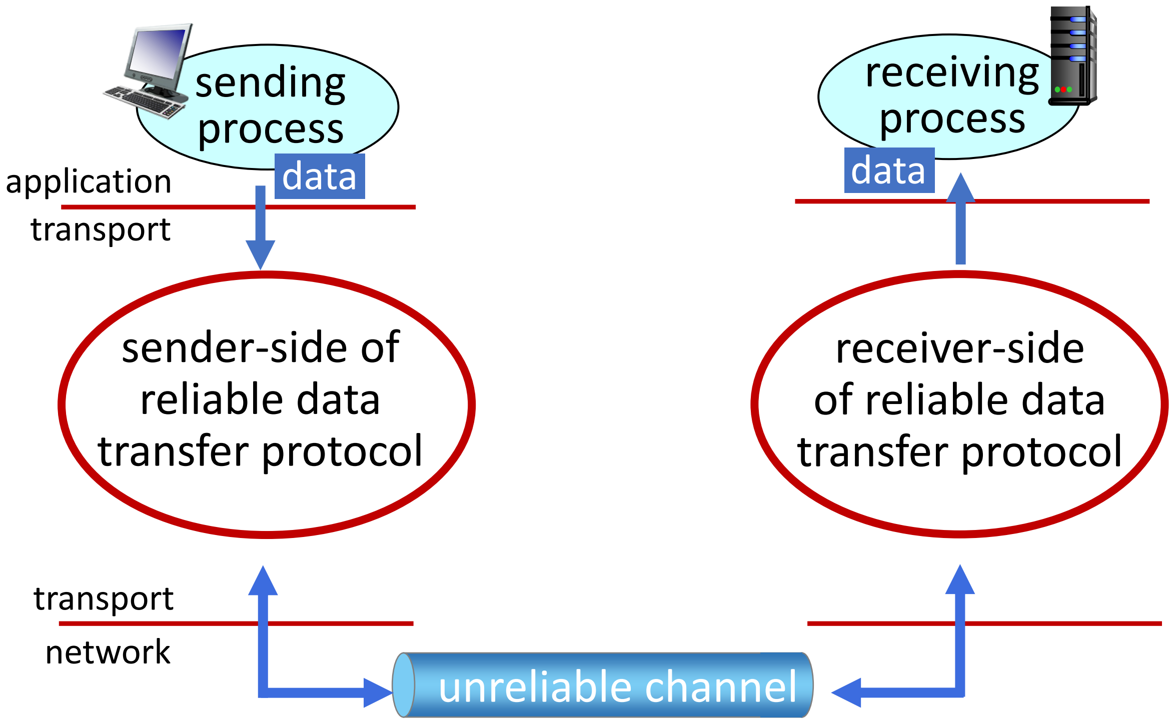

sending process
receiving process
data
data
application
transport
reliable channel

reliable service *abstraction*

Principles of reliable data transfer


sending process
receiving process
application
transport
data
data
reliable channel

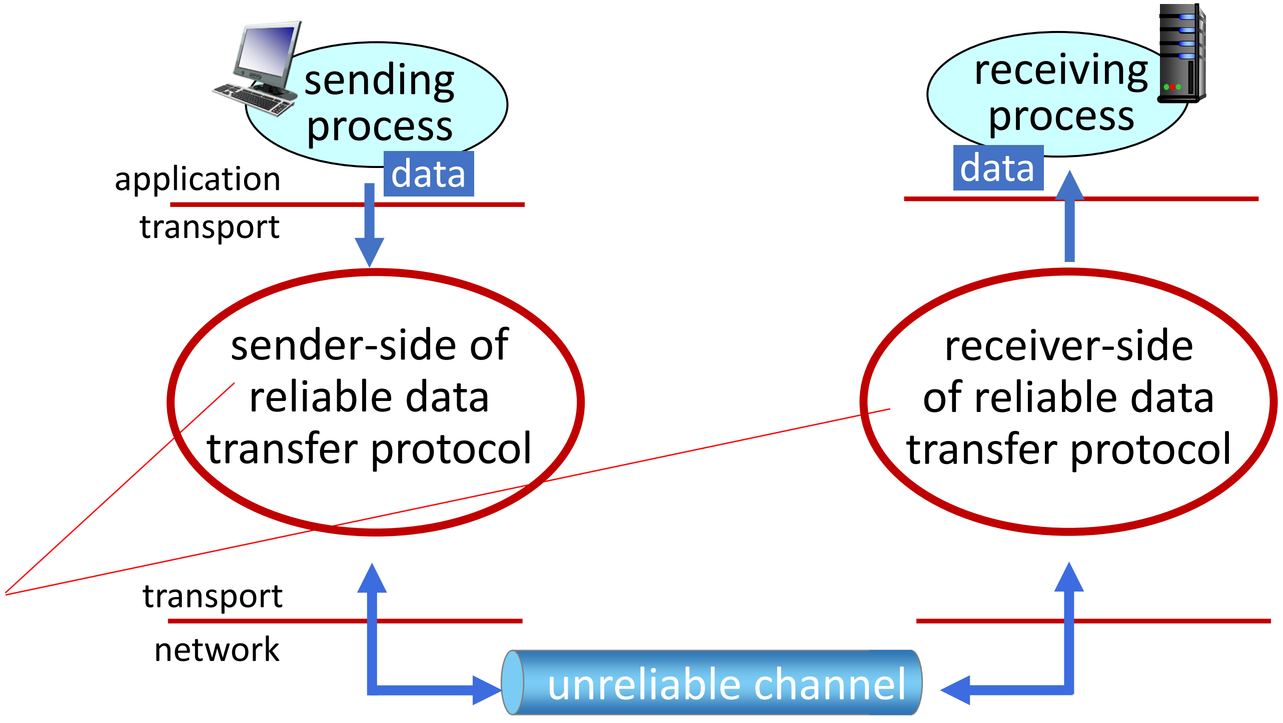
reliable service *abstraction*


sending process
receiving process
application
transport
data
data
sender-side of reliable data transfer protocol
receiver-side of reliable data transfer protocol
transport
network
unreliable channel

reliable service *implementation*

Principles of reliable data transfer

Complexity of reliable data transfer protocol will depend (strongly) on characteristics of unreliable channel (lose, corrupt, reorder data?)

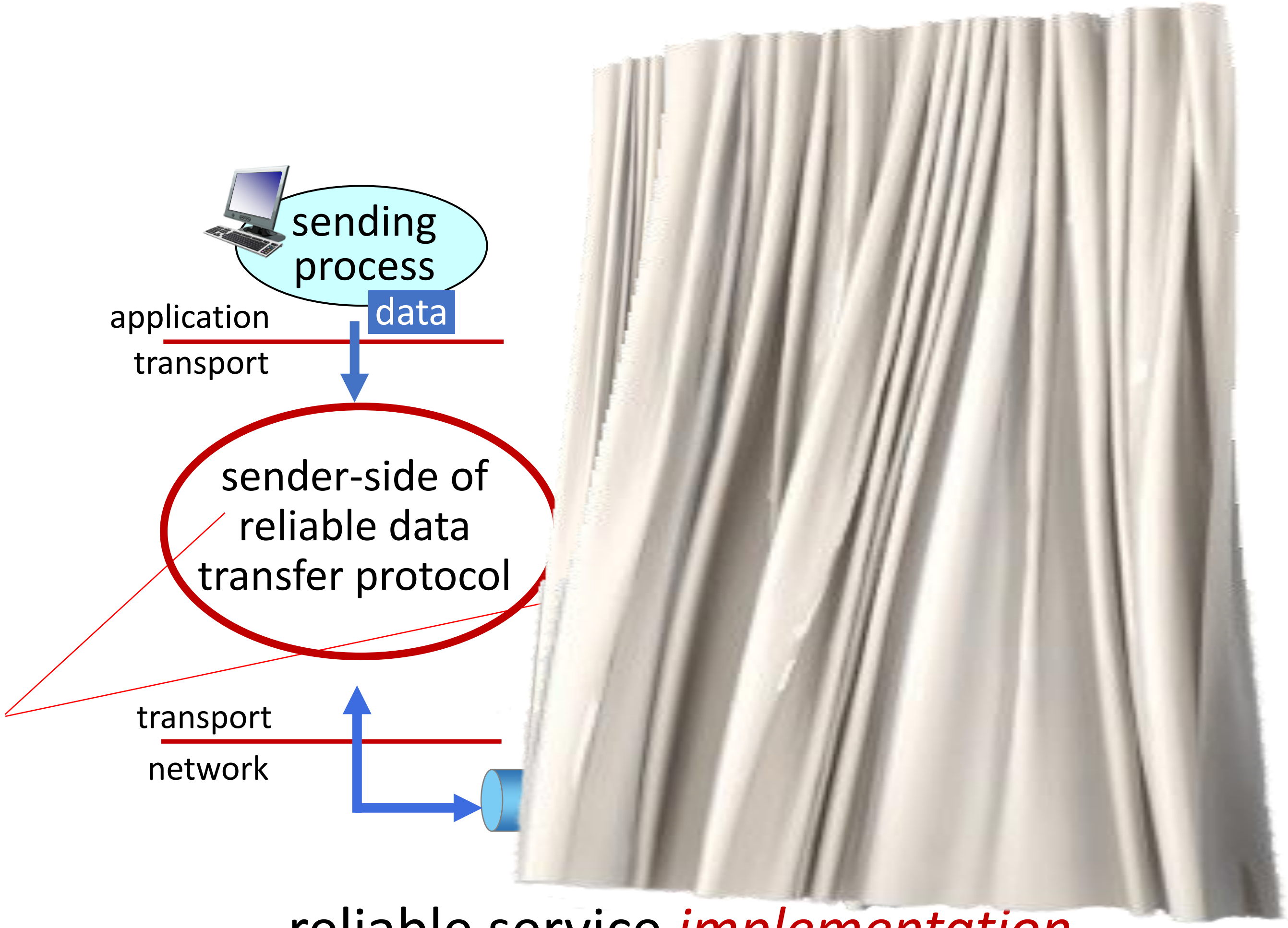

sending process
data
application
transport
sender-side of reliable data transfer protocol
transport
network
unreliable channel
receiving process
data
receiver-side of reliable data transfer protocol

reliable service *implementation*

Principles of reliable data transfer

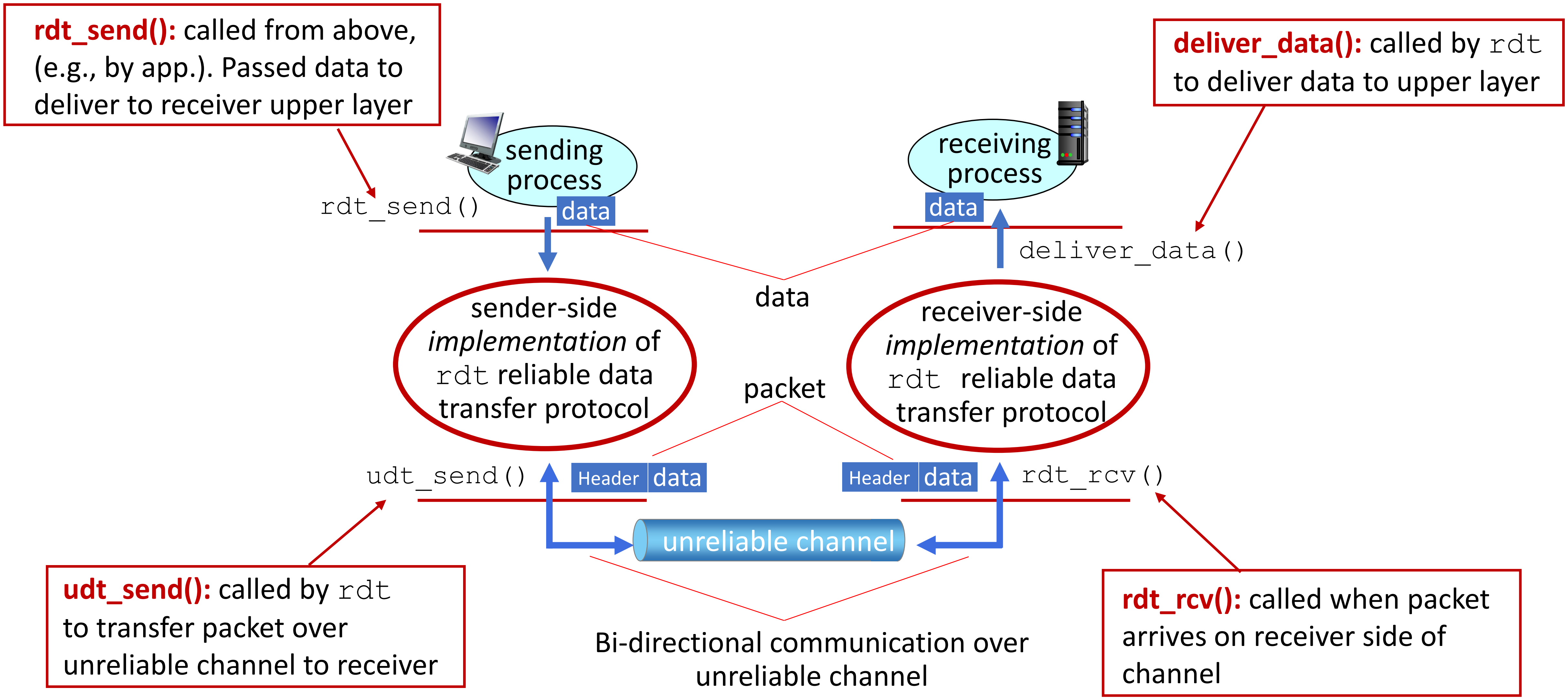
Sender, receiver do *not* know the “state” of each other, e.g., was a message received?

- unless communicated via a message


sending process
data
application
transport
sender-side of reliable data transfer protocol
transport
network

reliable service *implementation*

Reliable data transfer protocol (rdt): interfaces



rdt_send(): called from above, (e.g., by app.). Passed data to deliver to receiver upper layer

deliver_data(): called by `rdt` to deliver data to upper layer

sending process

receiving process

data

data

`rdt_send()`

`deliver_data()`

data

sender-side *implementation* of `rdt` reliable data transfer protocol

receiver-side *implementation* of `rdt` reliable data transfer protocol

packet

`udt_send()`

Header data

Header data

`rdt_rcv()`

unreliable channel

Bi-directional communication over unreliable channel

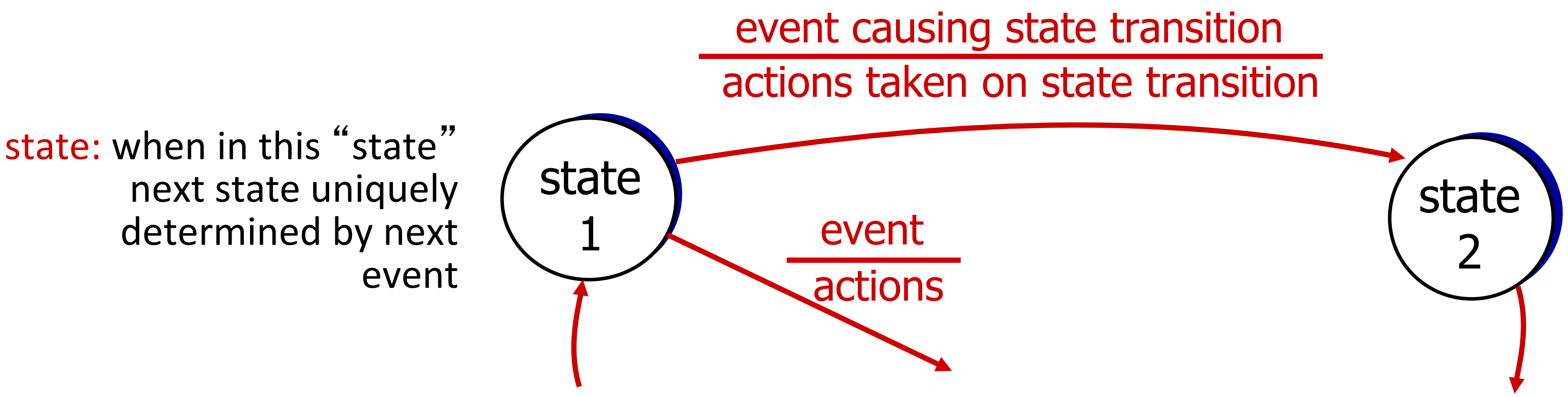
udt_send(): called by `rdt` to transfer packet over unreliable channel to receiver

rdt_rcv(): called when packet arrives on receiver side of channel

Reliable data transfer: getting started

We will:

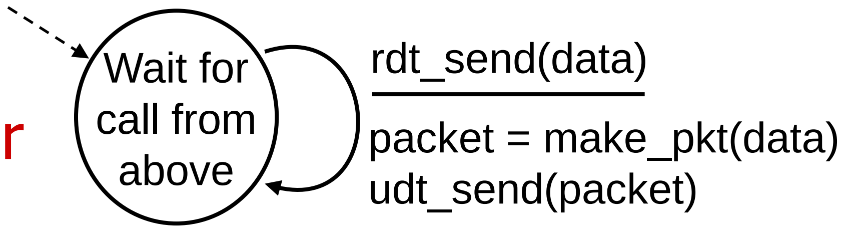
- incrementally develop sender, receiver sides of reliable data transfer protocol (`rdt`)
- consider only unidirectional data transfer
  - but control info will flow in both directions!
- use finite state machines (FSM) to specify sender, receiver


event causing state transition
actions taken on state transition
state: when in this "state" next state uniquely determined by next event
state 1
state 2
event
actions

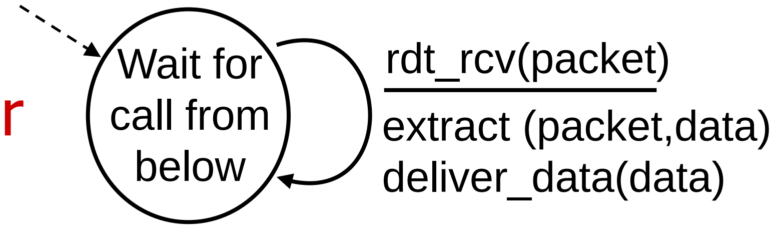
rdt1.0: reliable transfer over a reliable channel

- underlying channel perfectly reliable
  - no bit errors
  - no loss of packets
- *separate* FSMs for sender, receiver:
  - sender sends data into underlying channel
  - receiver reads data from underlying channel

sender


Wait for call from above
rdt_send(data)
packet = make_pkt(data)
udt_send(packet)

receiver


Wait for call from below
rdt_rcv(packet)
extract (packet,data)
deliver_data(data)

rdt2.0: channel with bit errors

- underlying channel may flip bits in packet
  - checksum (e.g., Internet checksum) to detect bit errors
- *the* question: how to recover from errors?

How do humans recover from "errors" during conversation?

rdt2.0: channel with bit errors

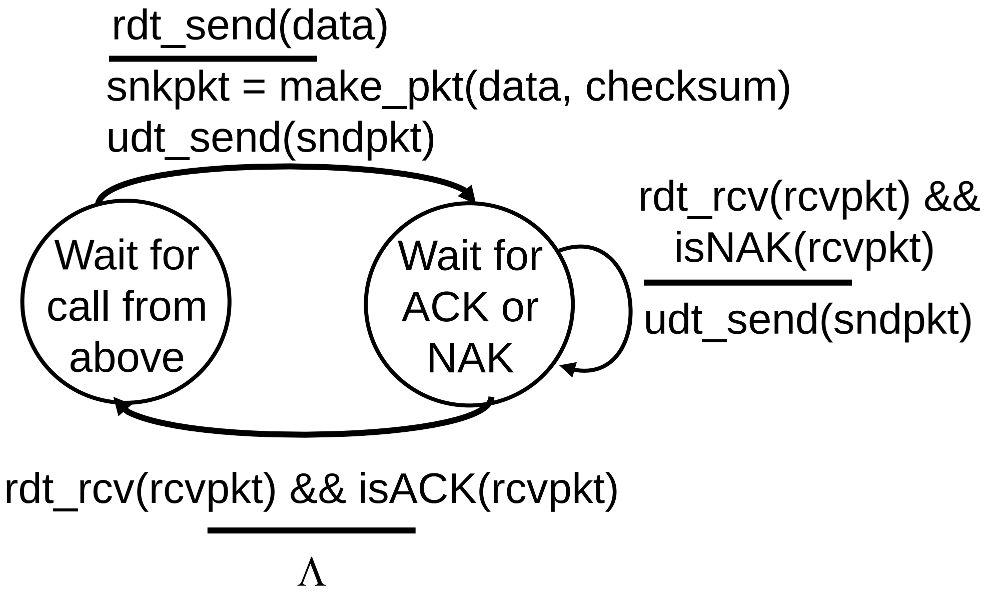
- underlying channel may flip bits in packet
  - checksum to detect bit errors
- *the* question: how to recover from errors?
  - *acknowledgements (ACKs):* receiver explicitly tells sender that pkt received OK
  - *negative acknowledgements (NAKs):* receiver explicitly tells sender that pkt had errors
  - sender *retransmits* pkt on receipt of NAK

stop and wait

sender sends one packet, then waits for receiver response

rdt2.0: FSM specifications

sender



rdt_send(data)

snkpkt = make_pkt(data, checksum)
udt_send(sndpkt)

Wait for call from above

Wait for ACK or NAK

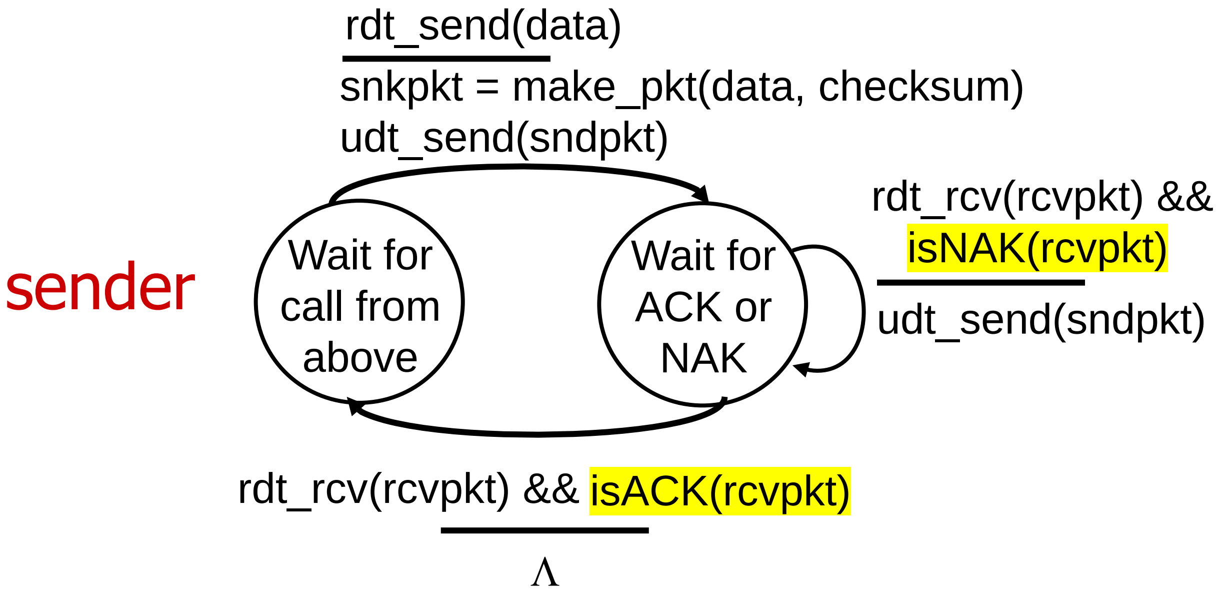
rdt_rcv(rcvpkt) &&
isNAK(rcvpkt)

udt_send(sndpkt)

rdt_rcv(rcvpkt) && isACK(rcvpkt)

Λ

rdt2.0: FSM specification



sender

rdt_send(data)

snkpkt = make_pkt(data, checksum)
udt_send(sndpkt)

Wait for call from above

Wait for ACK or NAK

rdt_rcv(rcvpkt) && isNAK(rcvpkt)

udt_send(sndpkt)

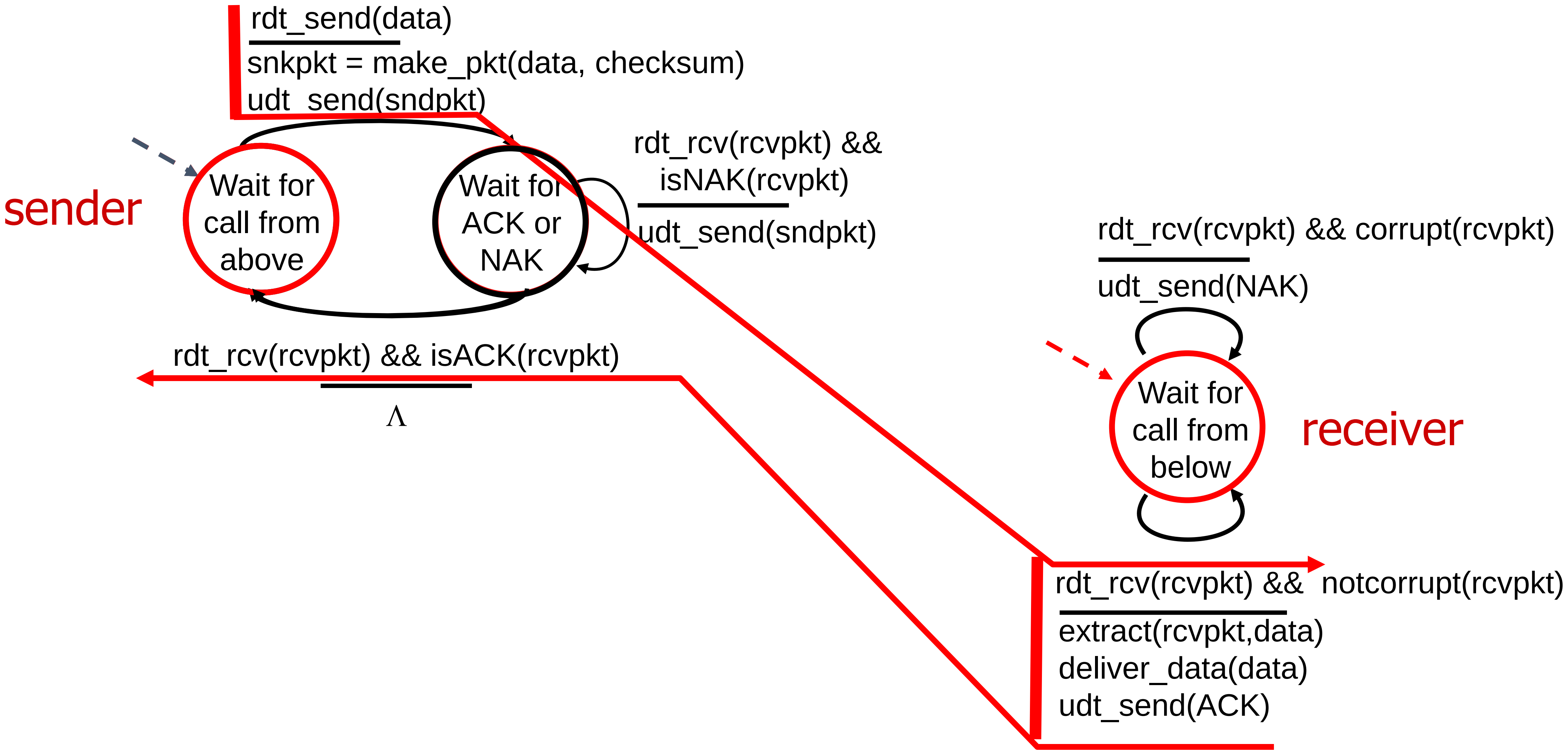
rdt_rcv(rcvpkt) && isACK(rcvpkt)

Λ

Note: “state” of receiver (did the receiver get my message correctly?) isn’t known to sender unless somehow communicated from receiver to sender

- that’s why we need a protocol!

rdt2.0: operation with no errors



sender

rdt_send(data)
snkpkt = make_pkt(data, checksum)
udt_send(sndpkt)

Wait for call from above

Wait for ACK or NAK

rdt_rcv(rcvpkt) && isNAK(rcvpkt)
udt_send(sndpkt)

rdt_rcv(rcvpkt) && isACK(rcvpkt)
Λ

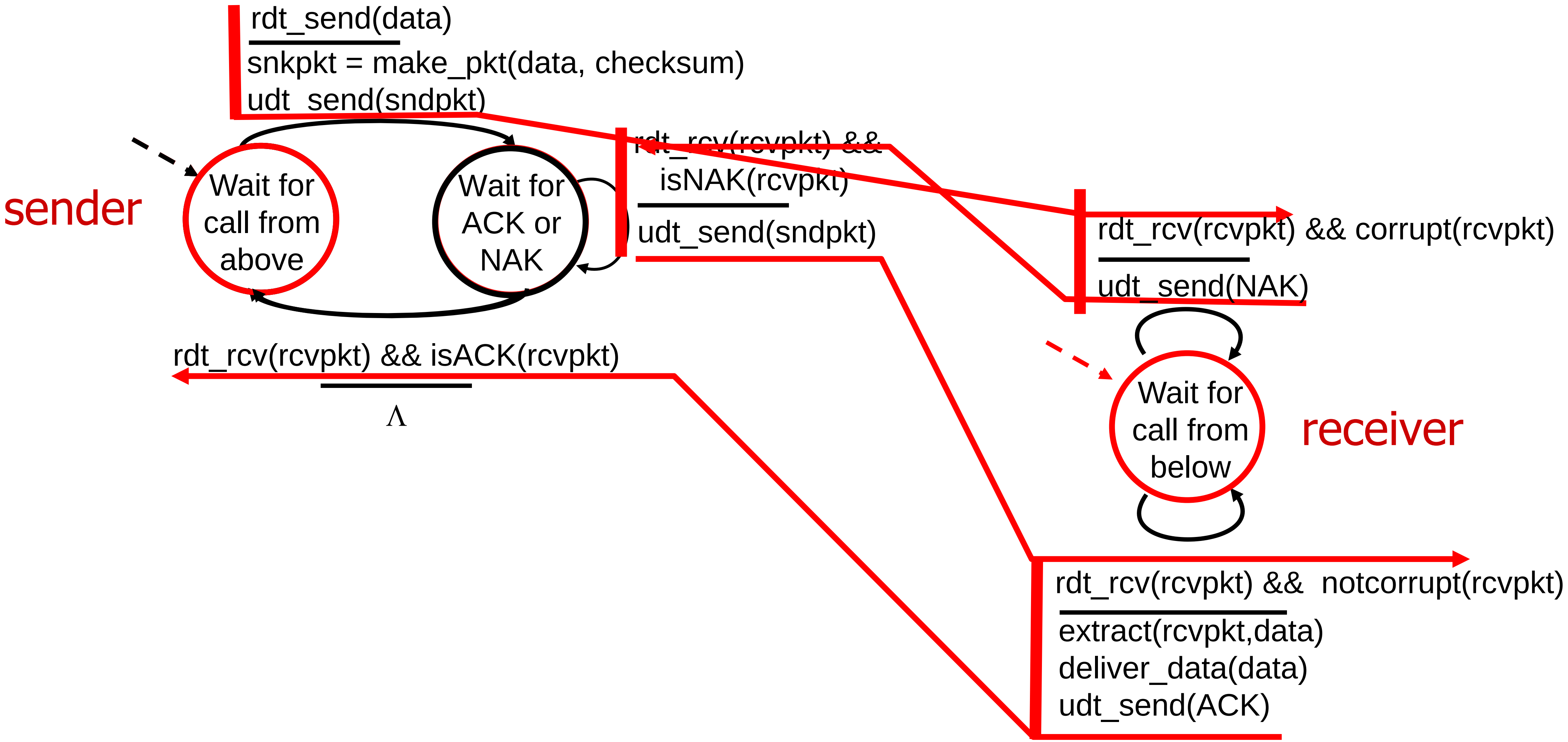
receiver

rdt_rcv(rcvpkt) && corrupt(rcvpkt)
udt_send(NAK)

Wait for call from below

rdt_rcv(rcvpkt) && notcorrupt(rcvpkt)
extract(rcvpkt,data)
deliver_data(data)
udt_send(ACK)

rdt2.0: corrupted packet scenario


rdt_send(data)
snkpkt = make_pkt(data, checksum)
udt_send(sndpkt)
sender
Wait for call from above
Wait for ACK or NAK
rdt_rcv(rcvpkt) && isNAK(rcvpkt)
udt_send(sndpkt)
rdt_rcv(rcvpkt) && isACK(rcvpkt)
Λ
rdt_rcv(rcvpkt) && corrupt(rcvpkt)
udt_send(NAK)
Wait for call from below
receiver
rdt_rcv(rcvpkt) && notcorrupt(rcvpkt)
extract(rcvpkt,data)
deliver_data(data)
udt_send(ACK)

rdt2.0 has a fatal flaw!

what happens if ACK/NAK corrupted?

- sender doesn't know what happened at receiver!
- can't just retransmit: possible duplicate

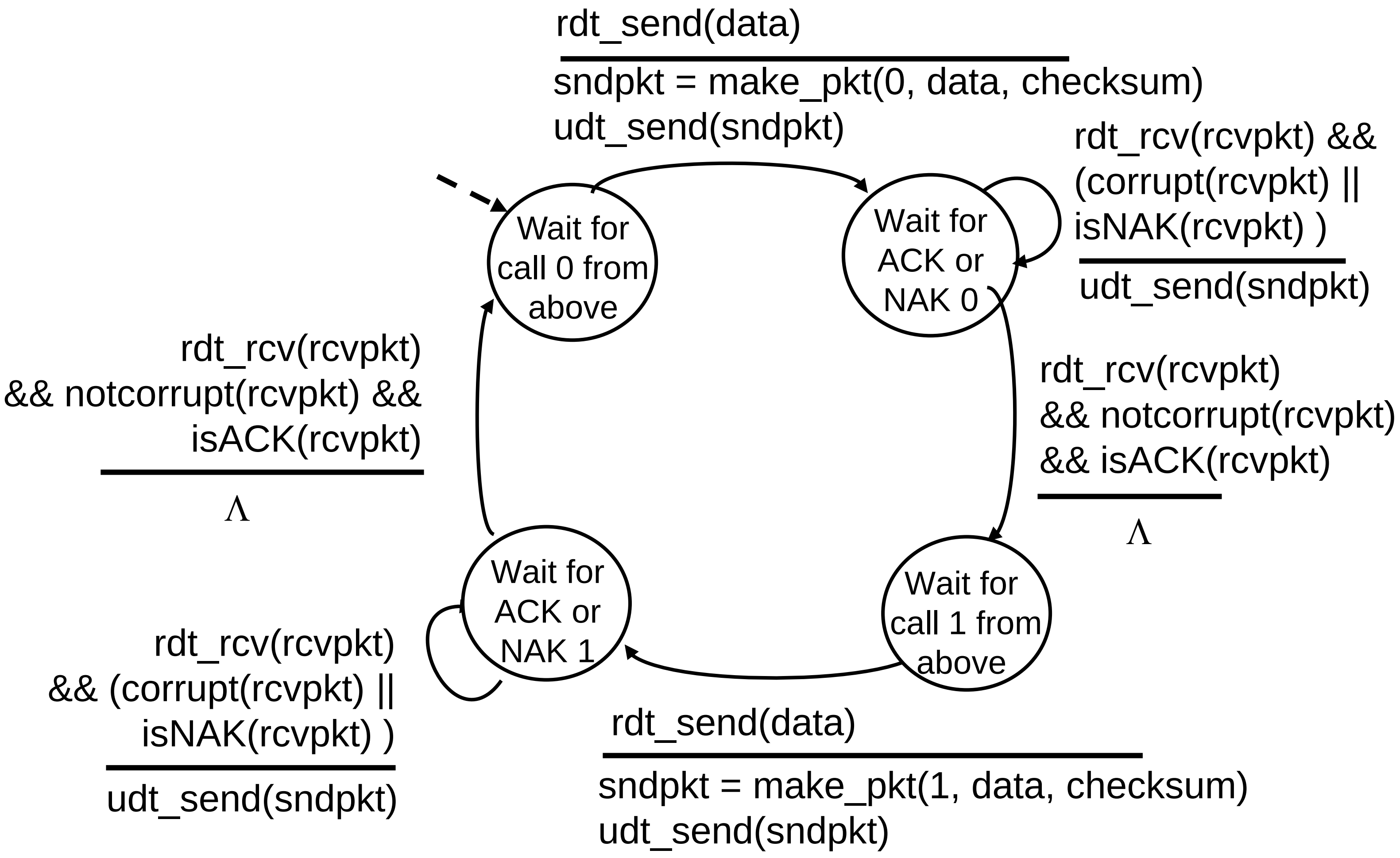
handling duplicates:

- sender retransmits current pkt if ACK/NAK corrupted
- sender adds *sequence number* to each pkt
- receiver discards (doesn't deliver up) duplicate pkt

stop and wait

sender sends one packet, then waits for receiver response

rdt2.1: sender, handling garbled ACK/NAKs



Wait for call 0 from above

rdt_send(data)

sndpkt = make_pkt(0, data, checksum)
udt_send(sndpkt)

Wait for ACK or NAK 0

rdt_rcv(rcvpkt) &&
(corrupt(rcvpkt) ||
isNAK(rcvpkt))

udt_send(sndpkt)

rdt_rcv(rcvpkt)
&& notcorrupt(rcvpkt)
&& isACK(rcvpkt)

Λ

Wait for call 1 from above

rdt_send(data)

sndpkt = make_pkt(1, data, checksum)
udt_send(sndpkt)

Wait for ACK or NAK 1

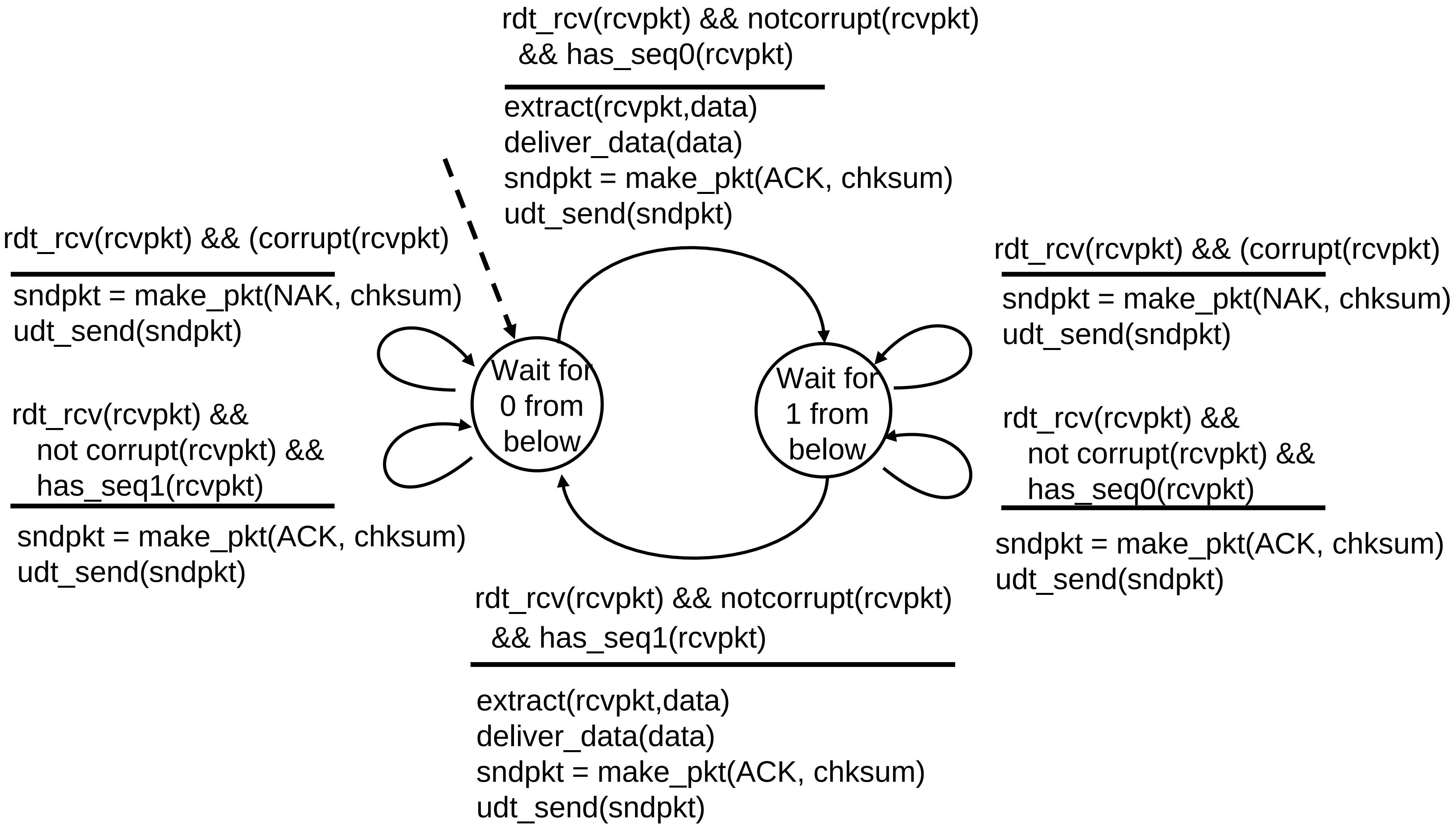
rdt_rcv(rcvpkt)
&& (corrupt(rcvpkt) ||
isNAK(rcvpkt))

udt_send(sndpkt)

rdt_rcv(rcvpkt)
&& notcorrupt(rcvpkt) &&
isACK(rcvpkt)

Λ

rdt2.1: receiver, handling garbled ACK/NAKs



rdt_rcv(rcvpkt) && notcorrupt(rcvpkt)
&& has_seq0(rcvpkt)

extract(rcvpkt,data)
deliver_data(data)
sndpkt = make_pkt(ACK, chksum)
udt_send(sndpkt)

rdt_rcv(rcvpkt) && (corrupt(rcvpkt)

sndpkt = make_pkt(NAK, chksum)
udt_send(sndpkt)

rdt_rcv(rcvpkt) &&
not corrupt(rcvpkt) &&
has_seq1(rcvpkt)

sndpkt = make_pkt(ACK, chksum)
udt_send(sndpkt)

rdt_rcv(rcvpkt) && (corrupt(rcvpkt)

sndpkt = make_pkt(NAK, chksum)
udt_send(sndpkt)

rdt_rcv(rcvpkt) &&
not corrupt(rcvpkt) &&
has_seq0(rcvpkt)

sndpkt = make_pkt(ACK, chksum)
udt_send(sndpkt)

rdt_rcv(rcvpkt) && notcorrupt(rcvpkt)
&& has_seq1(rcvpkt)

extract(rcvpkt,data)
deliver_data(data)
sndpkt = make_pkt(ACK, chksum)
udt_send(sndpkt)

rdt2.1: discussion

sender:

- seq # added to pkt
- two seq. #s (0,1) will suffice. Why?
- must check if received ACK/NAK corrupted
- twice as many states
    - state must "remember" whether "expected" pkt should have seq # of 0 or 1

receiver:

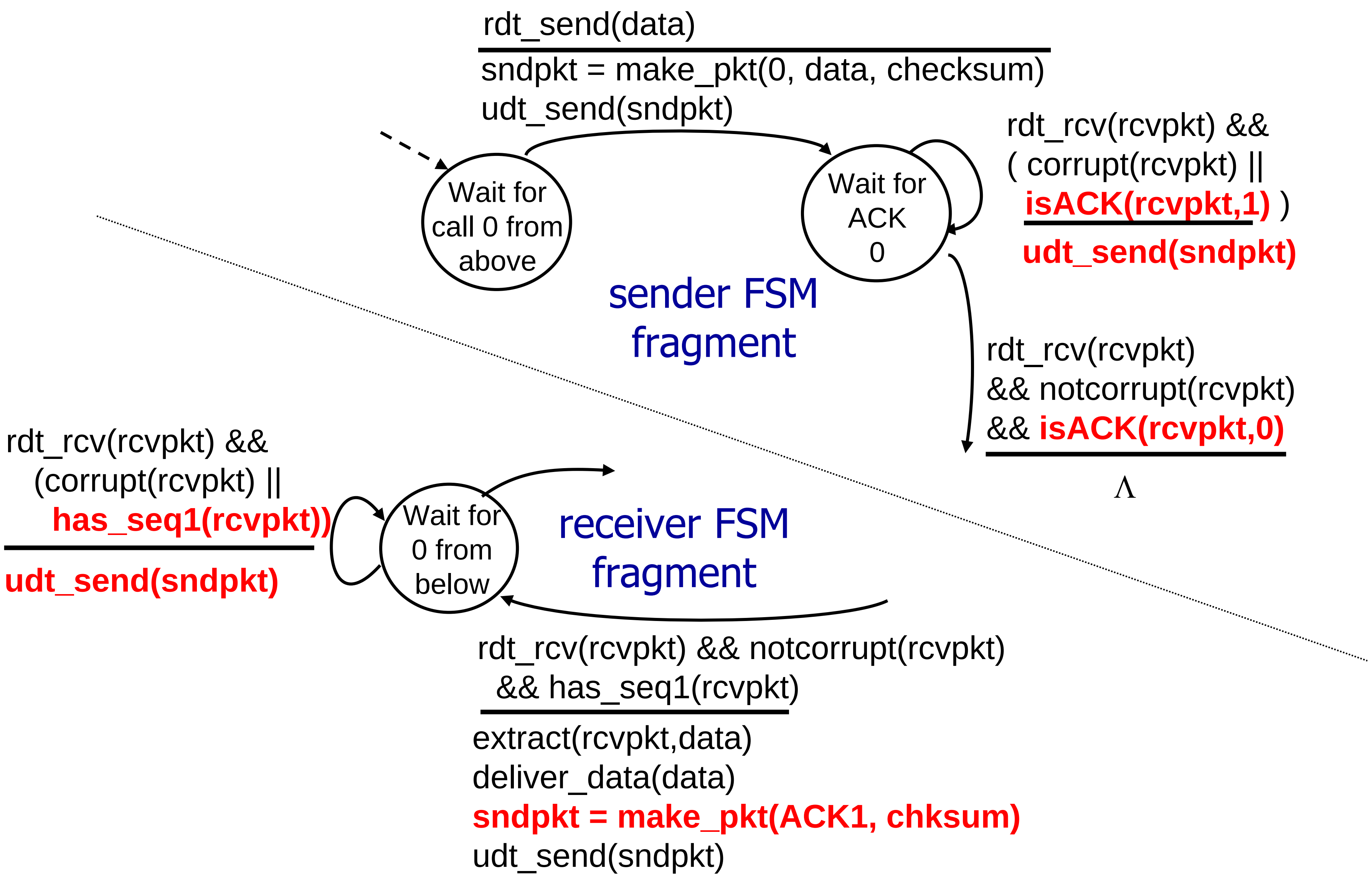
- must check if received packet is duplicate
    - state indicates whether 0 or 1 is expected pkt seq #
- note: receiver can *not* know if its last ACK/NAK received OK at sender

rdt2.2: a NAK-free protocol

- same functionality as rdt2.1, using ACKs only
- instead of NAK, receiver sends ACK for last pkt received OK
  - receiver must *explicitly* include seq # of pkt being ACKed
- duplicate ACK at sender results in same action as NAK: *retransmit current pkt*

As we will see, TCP uses this approach to be NAK-free

rdt2.2: sender, receiver fragments


Wait for call 0 from above
Wait for ACK 0
Wait for 0 from below

rdt_send(data)

sndpkt = make_pkt(0, data, checksum)
udt_send(sndpkt)

rdt_rcv(rcvpkt) &&
(corrupt(rcvpkt) ||
isACK(rcvpkt,1))

udt_send(sndpkt)

sender FSM fragment

rdt_rcv(rcvpkt)
&& notcorrupt(rcvpkt)
&& **isACK(rcvpkt,0)**

Λ

rdt_rcv(rcvpkt) &&
(corrupt(rcvpkt) ||
has_seq1(rcvpkt))

udt_send(sndpkt)

receiver FSM fragment

rdt_rcv(rcvpkt) && notcorrupt(rcvpkt)
&& has_seq1(rcvpkt)

extract(rcvpkt,data)
deliver_data(data)
sndpkt = make_pkt(ACK1, chksum)
udt_send(sndpkt)

rdt3.0: channels with errors *and* loss

New channel assumption: underlying channel can also *lose* packets (data, ACKs)

- checksum, sequence #s, ACKs, retransmissions will be of help … but not quite enough

Q: How do *humans* handle lost sender-to-receiver words in conversation?

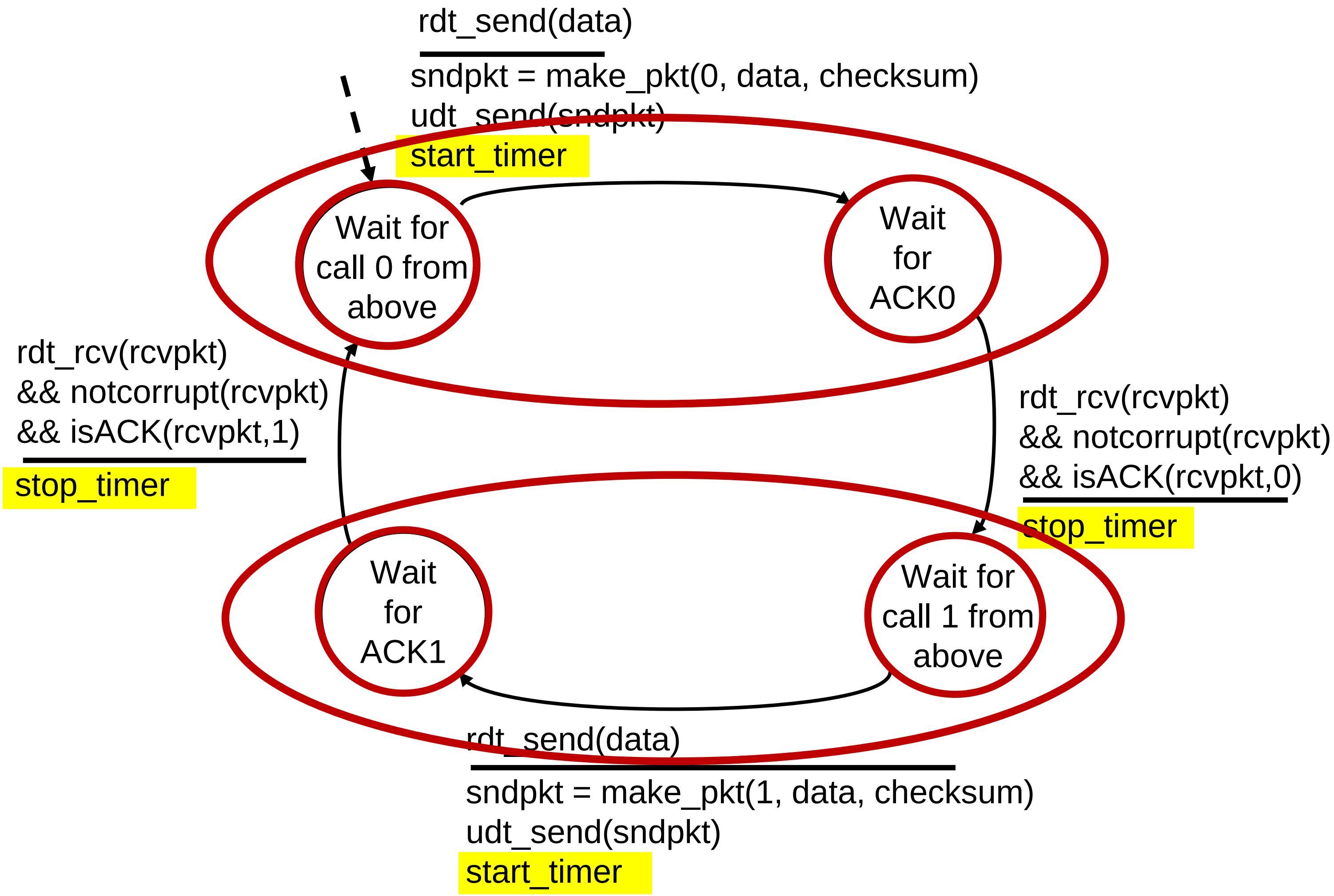
rdt3.0: channels with errors *and* loss

Approach: sender waits "reasonable" amount of time for ACK

- retransmits if no ACK received in this time
- if pkt (or ACK) just delayed (not lost):
  - retransmission will be duplicate, but seq #s already handles this!
  - receiver must specify seq # of packet being ACKed
- use countdown timer to interrupt after "reasonable" amount of time

timeout

rdt3.0 sender



rdt_send(data)

sndpkt = make_pkt(0, data, checksum)
udt_send(sndpkt)
start_timer

Wait for call 0 from above

Wait for ACK0

rdt_rcv(rcvpkt)
&& notcorrupt(rcvpkt)
&& isACK(rcvpkt,0)

stop_timer

Wait for call 1 from above

rdt_send(data)

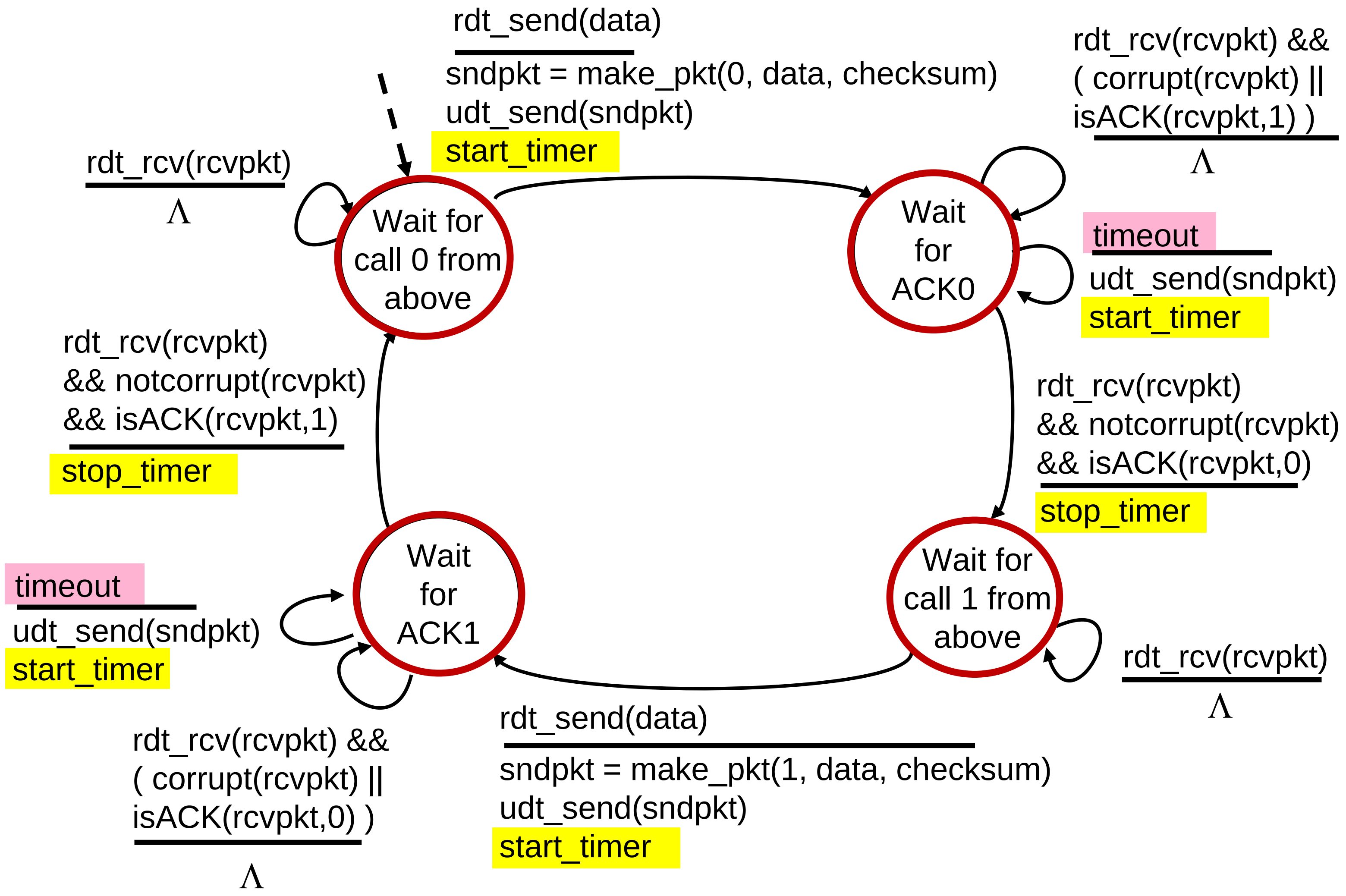
sndpkt = make_pkt(1, data, checksum)
udt_send(sndpkt)
start_timer

Wait for ACK1

rdt_rcv(rcvpkt)
&& notcorrupt(rcvpkt)
&& isACK(rcvpkt,1)

stop_timer

rdt3.0 sender



rdt_send(data)

sndpkt = make_pkt(0, data, checksum)
udt_send(sndpkt)
start_timer

rdt_rcv(rcvpkt)

Λ

Wait for call 0 from above

rdt_rcv(rcvpkt) &&
(corrupt(rcvpkt) ||
isACK(rcvpkt,1))

Λ

Wait for ACK0

timeout

udt_send(sndpkt)
start_timer

rdt_rcv(rcvpkt)
&& notcorrupt(rcvpkt)
&& isACK(rcvpkt,1)

stop_timer

rdt_rcv(rcvpkt)
&& notcorrupt(rcvpkt)
&& isACK(rcvpkt,0)

stop_timer

Wait for ACK1

Wait for call 1 from above

timeout

udt_send(sndpkt)
start_timer

rdt_rcv(rcvpkt)

Λ

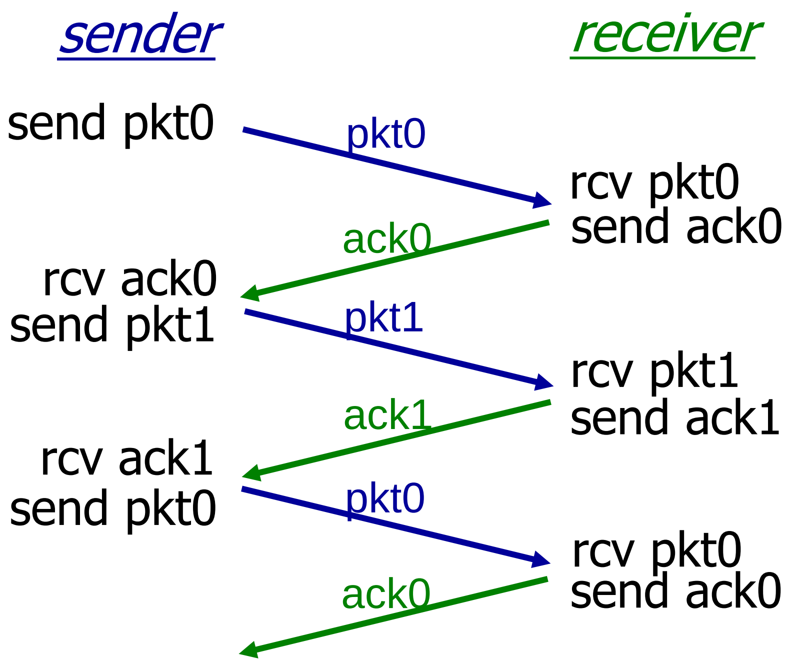
rdt_rcv(rcvpkt) &&
(corrupt(rcvpkt) ||
isACK(rcvpkt,0))

Λ

rdt_send(data)

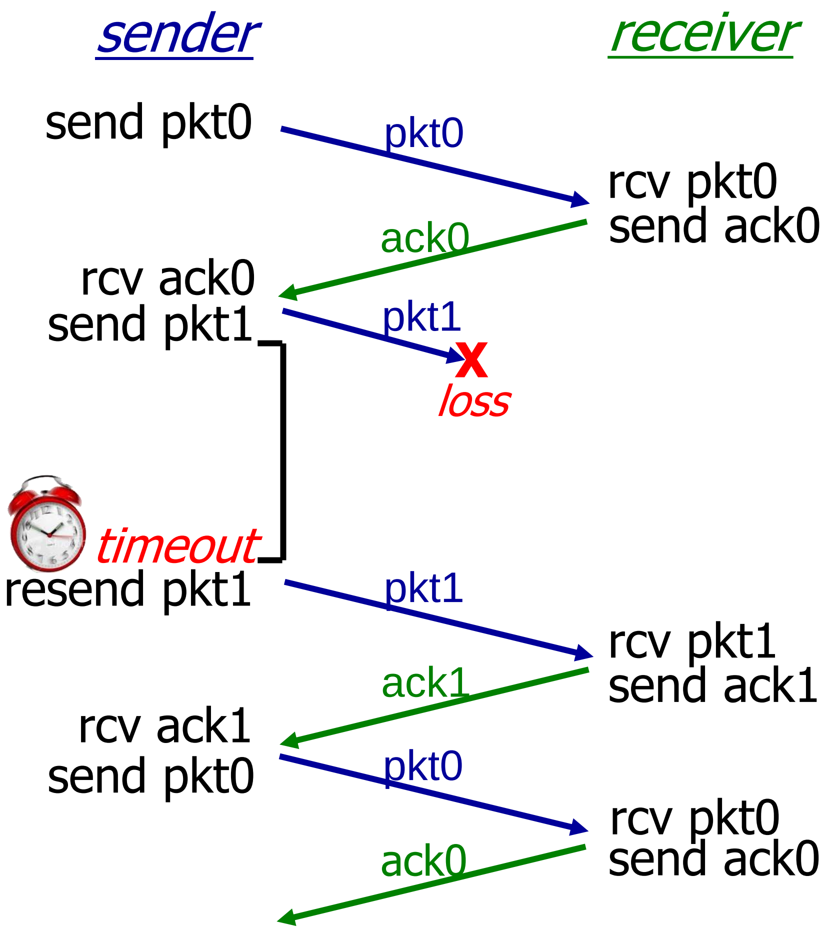
sndpkt = make_pkt(1, data, checksum)
udt_send(sndpkt)
start_timer

rdt3.0 in action



sender
receiver
send pkt0
pkt0
rcv pkt0
send ack0
ack0
rcv ack0
send pkt1
pkt1
rcv pkt1
send ack1
ack1
rcv ack1
send pkt0
pkt0
rcv pkt0
send ack0
ack0

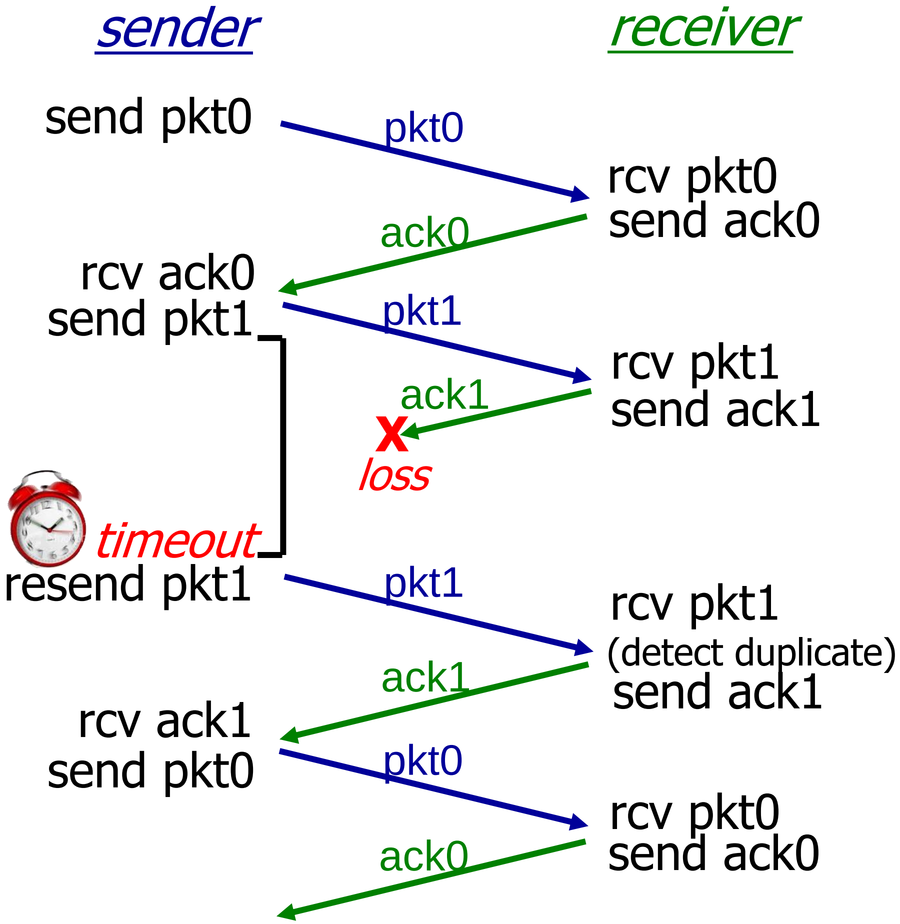
(a) no loss



sender
receiver
send pkt0
pkt0
rcv pkt0
send ack0
ack0
rcv ack0
send pkt1
pkt1
X
loss
timeout
resend pkt1
pkt1
rcv pkt1
send ack1
ack1
rcv ack1
send pkt0
pkt0
rcv pkt0
send ack0
ack0

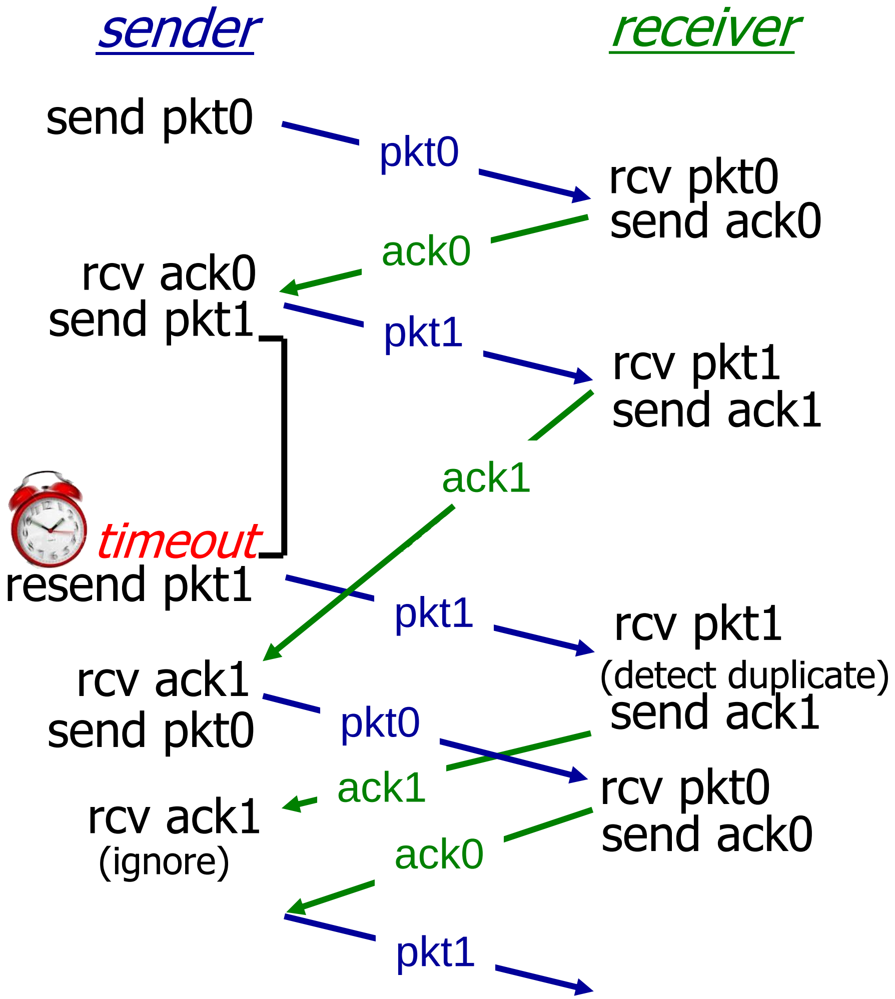
(b) packet loss

rdt3.0 in action



sender
receiver
send pkt0
pkt0
rcv pkt0
send ack0
ack0
rcv ack0
send pkt1
pkt1
rcv pkt1
send ack1
ack1
X
loss
timeout
resend pkt1
pkt1
rcv pkt1
(detect duplicate)
send ack1
ack1
rcv ack1
send pkt0
pkt0
rcv pkt0
send ack0
ack0

(c) ACK loss



sender
receiver
send pkt0
pkt0
rcv pkt0
send ack0
ack0
rcv ack0
send pkt1
pkt1
rcv pkt1
send ack1
ack1
timeout
resend pkt1
pkt1
rcv pkt1
(detect duplicate)
send ack1
rcv ack1
send pkt0
pkt0
rcv pkt0
send ack0
ack1
rcv ack1
(ignore)
ack0
pkt1

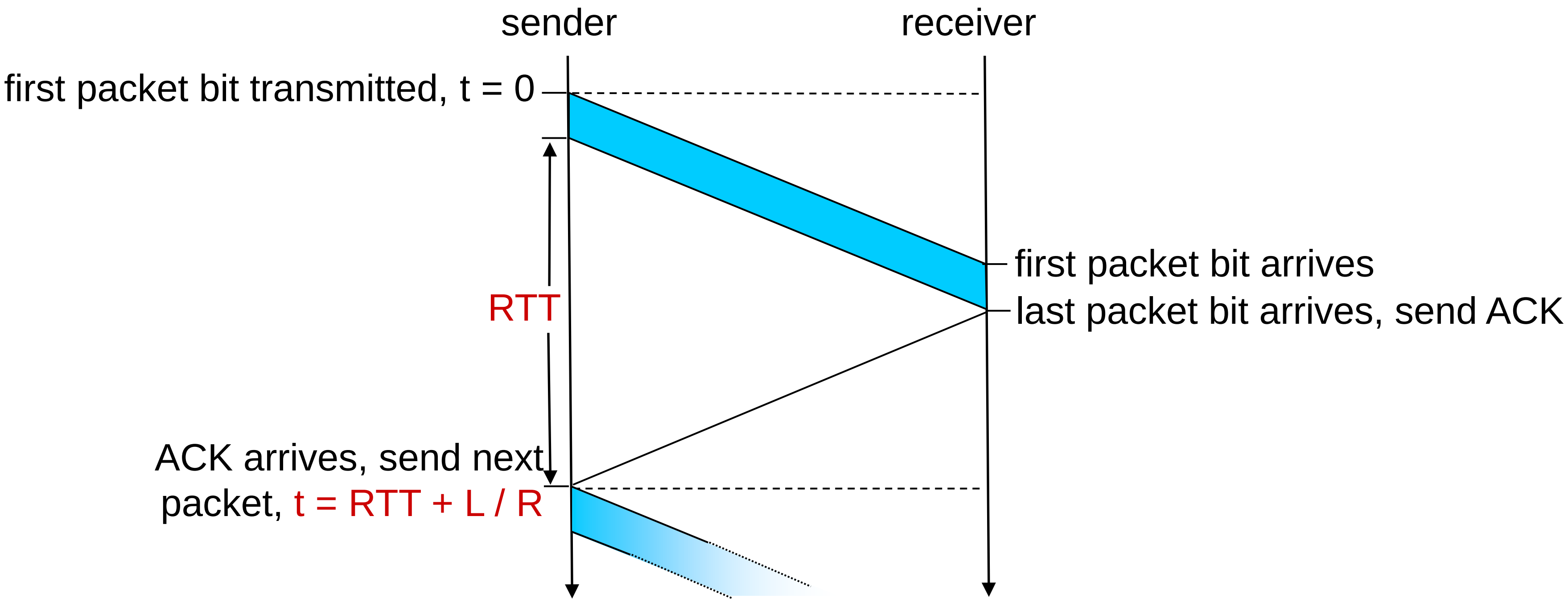
(d) premature timeout/ delayed ACK

Performance of rdt3.0 (stop-and-wait)

- U_{sender}: *utilization* – fraction of time sender busy sending
- example: 1 Gbps link, 15 ms prop. delay, 8000 bit packet
  - time to transmit packet into channel:

$$D_{trans} = \frac{L}{R} = \frac{8000 \text{ bits}}{10^9 \text{ bits/sec}} = 8 \text{ microsecs}$$

rdt3.0: stop-and-wait operation

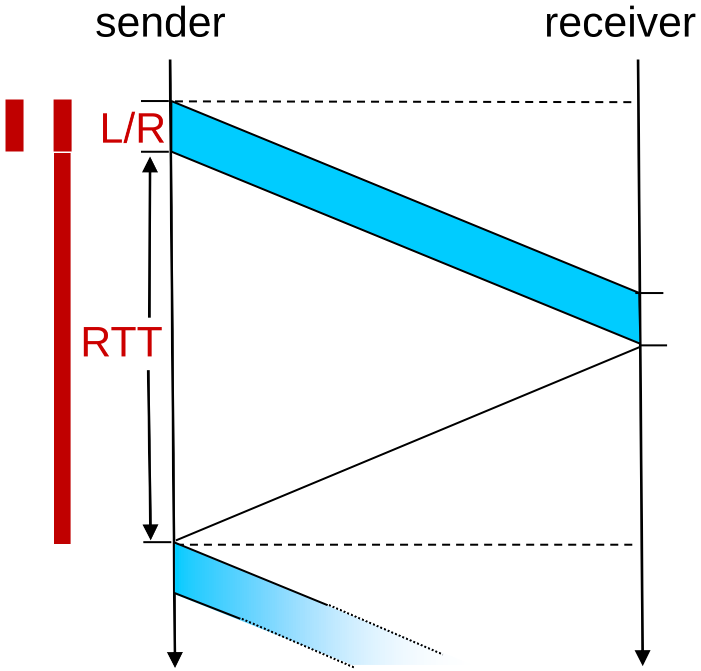

sender
receiver
first packet bit transmitted, t = 0
first packet bit arrives
last packet bit arrives, send ACK
RTT
ACK arrives, send next
packet, t = RTT + L / R

rdt3.0: stop-and-wait operation

$$U_{\text{sender}} = \frac{L / R}{RTT + L / R}$$

$$= \frac{.008}{30.008}$$

$$= 0.00027$$

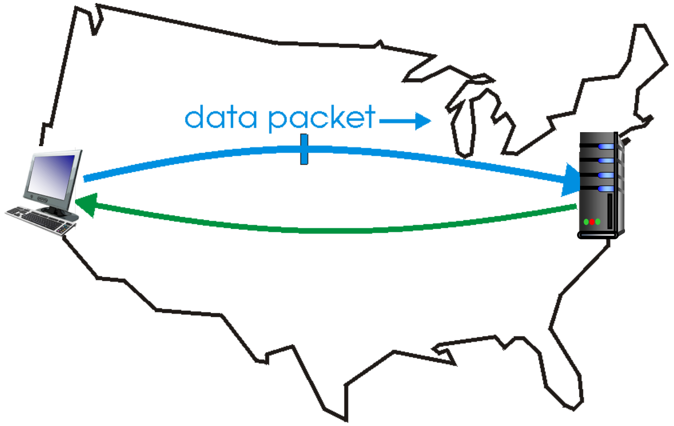

sender
receiver
L/R
RTT

- rdt 3.0 protocol performance stinks!
- Protocol limits performance of underlying infrastructure (channel)

rdt3.0: pipelined protocols operation

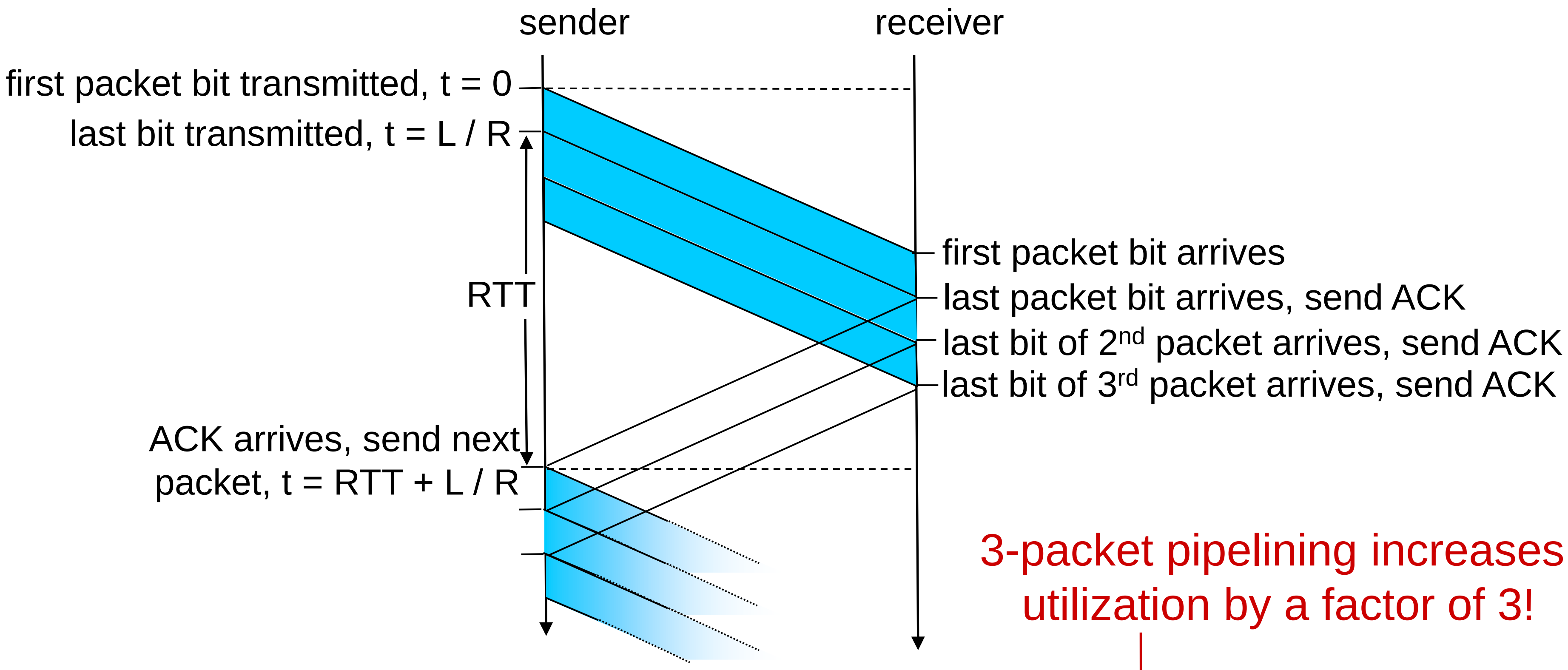
pipelining: sender allows multiple, “in-flight”, yet-to-be-acknowledged packets

- range of sequence numbers must be increased
- buffering at sender and/or receiver



(a) a stop-and-wait protocol in operation

Pipelining: increased utilization

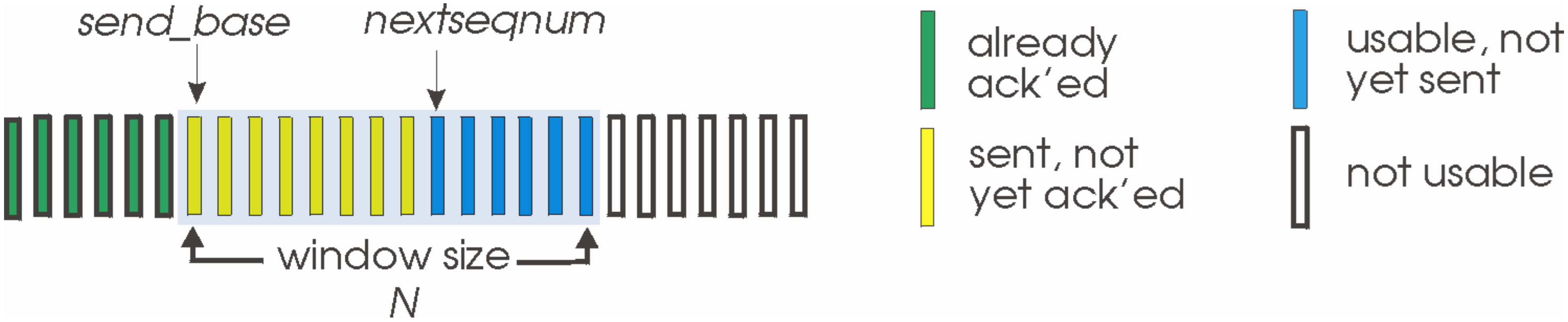

sender
receiver
first packet bit transmitted, t = 0
last bit transmitted, t = L / R
RTT
first packet bit arrives
last packet bit arrives, send ACK
last bit of 2nd packet arrives, send ACK
last bit of 3rd packet arrives, send ACK
ACK arrives, send next packet, t = RTT + L / R

3-packet pipelining increases utilization by a factor of 3!

$$U_{sender} = \frac{3L / R}{RTT + L / R} = \frac{.0024}{30.008} = 0.00081$$

Go-Back-N: sender

- sender: "window" of up to N, consecutive transmitted but unACKed pkts
  - k-bit seq # in pkt header

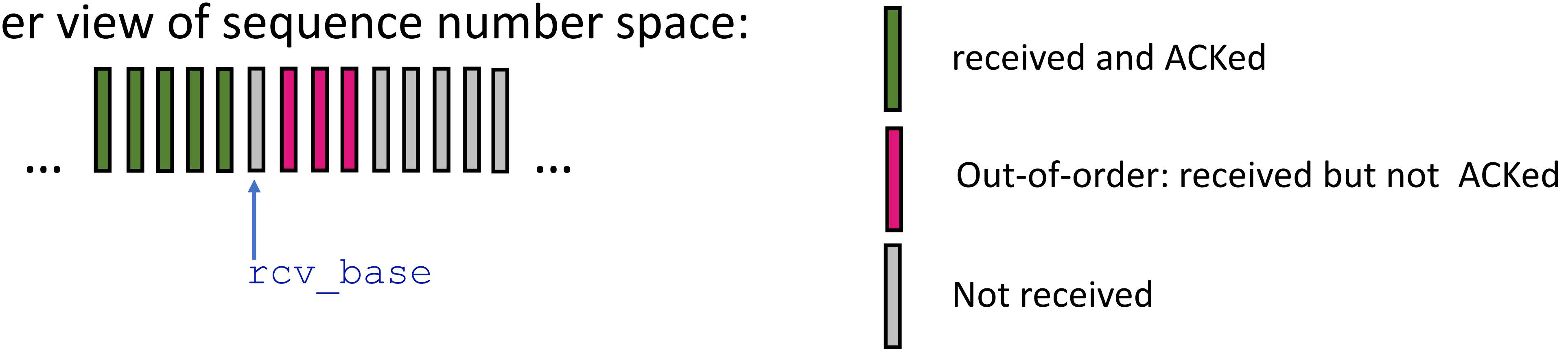

send_base
nextseqnum
window size
N
already ack'ed
sent, not yet ack'ed
usable, not yet sent
not usable

- *cumulative ACK:* ACK(*n*): ACKs all packets up to, including seq # *n*
  - on receiving ACK(*n*): move window forward to begin at *n+1*
- timer for oldest in-flight packet
- *timeout(n):* retransmit packet n and all higher seq # packets in window

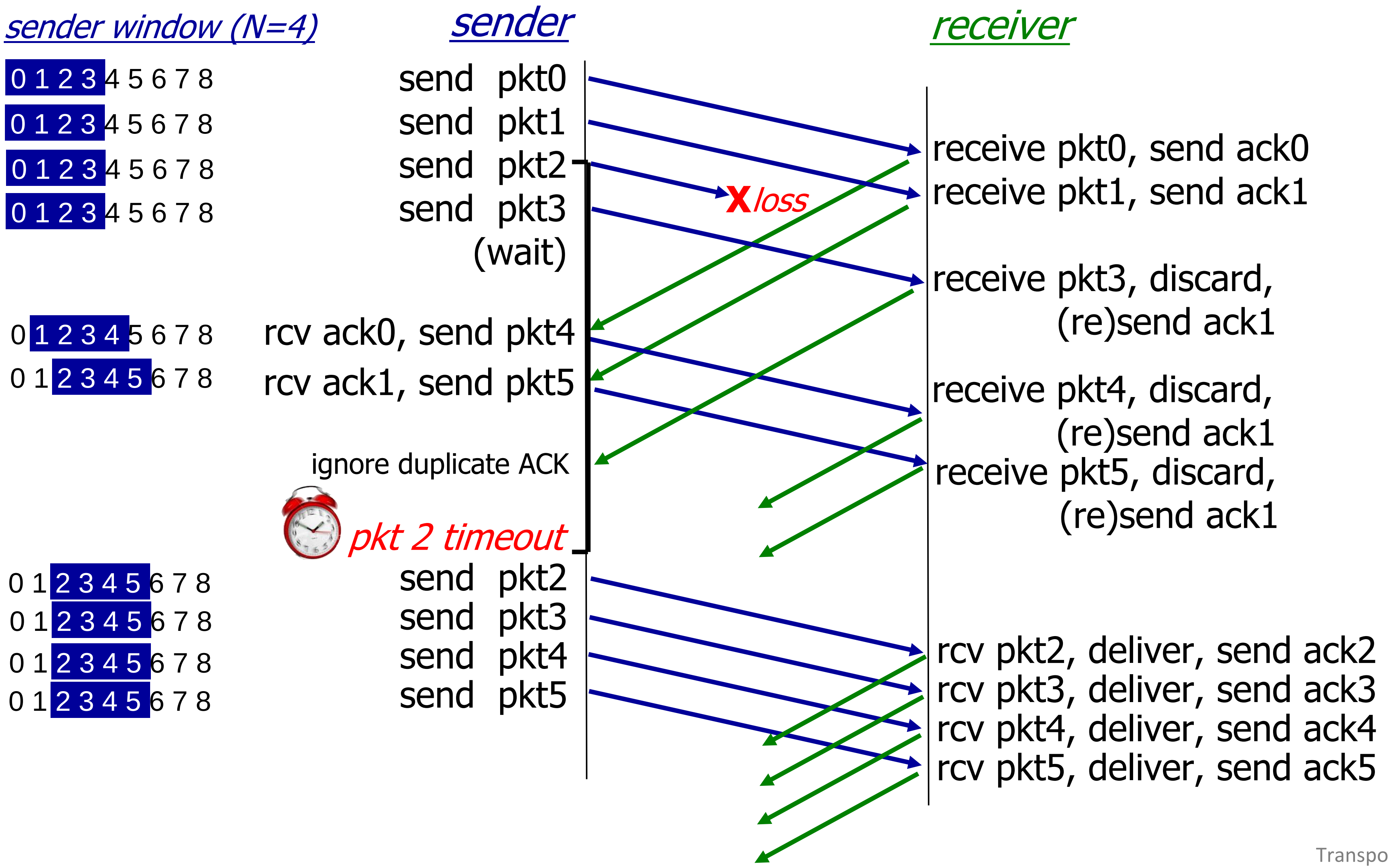
Go-Back-N: receiver

- ACK-only: always send ACK for correctly-received packet so far, with highest *in-order* seq #
  - may generate duplicate ACKs
  - need only remember `rcv_base`
- on receipt of out-of-order packet:
  - can discard (don't buffer) or buffer: an implementation decision
  - re-ACK pkt with highest in-order seq #

Receiver view of sequence number space:


…
…
rcv_base
received and ACKed
Out-of-order: received but not ACKed
Not received

Go-Back-N in action



sender window (N=4)

0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8

0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8

0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8

sender

send pkt0
send pkt1
send pkt2
send pkt3
(wait)

rcv ack0, send pkt4
rcv ack1, send pkt5

ignore duplicate ACK

pkt 2 timeout

send pkt2
send pkt3
send pkt4
send pkt5

receiver

receive pkt0, send ack0
receive pkt1, send ack1

receive pkt3, discard,
(re)send ack1

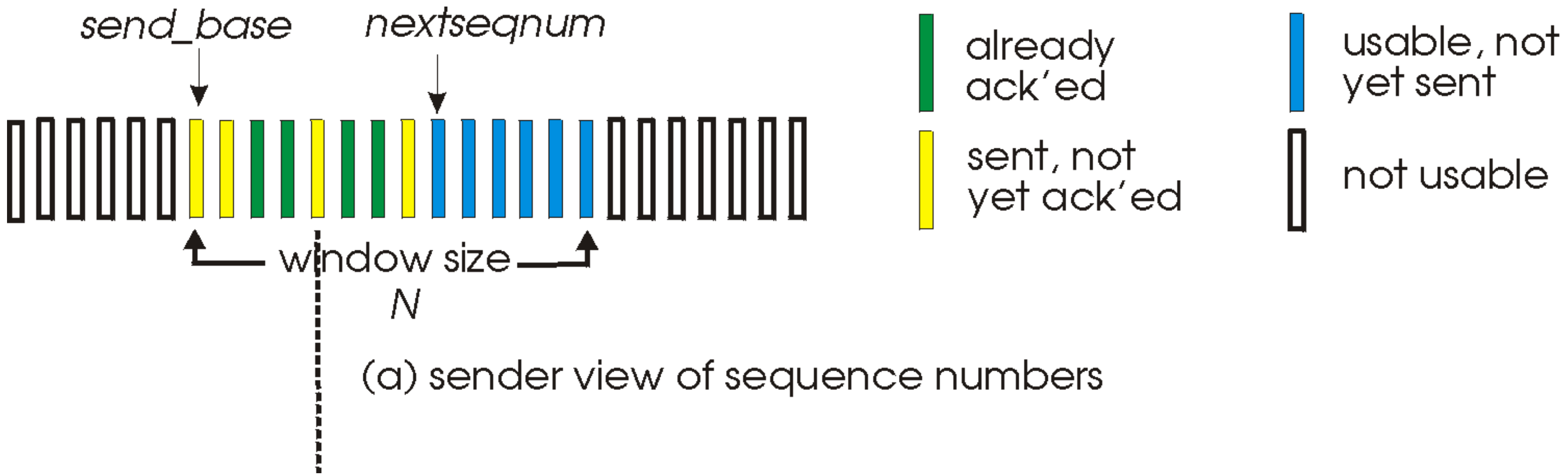
receive pkt4, discard,
(re)send ack1
receive pkt5, discard,
(re)send ack1

rcv pkt2, deliver, send ack2
rcv pkt3, deliver, send ack3
rcv pkt4, deliver, send ack4
rcv pkt5, deliver, send ack5

Selective repeat

- receiver *individually* acknowledges all correctly received packets
  - buffers packets, as needed, for eventual in-order delivery to upper layer
- sender times-out/retransmits individually for unACKed packets
  - sender maintains timer for each unACKed pkt
- sender window
  - *N* consecutive seq #s
  - limits seq #s of sent, unACKed packets

Selective repeat: sender, receiver windows


send_base
nextseqnum
window size
N
already ack'ed
usable, not yet sent
sent, not yet ack'ed
not usable

(a) sender view of sequence numbers

Selective repeat: sender and receiver

sender

data from above:

- if next available seq # in window, send packet

timeout(*n*):

- resend packet *n*, restart timer

ACK(*n*) in [sendbase,sendbase+N]:

- mark packet *n* as received
- if n smallest unACKed packet, advance window base to next unACKed seq #

receiver

packet *n* in [rcvbase, rcvbase+N-1]

- send ACK(*n*)
- out-of-order: buffer
- in-order: deliver (also deliver buffered, in-order packets), advance window to next not-yet-received packet

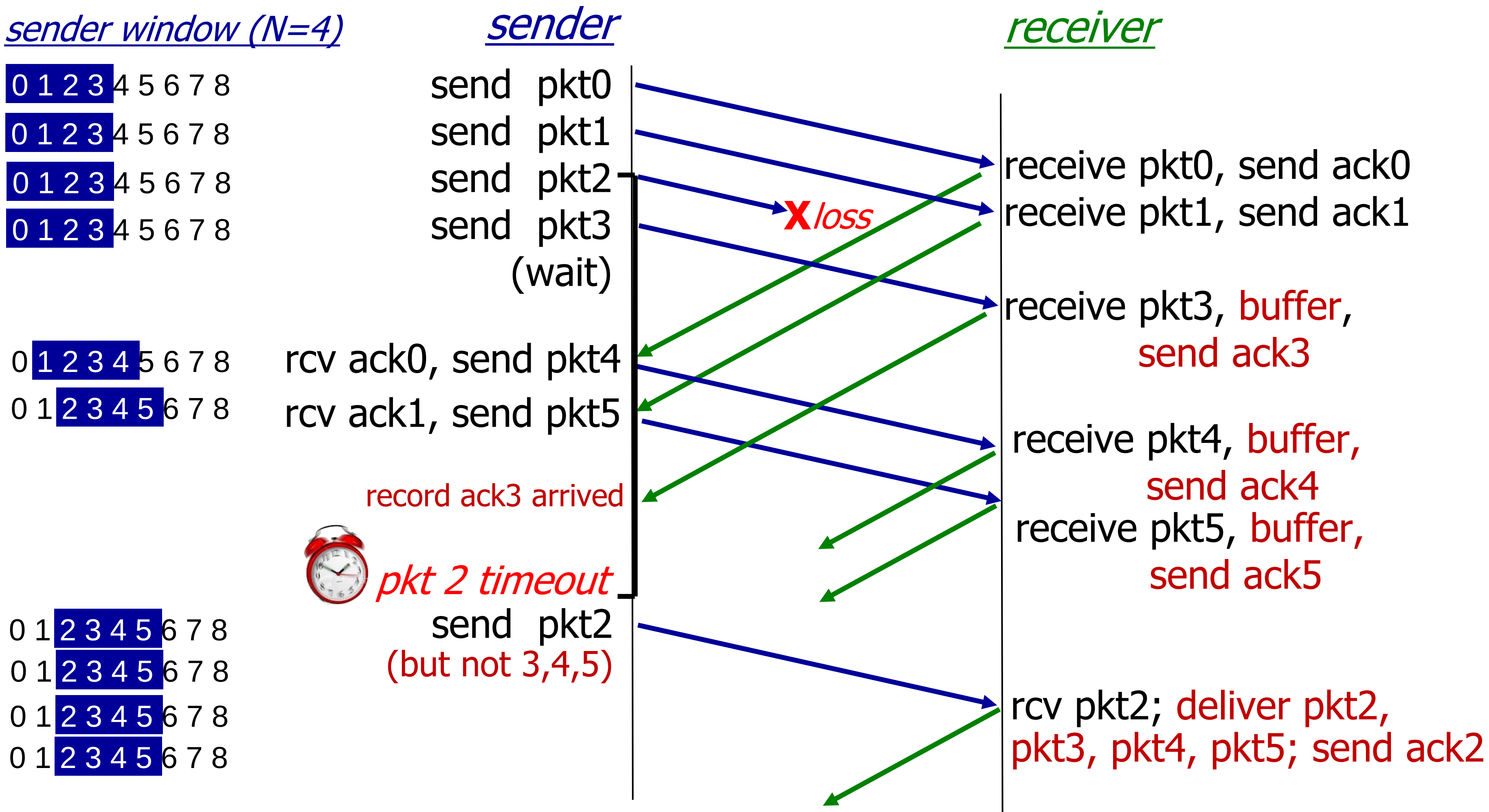
packet *n* in [rcvbase-N,rcvbase-1]

- ACK(*n*)

otherwise:

- ignore

Selective Repeat in action

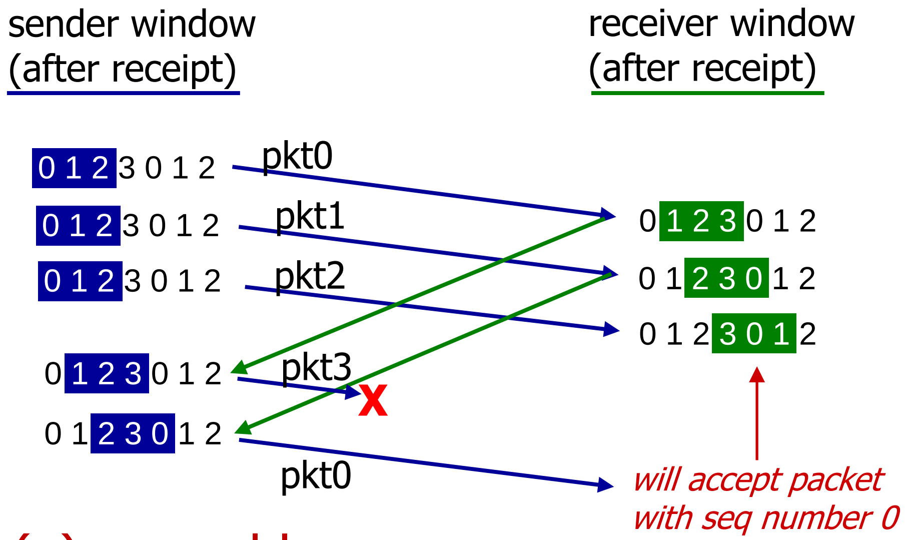

sender window (N=4)
sender
receiver
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
send pkt0
send pkt1
send pkt2
send pkt3
(wait)
receive pkt0, send ack0
receive pkt1, send ack1
X loss
receive pkt3, buffer,
send ack3
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
rcv ack0, send pkt4
rcv ack1, send pkt5
receive pkt4, buffer,
send ack4
receive pkt5, buffer,
send ack5
record ack3 arrived
pkt 2 timeout
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
send pkt2
(but not 3,4,5)
rcv pkt2; deliver pkt2,
pkt3, pkt4, pkt5; send ack2

Q: what happens when ack2 arrives?

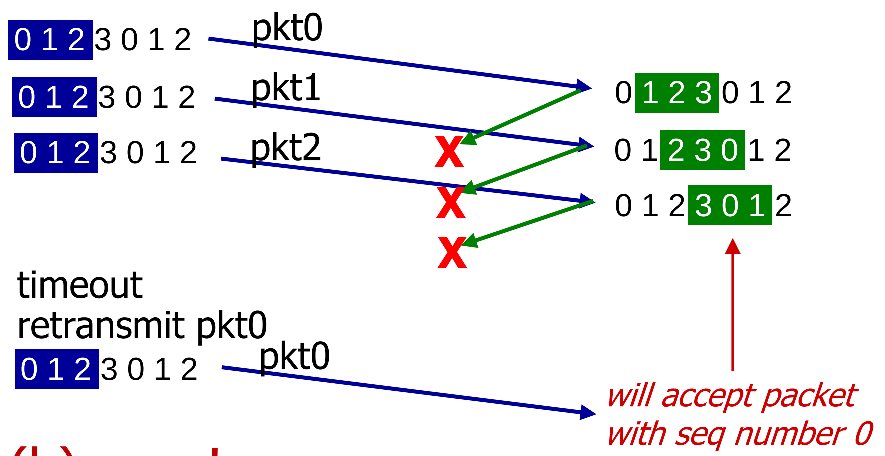
Selective repeat: a dilemma!

example:

- seq #s: 0, 1, 2, 3 (base 4 counting)
- window size=3


sender window (after receipt)
receiver window (after receipt)
0 1 2 3 0 1 2
pkt0
0 1 2 3 0 1 2
pkt1
0 1 2 3 0 1 2
pkt2
0 1 2 3 0 1 2
pkt3
X
0 1 2 3 0 1 2
pkt0
0 1 2 3 0 1 2
0 1 2 3 0 1 2
0 1 2 3 0 1 2
will accept packet with seq number 0

(a) no problem

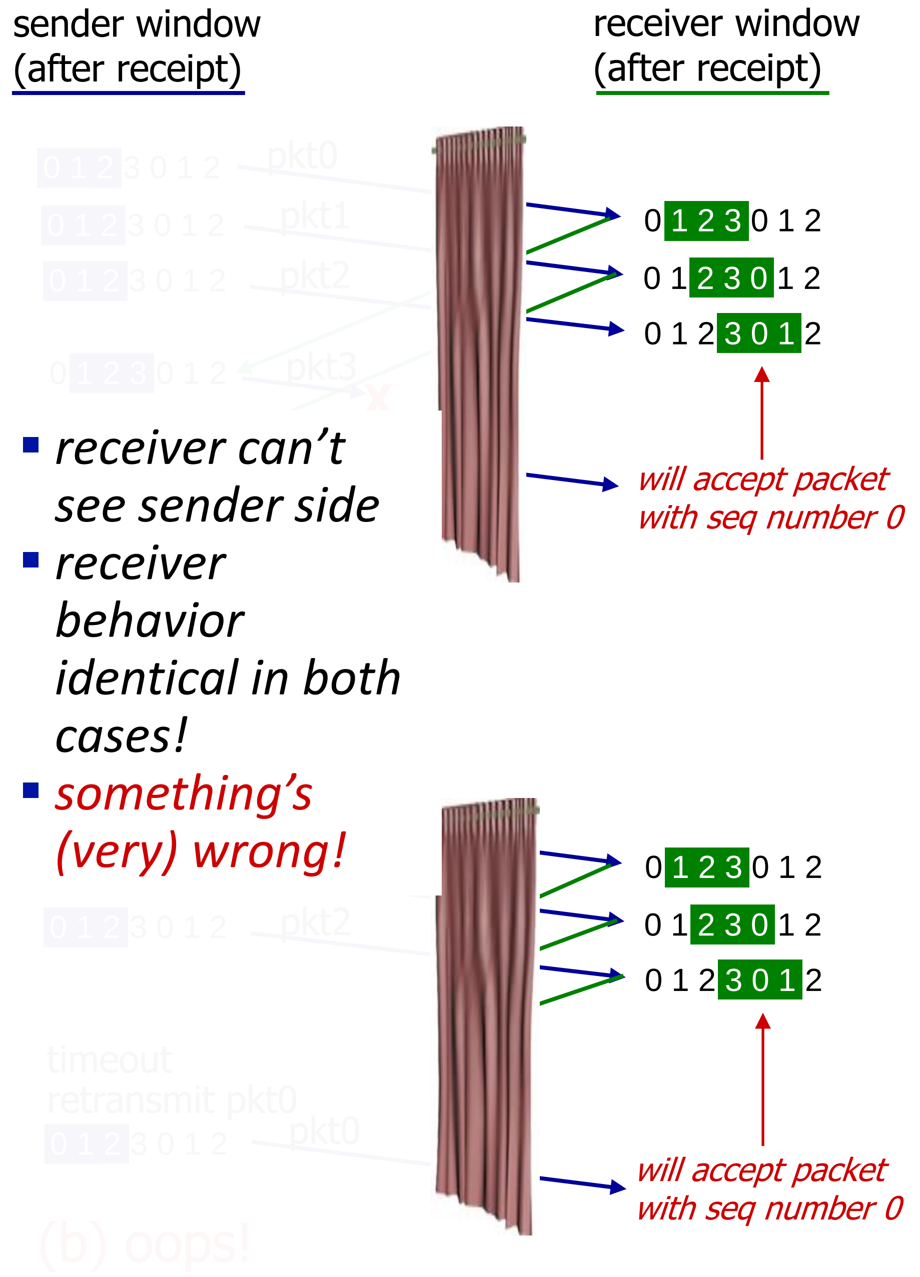

0 1 2 3 0 1 2
pkt0
0 1 2 3 0 1 2
pkt1
0 1 2 3 0 1 2
pkt2
X
X
X
timeout
retransmit pkt0
0 1 2 3 0 1 2
pkt0
0 1 2 3 0 1 2
0 1 2 3 0 1 2
0 1 2 3 0 1 2
will accept packet with seq number 0

(b) oops!

Selective repeat: a dilemma!

example:

- seq #s: 0, 1, 2, 3 (base 4 counting)
- window size=3


sender window
(after receipt)
receiver window
(after receipt)
0 1 2 3 0 1 2
pkt0
0 1 2 3 0 1 2
pkt1
0 1 2 3 0 1 2
pkt2
0 1 2 3 0 1 2
pkt3
0 1 2 3 0 1 2
0 1 2 3 0 1 2
0 1 2 3 0 1 2
0 1 2 3 0 1 2
pkt2
timeout
retransmit pkt0
0 1 2 3 0 1 2
pkt0
0 1 2 3 0 1 2
0 1 2 3 0 1 2
0 1 2 3 0 1 2
(b) oops!

will accept packet with seq number 0

- *receiver can't see sender side*
- *receiver behavior identical in both cases!*
- *something's (very) wrong!*

will accept packet with seq number 0

Q: what relationship is needed between sequence # size and window size to avoid problem in scenario (b)?

Chapter 3: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- Principles of reliable data transfer
- Connection-oriented transport: TCP
  - segment structure
  - reliable data transfer
  - flow control
  - connection management
- Principles of congestion control
- TCP congestion control

TCP: overview RFCs: 793,1122, 2018, 5681, 7323

- point-to-point:
  - one sender, one receiver
- reliable, in-order *byte steam:*
  - no "message boundaries"
- full duplex data:
  - bi-directional data flow in same connection
  - MSS: maximum segment size
- cumulative ACKs
- pipelining:
  - TCP congestion and flow control set window size
- connection-oriented:
  - handshaking (exchange of control messages) initializes sender, receiver state before data exchange
- flow controlled:
  - sender will not overwhelm receiver

TCP segment structure

ACK: seq # of next expected byte; A bit: this is an ACK

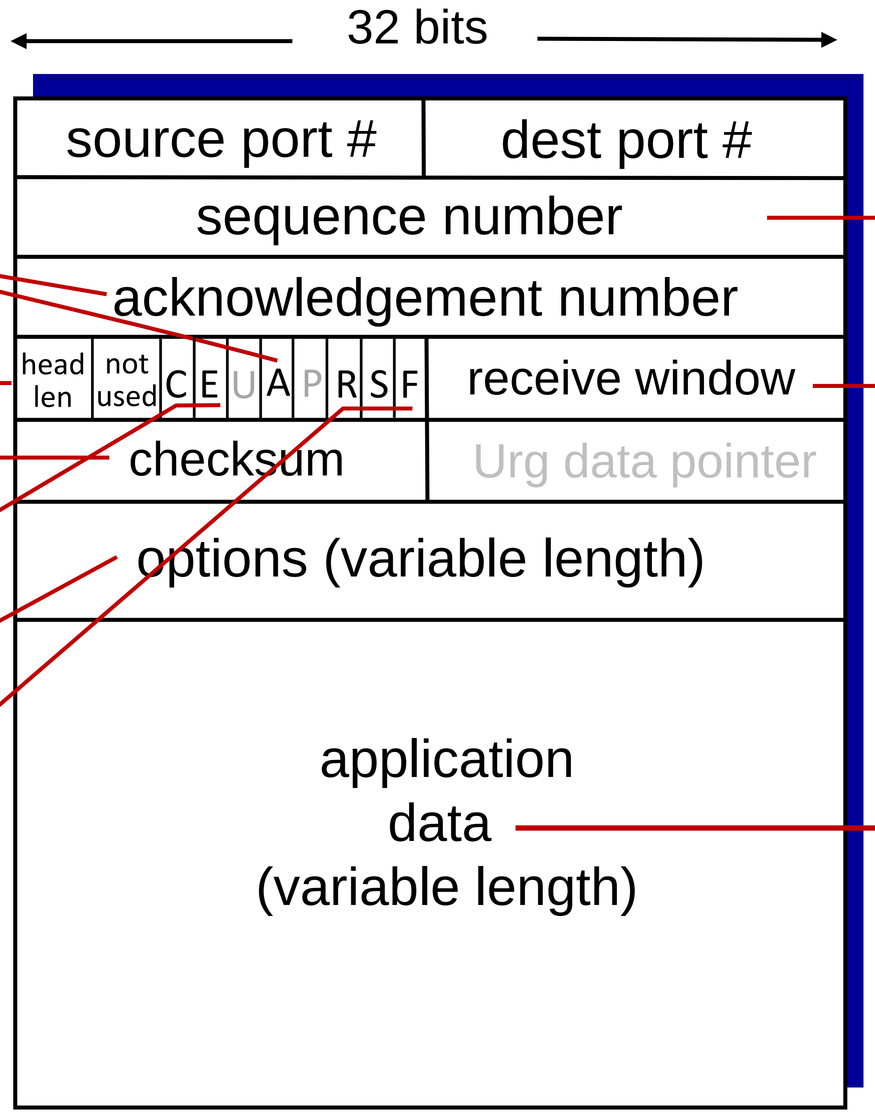
length (of TCP header)

Internet checksum

C, E: congestion notification

TCP options

RST, SYN, FIN: connection management


32 bits
source port #
dest port #
sequence number
acknowledgement number
head len
not used
C
E
U
A
P
R
S
F
receive window
checksum
Urg data pointer
options (variable length)
application data (variable length)

segment seq #: counting bytes of data into bytestream (not segments!)

flow control: # bytes receiver willing to accept

data sent by application into TCP socket

TCP sequence numbers, ACKs

Sequence numbers:

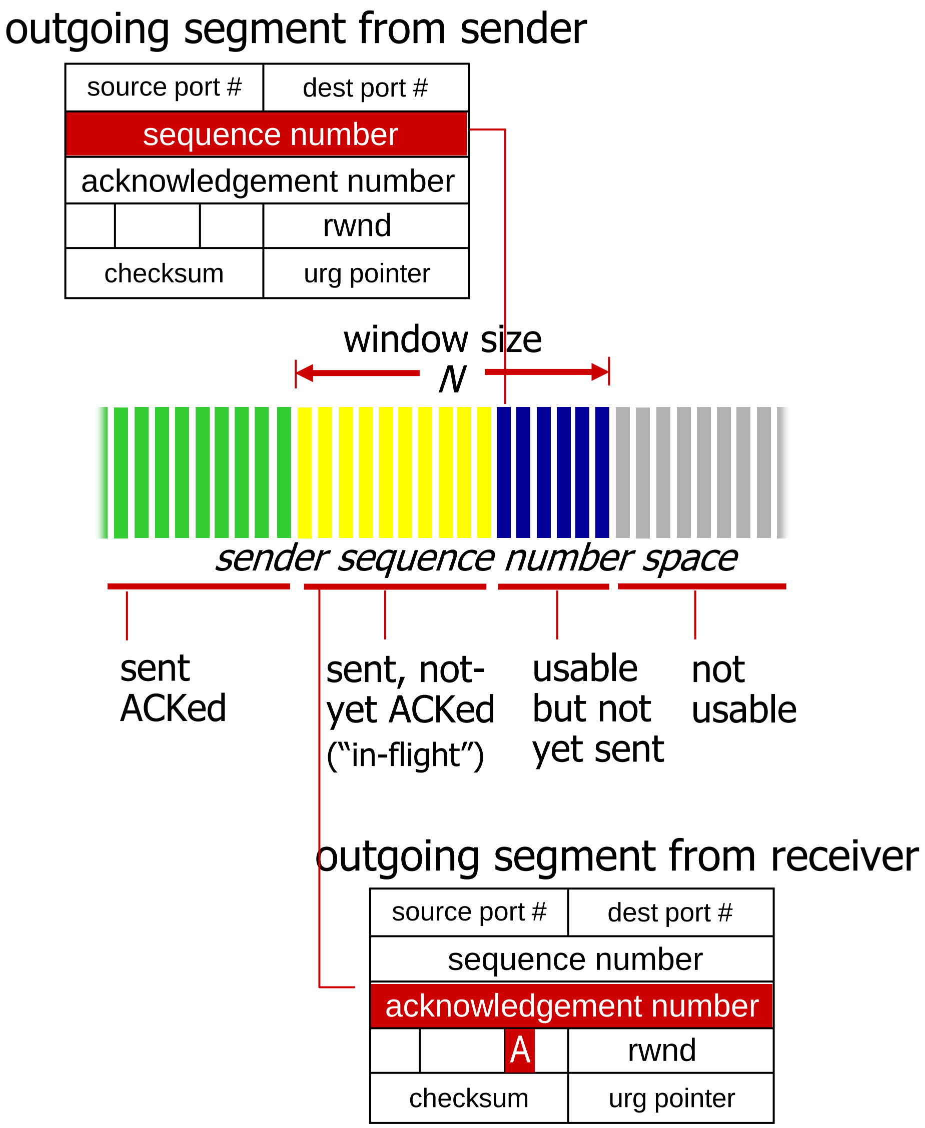
- byte stream "number" of first byte in segment's data

Acknowledgements:

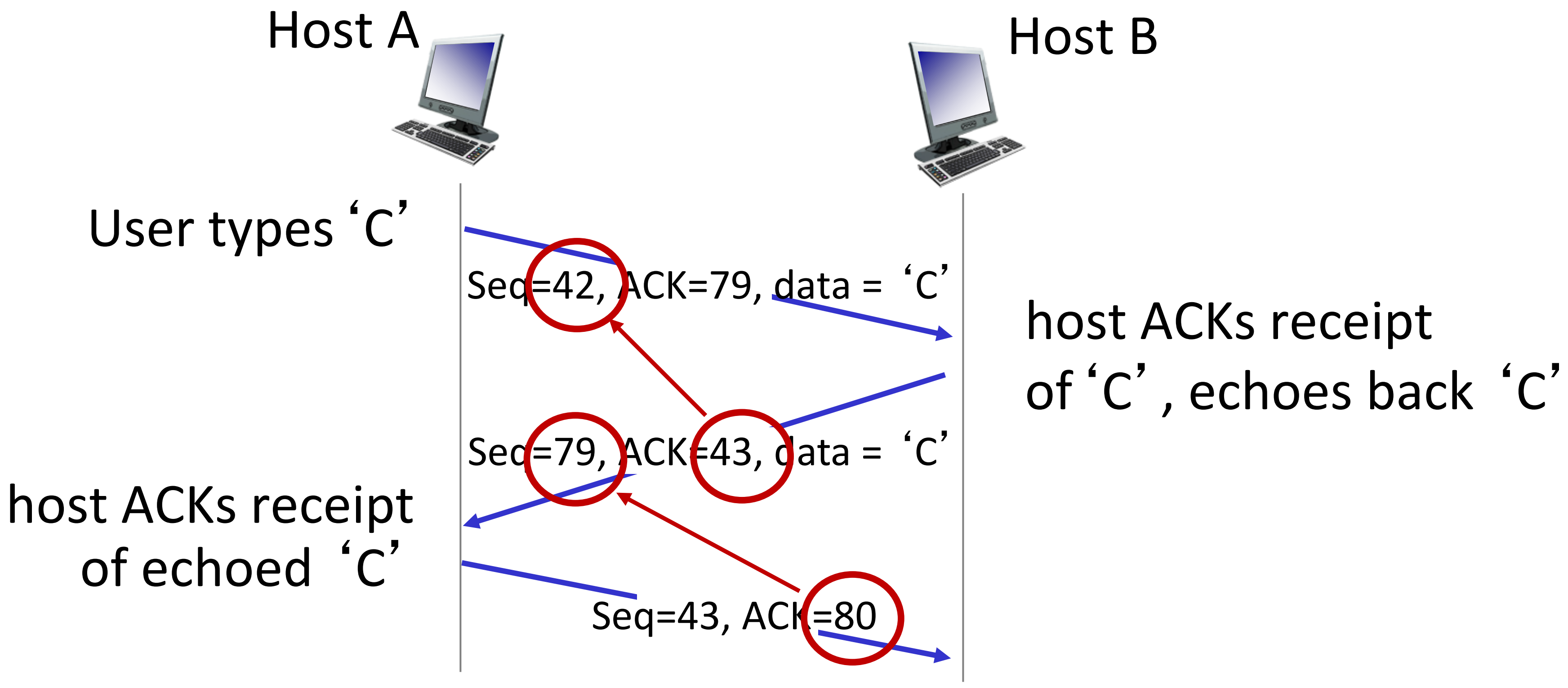
- seq # of next byte expected from other side
- cumulative ACK

Q: how receiver handles out-of-order segments

- *A:* TCP spec doesn't say, - up to implementor


outgoing segment from sender
source port #
dest port #
sequence number
acknowledgement number
rwnd
checksum
urg pointer
window size
N
sender sequence number space
sent ACKed
sent, not-yet ACKed ("in-flight")
usable but not yet sent
not usable
outgoing segment from receiver
source port #
dest port #
sequence number
acknowledgement number
A
rwnd
checksum
urg pointer

TCP sequence numbers, ACKs



Host A

Host B

User types ‘C’

Seq=42, ACK=79, data = ‘C’

host ACKs receipt of ‘C’, echoes back ‘C’

Seq=79, ACK=43, data = ‘C’

host ACKs receipt of echoed ‘C’

Seq=43, ACK=80

simple telnet scenario

TCP round trip time, timeout

Q: how to set TCP timeout value?

- longer than RTT, but RTT varies!
- *too short:* premature timeout, unnecessary retransmissions
- *too long:* slow reaction to segment loss

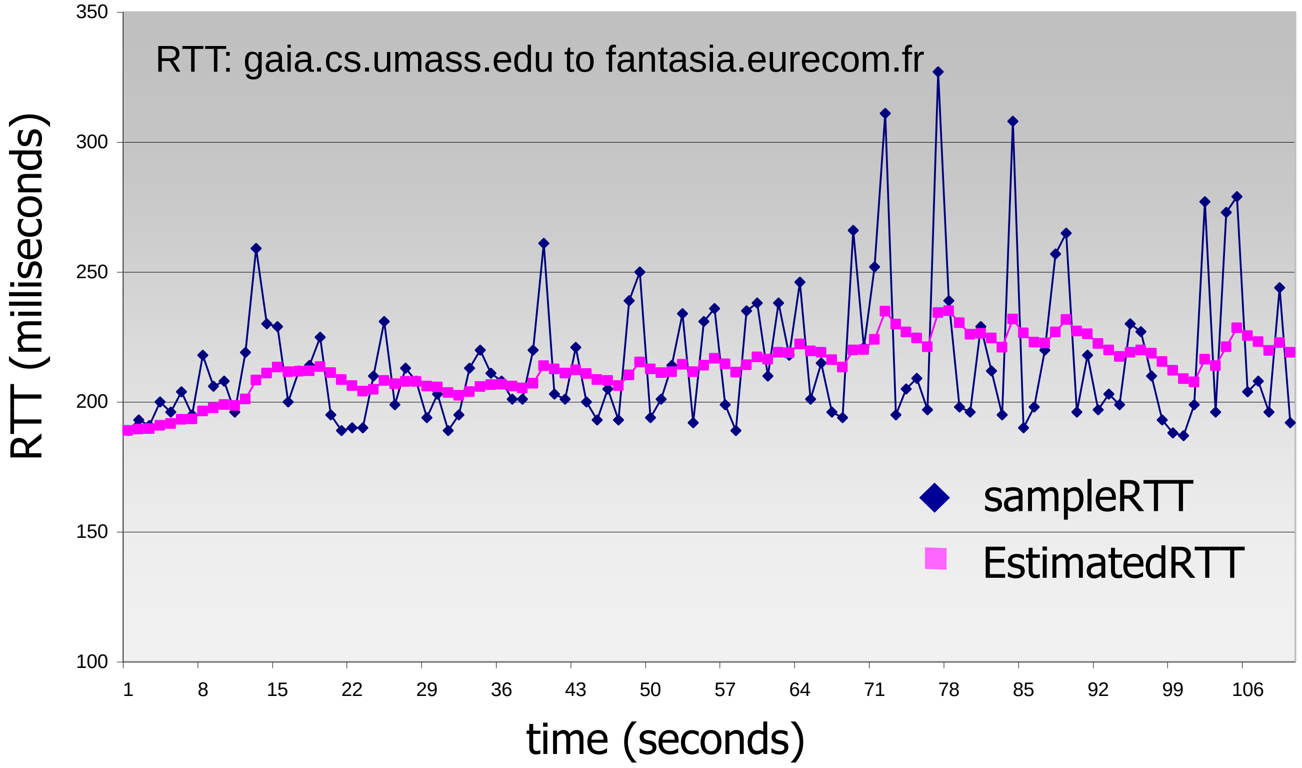
Q: how to estimate RTT?

- `SampleRTT`: measured time from segment transmission until ACK receipt
  - ignore retransmissions
- `SampleRTT` will vary, want estimated RTT "smoother"
  - average several *recent* measurements, not just current `SampleRTT`

TCP round trip time, timeout

`EstimatedRTT = (1- α)*EstimatedRTT + α*SampleRTT`

- exponential weighted moving average (EWMA)
- influence of past sample decreases exponentially fast
- typical value: α = 0.125



TCP round trip time, timeout

- timeout interval: **`EstimatedRTT`** plus "safety margin"
  - large variation in **`EstimatedRTT:`** want a larger safety margin

```
TimeoutInterval = EstimatedRTT + 4*DevRTT
```

estimated RTT

"safety margin"

- **`DevRTT`**: EWMA of **`SampleRTT`** deviation from **`EstimatedRTT`**:

```
DevRTT = (1-β)*DevRTT + β*|SampleRTT-EstimatedRTT|
```

(typically, β = 0.25)

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

TCP Sender (simplified)

event: data received from application

- create segment with seq #
- seq # is byte-stream number of first data byte in segment
- start timer if not already running
  - think of timer as for oldest unACKed segment
  - expiration interval: **`TimeOutInterval`**

event: timeout

- retransmit segment that caused timeout
- restart timer

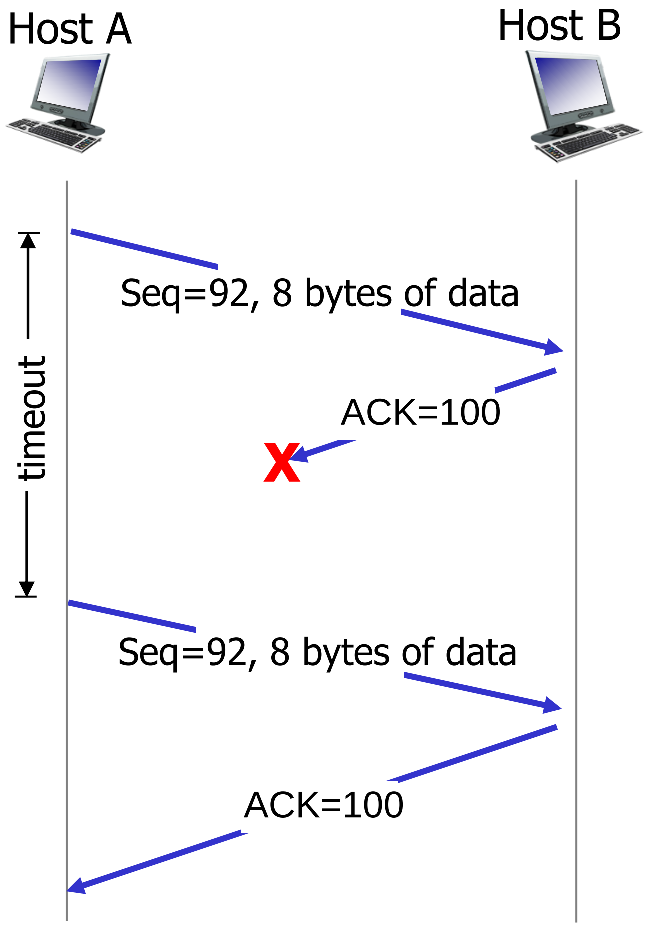
event: ACK received

- if ACK acknowledges previously unACKed segments
  - update what is known to be ACKed
  - start timer if there are still unACKed segments

TCP Receiver: ACK generation [RFC 5681]

Event at receiver	*TCP receiver action*

TCP: retransmission scenarios



Host A

Host B

timeout

Seq=92, 8 bytes of data

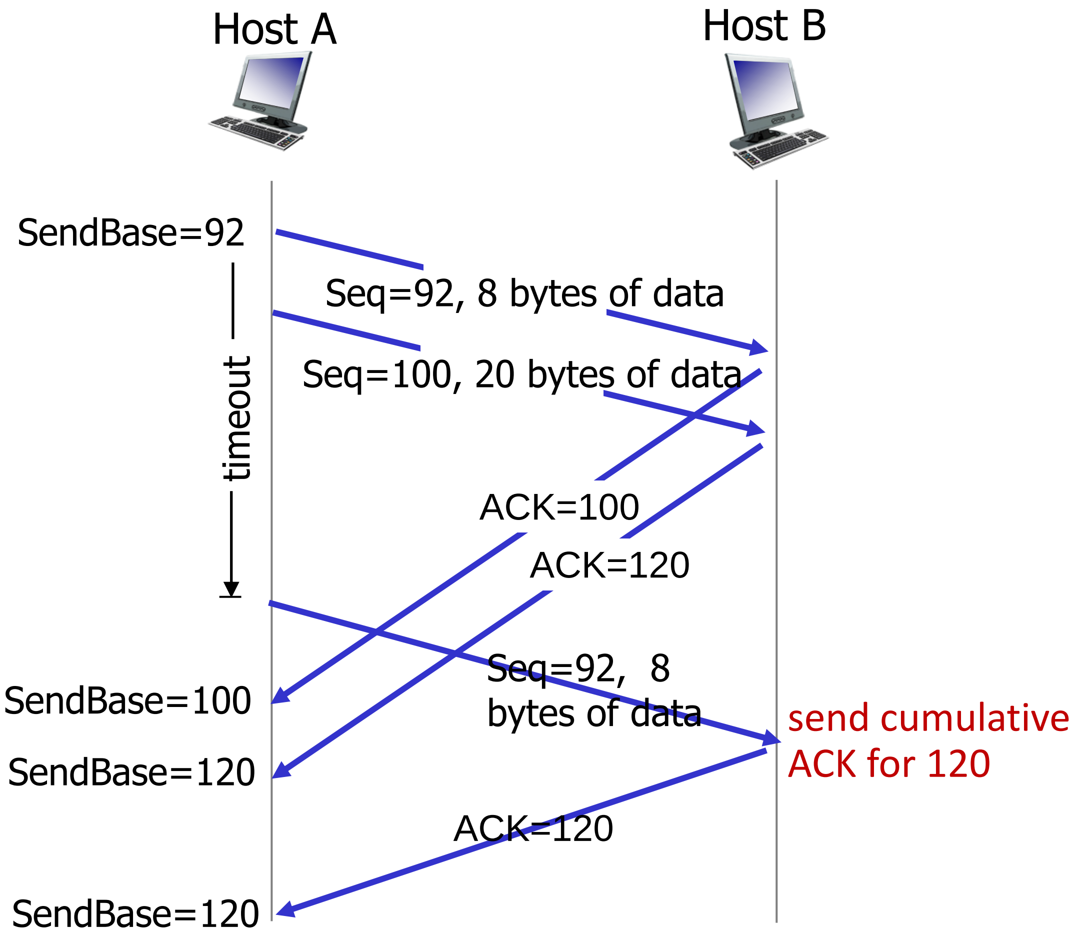
ACK=100

X

Seq=92, 8 bytes of data

ACK=100

lost ACK scenario



Host A

Host B

SendBase=92

timeout

Seq=92, 8 bytes of data

Seq=100, 20 bytes of data

ACK=100

ACK=120

SendBase=100

SendBase=120

Seq=92, 8 bytes of data

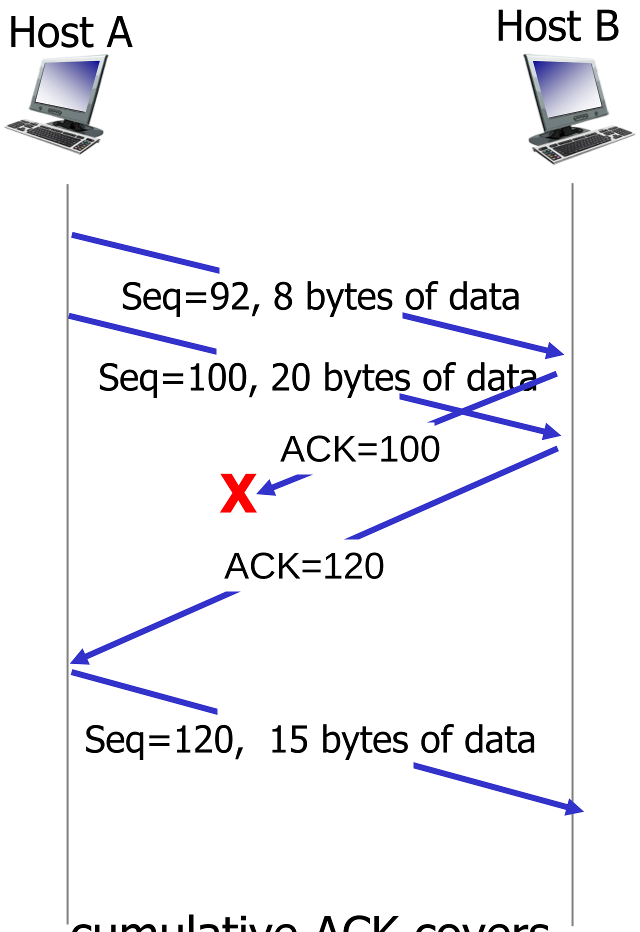
send cumulative ACK for 120

ACK=120

SendBase=120

premature timeout

TCP: retransmission scenarios


Host A
Host B
Seq=92, 8 bytes of data
Seq=100, 20 bytes of data
ACK=100
X
ACK=120
Seq=120, 15 bytes of data

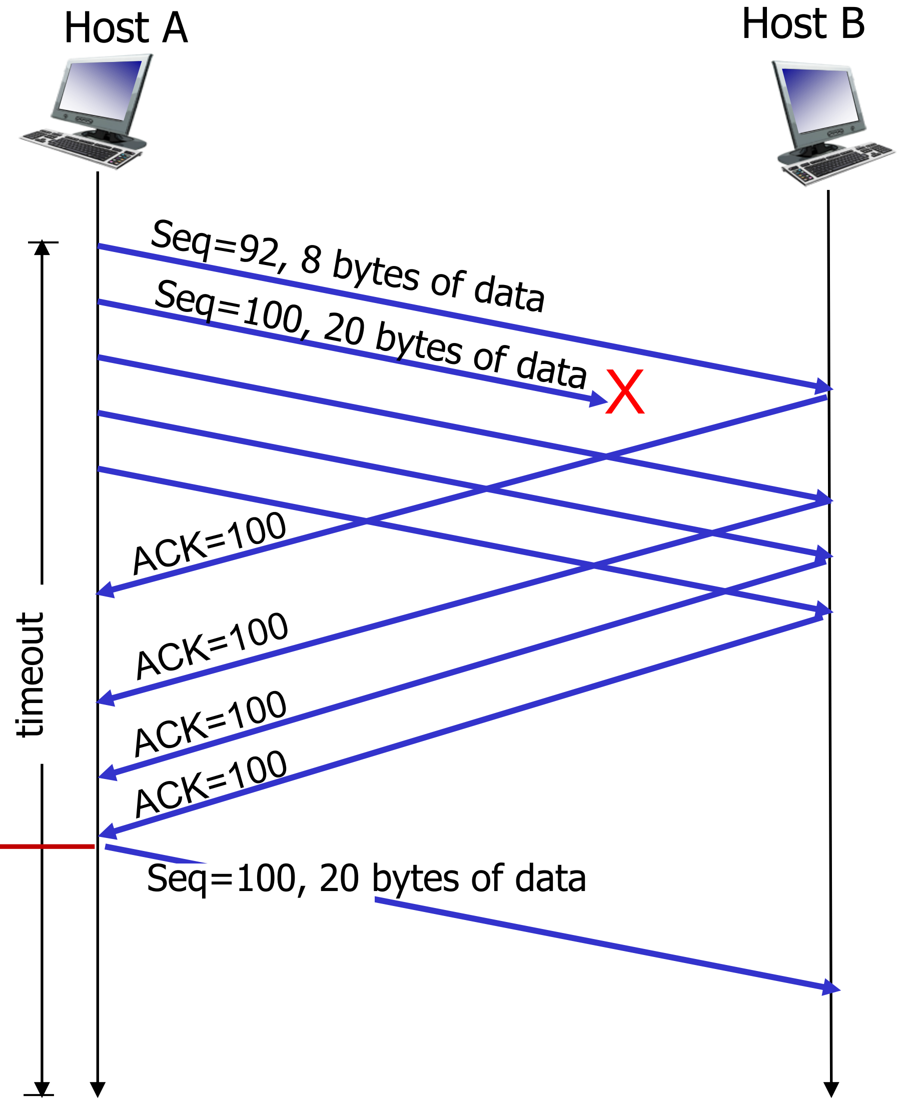
cumulative ACK covers for earlier lost ACK

TCP fast retransmit

TCP fast retransmit

if sender receives 3 additional ACKs for same data ("triple duplicate ACKs"), resend unACKed segment with smallest seq #

- likely that unACKed segment lost, so don't wait for timeout



Host A
Host B
Seq=92, 8 bytes of data
Seq=100, 20 bytes of data
X
ACK=100
ACK=100
ACK=100
ACK=100
timeout
Seq=100, 20 bytes of data

Receipt of three duplicate ACKs indicates 3 segments received after a missing segment – lost segment is likely. So retransmit!

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- Transport-layer services
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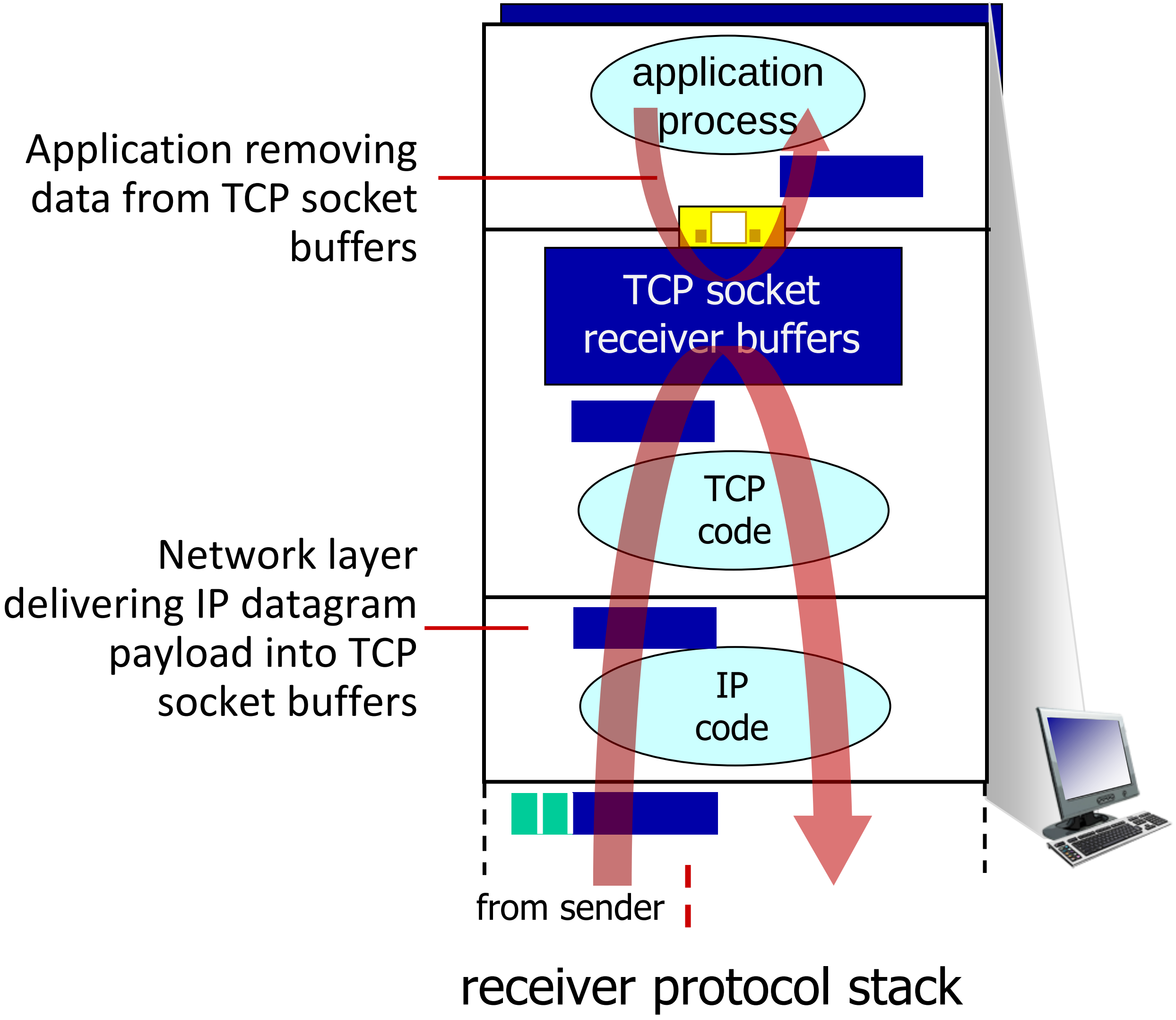
- Connection-oriented transport: TCP
  - segment structure
  - reliable data transfer
  - flow control
  - connection management
- Principles of congestion control
- TCP congestion control

TCP flow control

Q: What happens if network layer delivers data faster than application layer removes data from socket buffers?

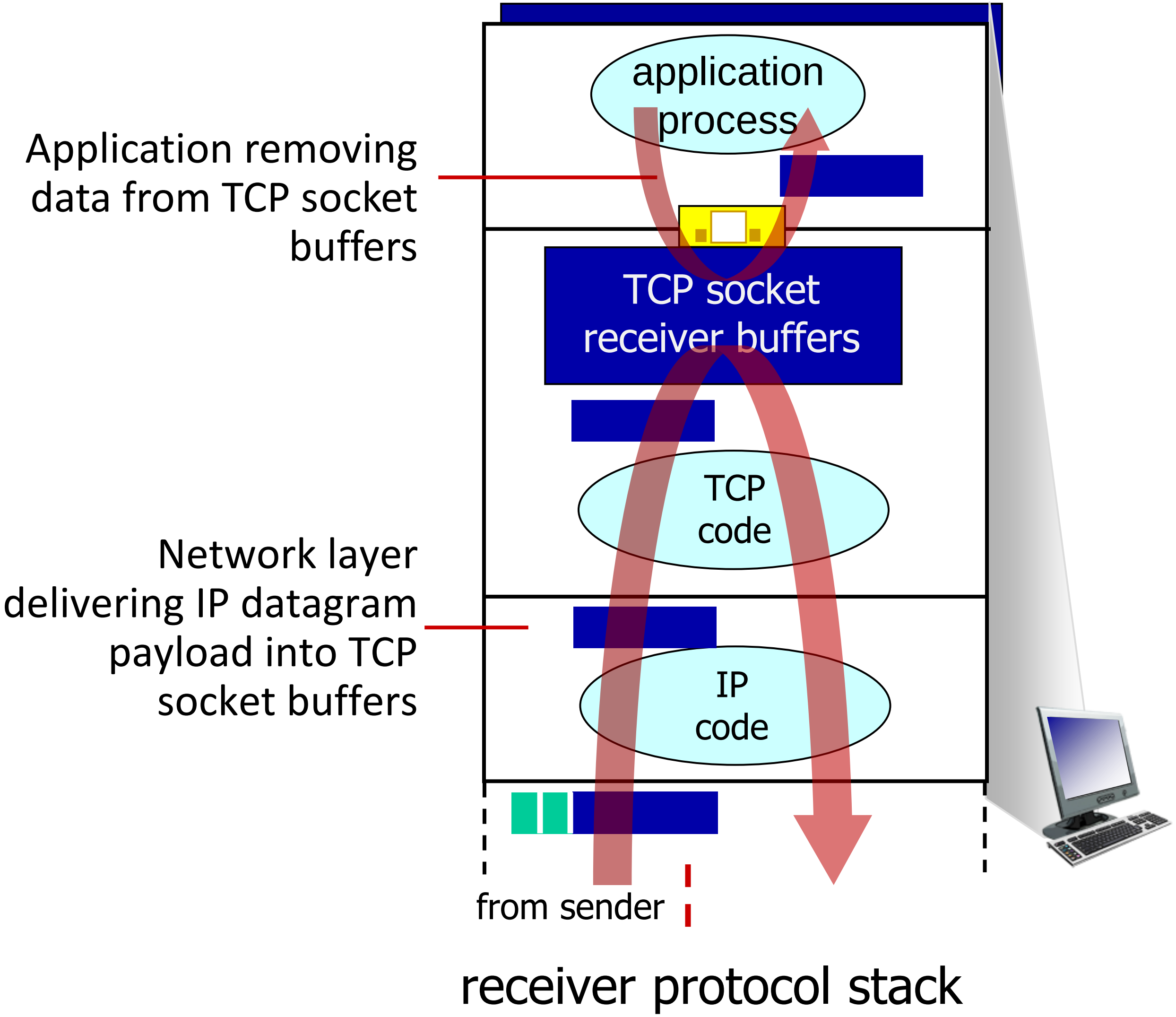

Application removing data from TCP socket buffers

Network layer delivering IP datagram payload into TCP socket buffers

application process
TCP socket receiver buffers
TCP code
IP code
from sender
receiver protocol stack

TCP flow control

Q: What happens if network layer delivers data faster than application layer removes data from socket buffers?

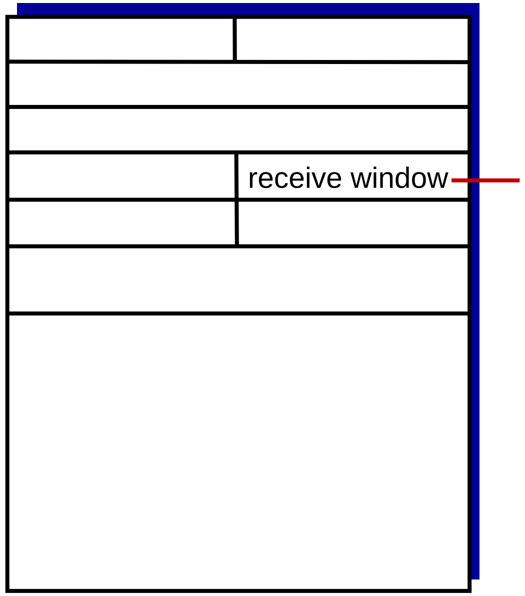


application process
Application removing data from TCP socket buffers
TCP socket receiver buffers
TCP code
Network layer delivering IP datagram payload into TCP socket buffers
IP code
from sender

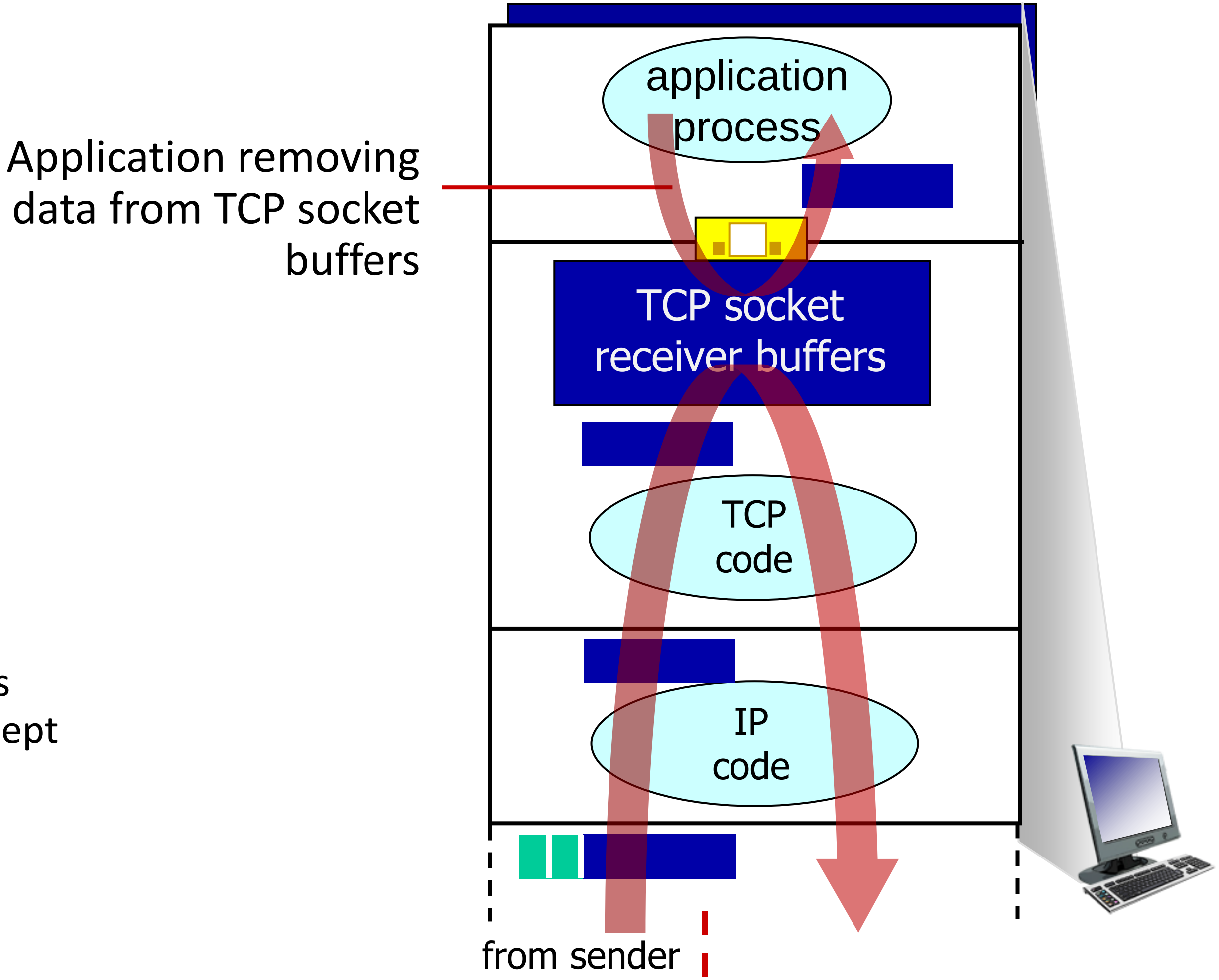
receiver protocol stack

TCP flow control

Q: What happens if network layer delivers data faster than application layer removes data from socket buffers?


receive window

flow control: # bytes receiver willing to accept


application process
TCP socket receiver buffers
TCP code
IP code
from sender

Application removing data from TCP socket buffers

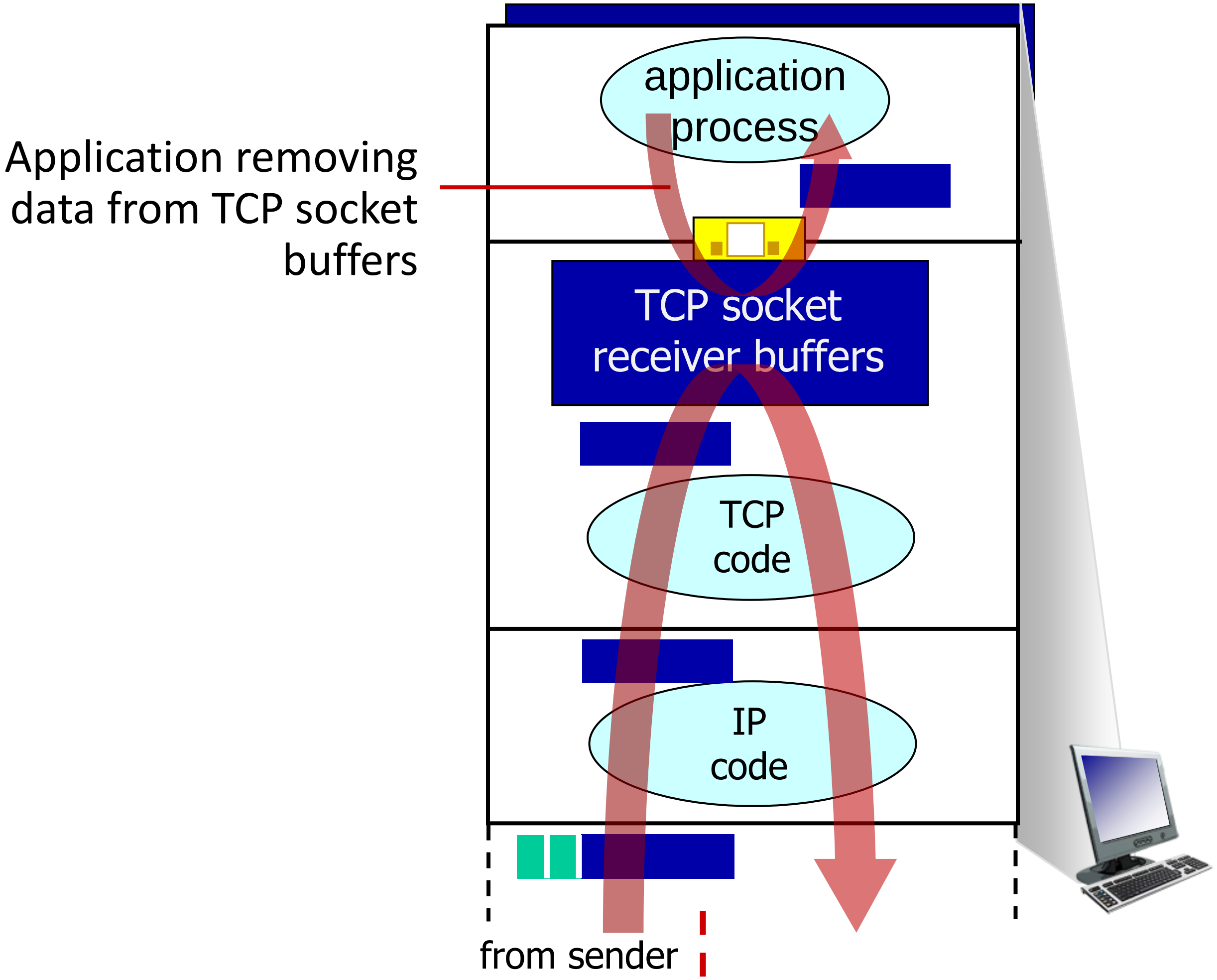
receiver protocol stack

TCP flow control

Q: What happens if network layer delivers data faster than application layer removes data from socket buffers?

flow control

receiver controls sender, so sender won't overflow receiver's buffer by transmitting too much, too fast

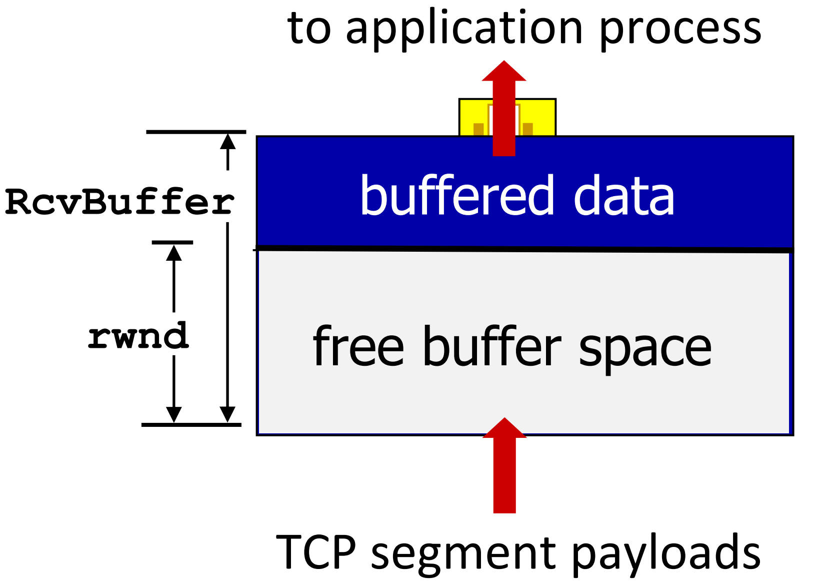


Application removing data from TCP socket buffers
application process
TCP socket receiver buffers
TCP code
IP code
from sender

receiver protocol stack

TCP flow control

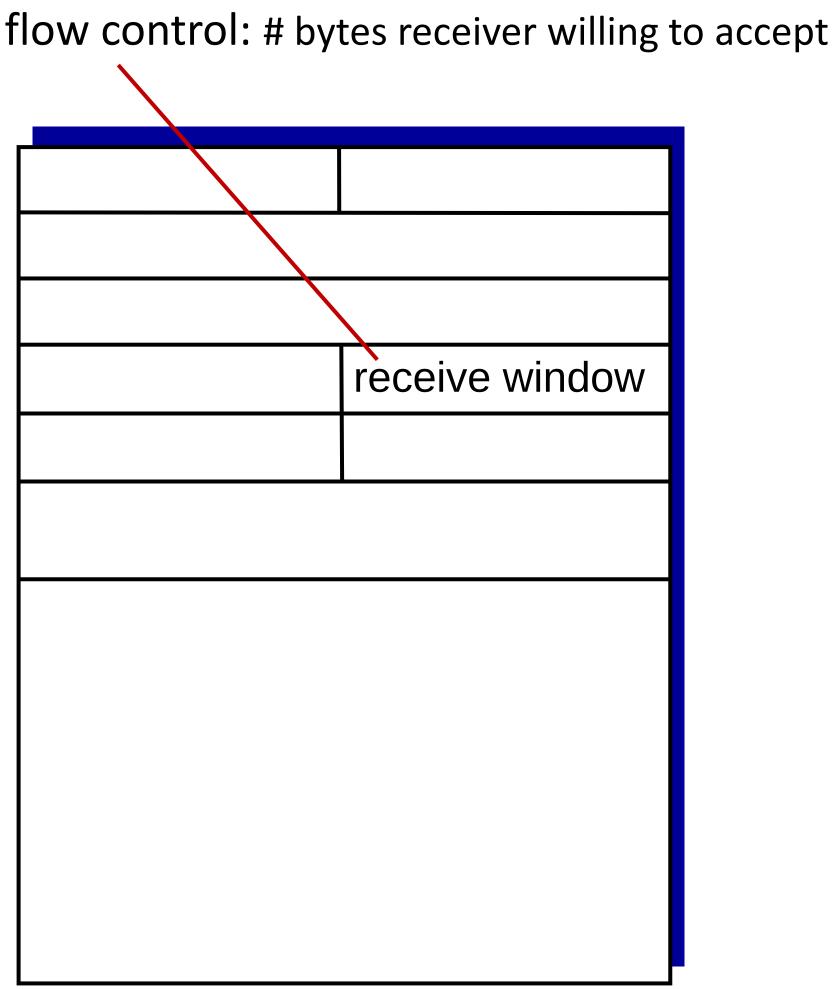
- TCP receiver "advertises" free buffer space in **`rwnd`** field in TCP header
  - **`RcvBuffer`** size set via socket options (typical default is 4096 bytes)
  - many operating systems autoadjust **`RcvBuffer`**
- sender limits amount of unACKed ("in-flight") data to received **`rwnd`**
- guarantees receive buffer will not overflow


to application process
RcvBuffer
buffered data
rwnd
free buffer space
TCP segment payloads

TCP receiver-side buffering

TCP flow control

- TCP receiver "advertises" free buffer space in **`rwnd`** field in TCP header
  - **`RcvBuffer`** size set via socket options (typical default is 4096 bytes)
  - many operating systems autoadjust **`RcvBuffer`**
- sender limits amount of unACKed ("in-flight") data to received **`rwnd`**
- guarantees receive buffer will not overflow

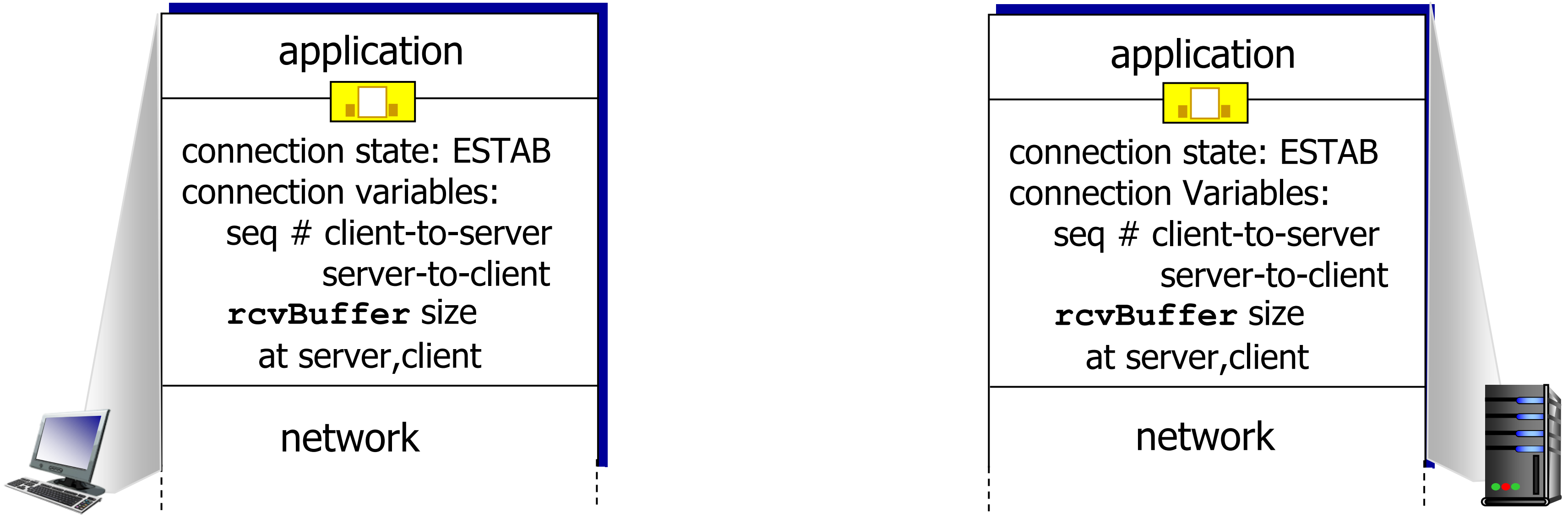

flow control: # bytes receiver willing to accept
receive window

TCP segment format

TCP connection management

before exchanging data, sender/receiver "handshake":

- agree to establish connection (each knowing the other willing to establish connection)
- agree on connection parameters (e.g., starting seq #s)

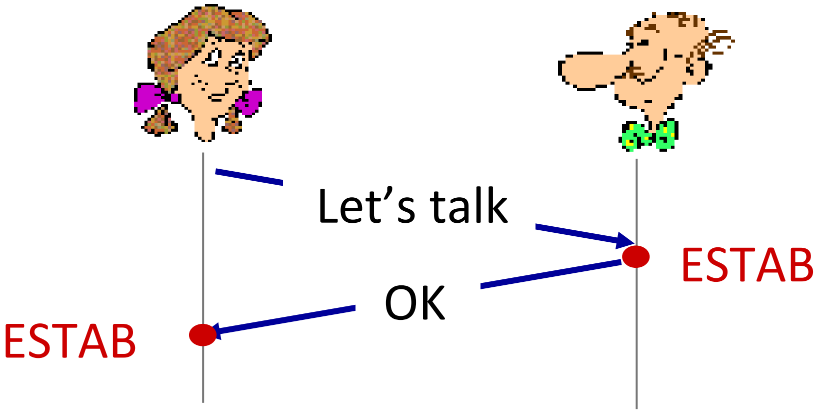

application
connection state: ESTAB
connection variables:
seq # client-to-server
server-to-client
rcvBuffer size
at server,client
network
application
connection state: ESTAB
connection Variables:
seq # client-to-server
server-to-client
rcvBuffer size
at server,client
network

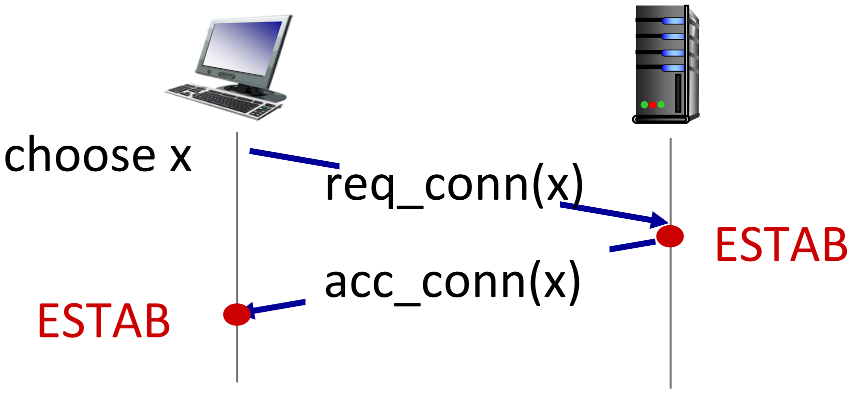
```
Socket clientSocket =
  newSocket("hostname","port number");
```

```
Socket connectionSocket =
  welcomeSocket.accept();
```

Agreeing to establish a connection

2-way handshake:

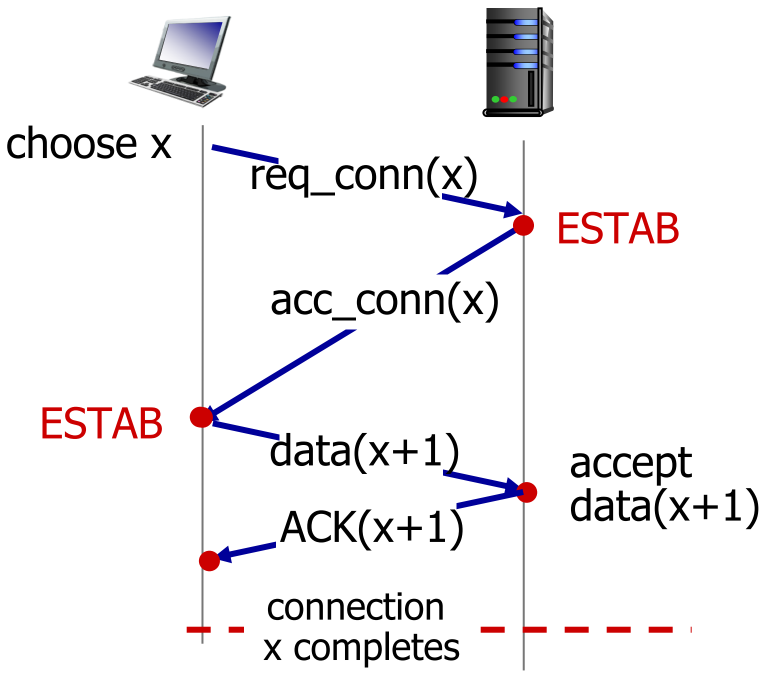

Let's talk
ESTAB
OK
ESTAB


choose x
req_conn(x)
ESTAB
acc_conn(x)
ESTAB

Q: will 2-way handshake always work in network?

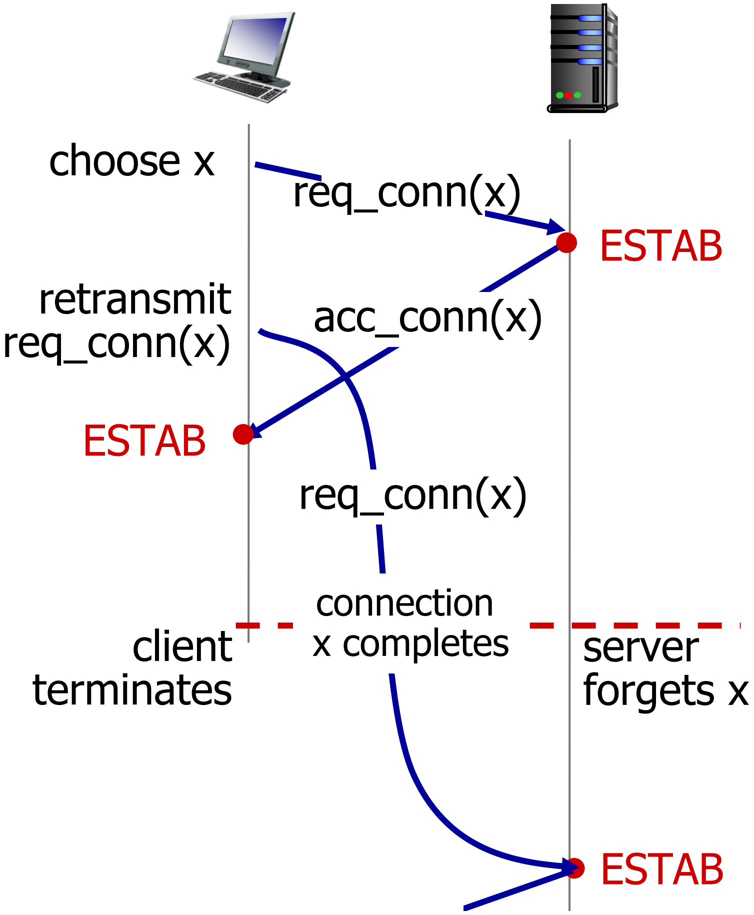
- variable delays
- retransmitted messages (e.g. req_conn(x)) due to message loss
- message reordering
- can't "see" other side

2-way handshake scenarios


choose x
req_conn(x)
ESTAB
acc_conn(x)
ESTAB
data(x+1)
accept data(x+1)
ACK(x+1)
connection x completes

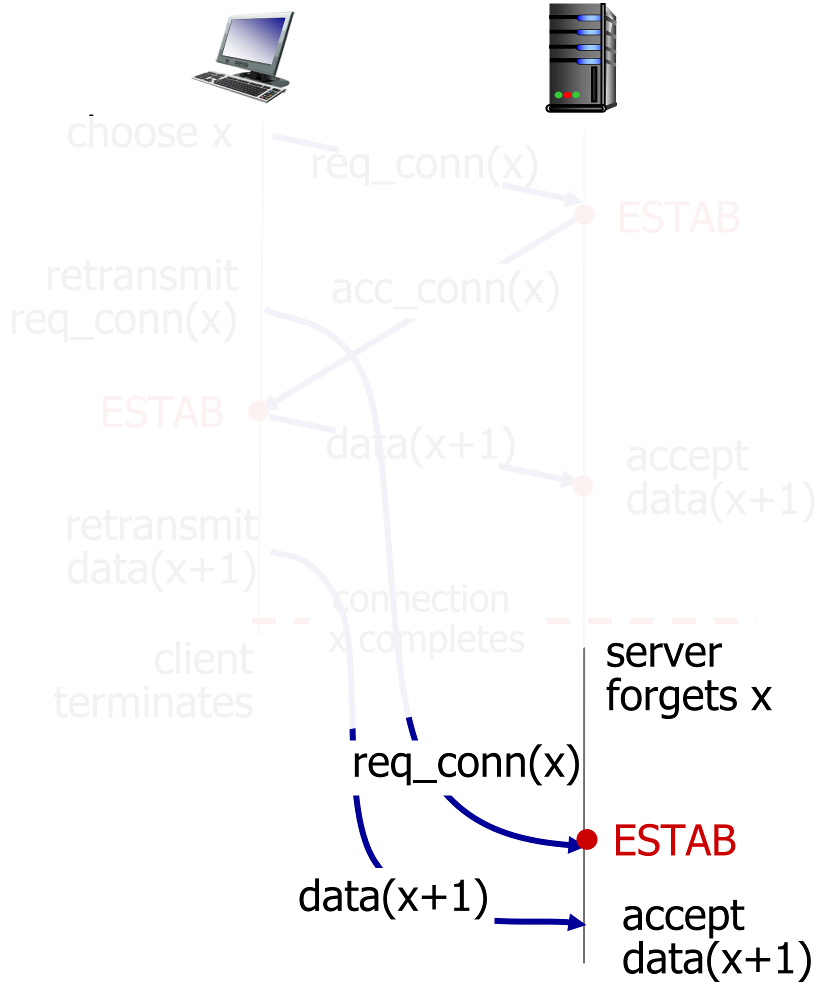
No problem! ✓

2-way handshake scenarios


choose x
req_conn(x)
ESTAB
retransmit req_conn(x)
acc_conn(x)
ESTAB
req_conn(x)
connection x completes
client terminates
server forgets x
ESTAB

❌ Problem: half open connection! (no client)

2-way handshake scenarios



choose x

req_conn(x)

ESTAB

retransmit req_conn(x)

acc_conn(x)

ESTAB

data(x+1)

accept data(x+1)

retransmit data(x+1)

connection x completes

client terminates

server forgets x

req_conn(x)

ESTAB

data(x+1)

accept data(x+1)

❌ Problem: dup data accepted!

TCP 3-way handshake

Client state

```
clientSocket = socket(AF_INET, SOCK_STREAM)
```

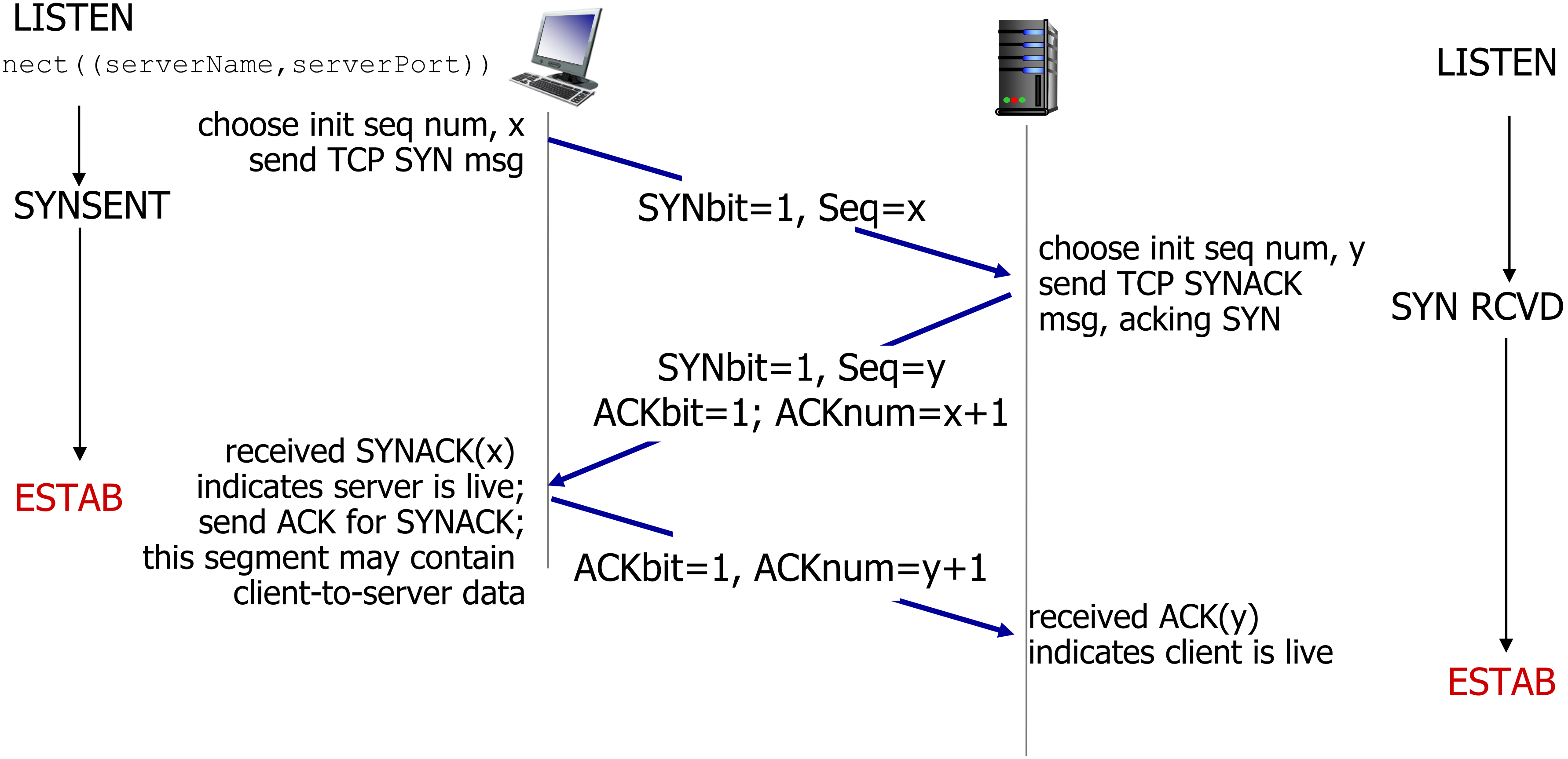
LISTEN

```
clientSocket.connect((serverName,serverPort))
```

Server state

```
serverSocket = socket(AF_INET,SOCK_STREAM)
serverSocket.bind((‘’,serverPort))
serverSocket.listen(1)
connectionSocket, addr = serverSocket.accept()
```

LISTEN



choose init seq num, x
send TCP SYN msg

SYNbit=1, Seq=x

SYNSENT

choose init seq num, y
send TCP SYNACK
msg, acking SYN

SYN RCVD

SYNbit=1, Seq=y
ACKbit=1; ACKnum=x+1

received SYNACK(x)
indicates server is live;
send ACK for SYNACK;
this segment may contain
client-to-server data

ESTAB

ACKbit=1, ACKnum=y+1

received ACK(y)
indicates client is live

ESTAB

A human 3-way handshake protocol

1. On belay?
2. Belay on.
3. Climbing.

Closing a TCP connection

- client, server each close their side of connection
  - send TCP segment with FIN bit = 1
- respond to received FIN with ACK
  - on receiving FIN, ACK can be combined with own FIN
- simultaneous FIN exchanges can be handled

Chapter 3: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- Principles of reliable data transfer
- Connection-oriented transport: TCP
- Principles of congestion control
- TCP congestion control
- Evolution of transport-layer functionality

Principles of congestion control

Congestion:

- informally: “too many sources sending too much data too fast for *network* to handle”
- manifestations:
  - long delays (queueing in router buffers)
  - packet loss (buffer overflow at routers)
- different from flow control!
- a top-10 problem!

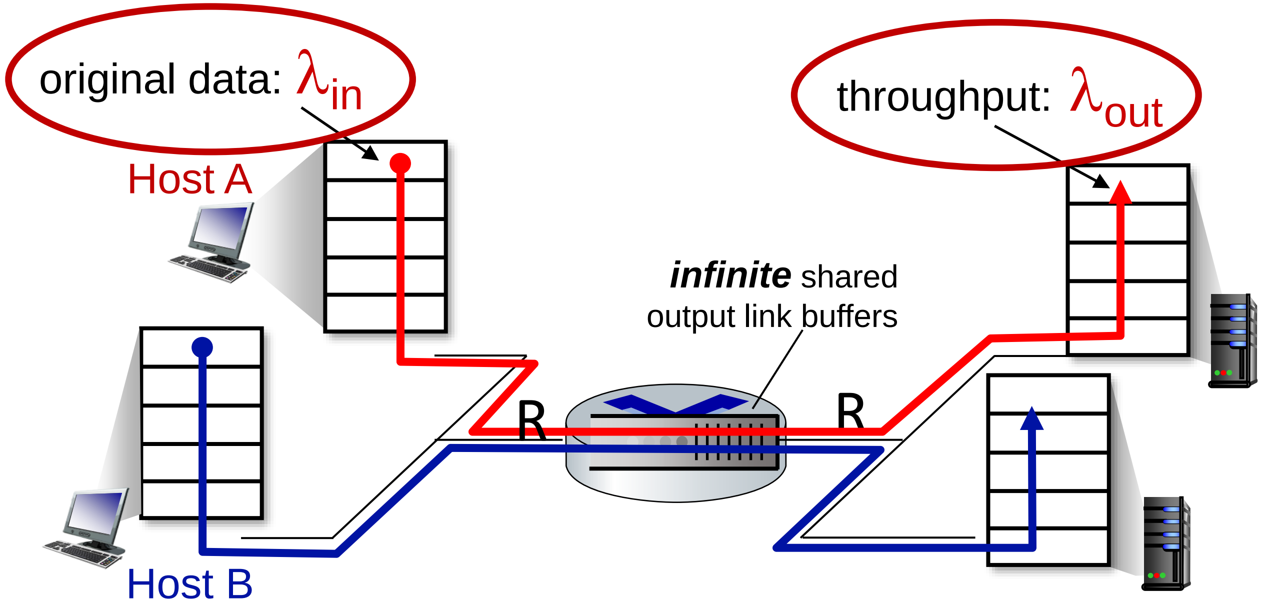
congestion control: too many senders, sending too fast

flow control: one sender too fast for one receiver

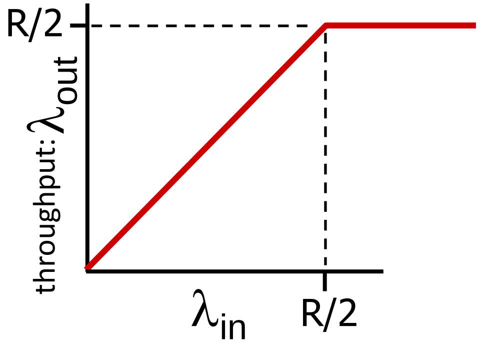
Causes/costs of congestion: scenario 1

Simplest scenario:

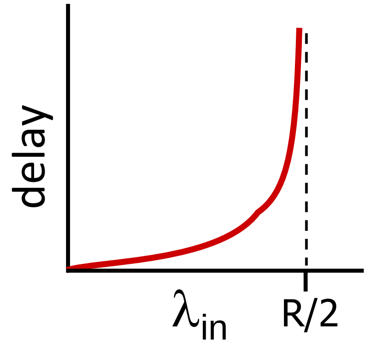
- one router, infinite buffers
- input, output link capacity: R
- two flows
- no retransmissions needed


original data: λ_{in}
throughput: λ_{out}
Host A
Host B
infinite shared output link buffers
R
R

Q: What happens as arrival rate λ_{in} approaches R/2?



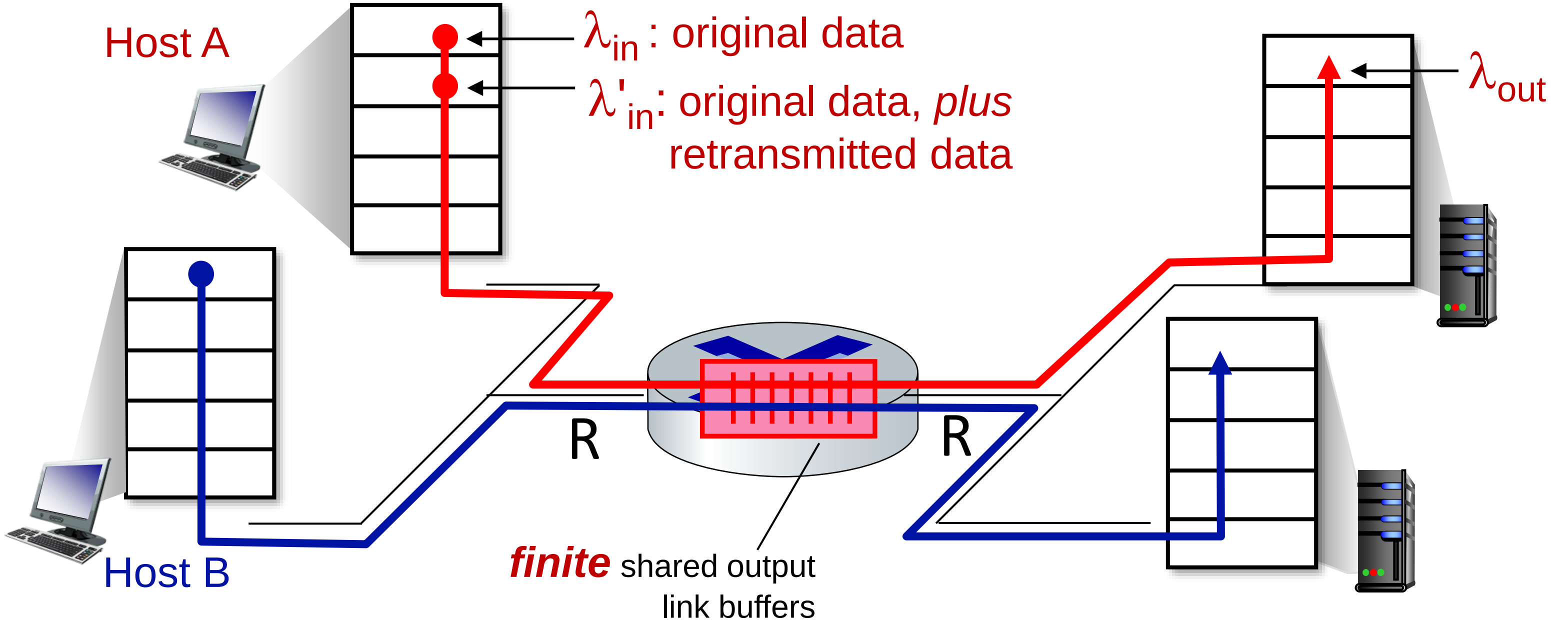
maximum per-connection throughput: R/2



large delays as arrival rate λ_{in} approaches capacity

Causes/costs of congestion: scenario 2

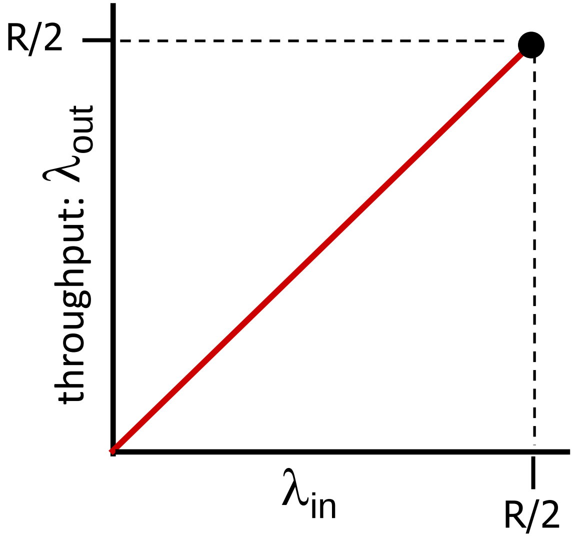
- one router, *finite* buffers
- sender retransmits lost, timed-out packet
  - application-layer input = application-layer output: $\lambda_{in} = \lambda_{out}$
  - transport-layer input includes *retransmissions* : $\lambda'_{in} \geq \lambda_{in}$

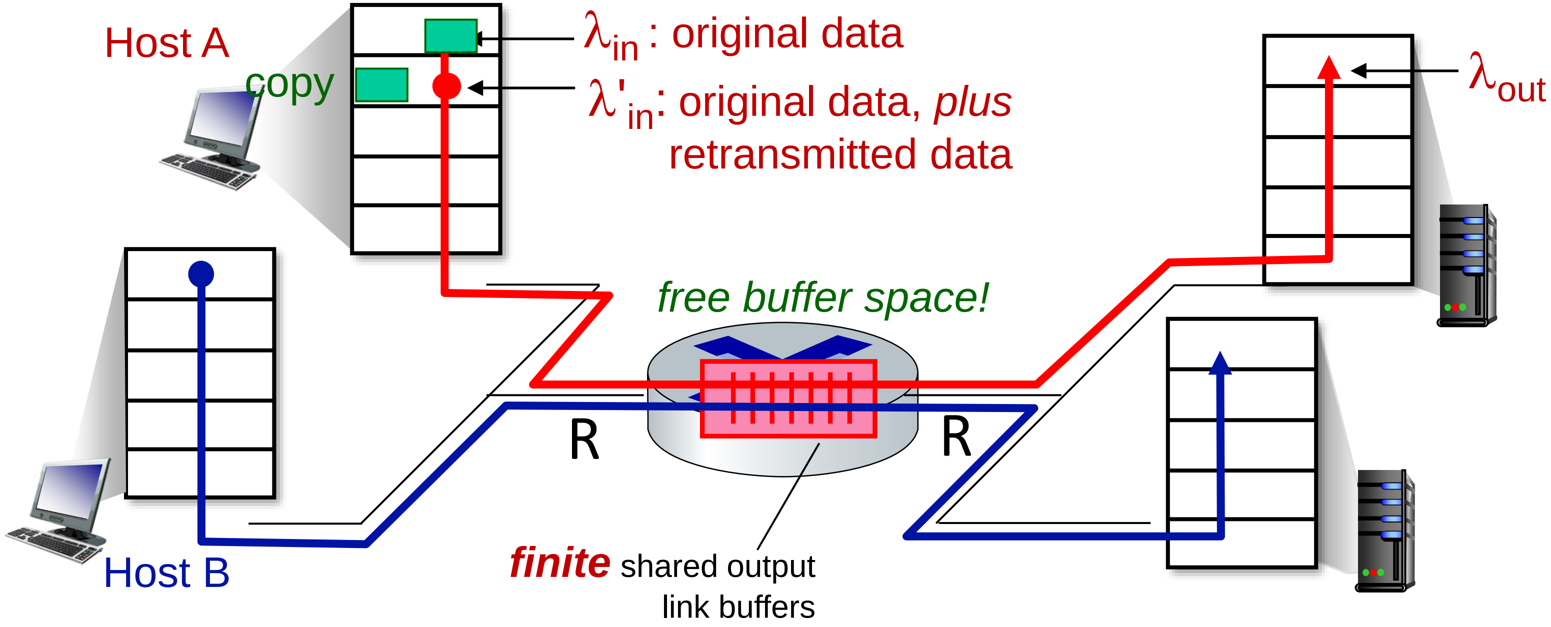

Host A
λ_{in} : original data
λ'_{in}: original data, *plus* retransmitted data
λ_{out}
R
R
Host B
finite shared output link buffers

Causes/costs of congestion: scenario 2

Idealization: perfect knowledge

- sender sends only when router buffers available

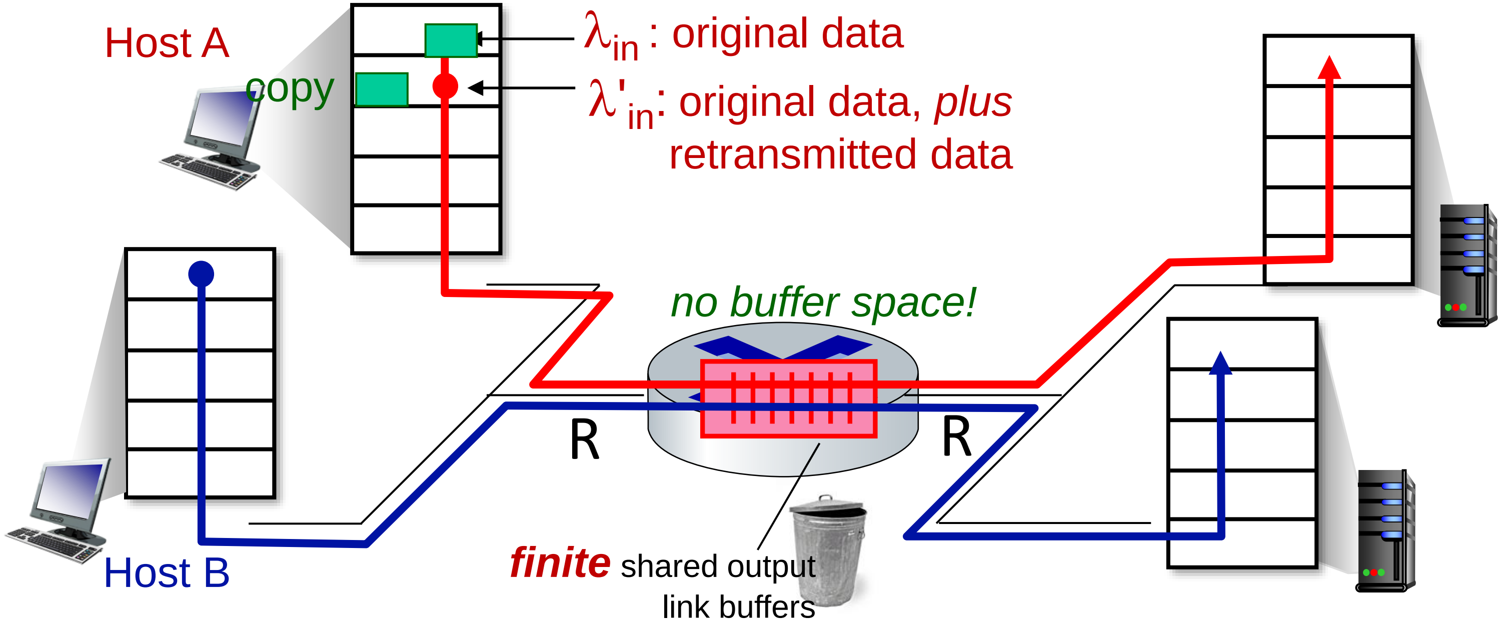



Host A
copy
λ_{in} : original data
λ'_{in}: original data, *plus* retransmitted data
λ_{out}
free buffer space!
R
R
finite shared output link buffers
Host B

Causes/costs of congestion: scenario 2

Idealization: *some* perfect knowledge

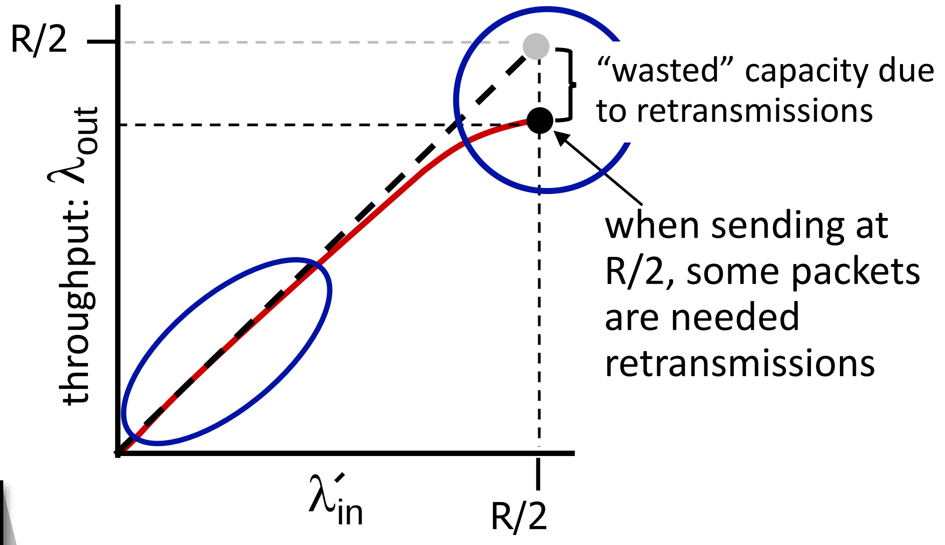
- packets can be lost (dropped at router) due to full buffers
- sender knows when packet has been dropped: only resends if packet *known* to be lost


Host A
copy
λ_{in} : original data
λ'_{in}: original data, *plus* retransmitted data
no buffer space!
R
R
Host B
finite shared output link buffers

Causes/costs of congestion: scenario 2

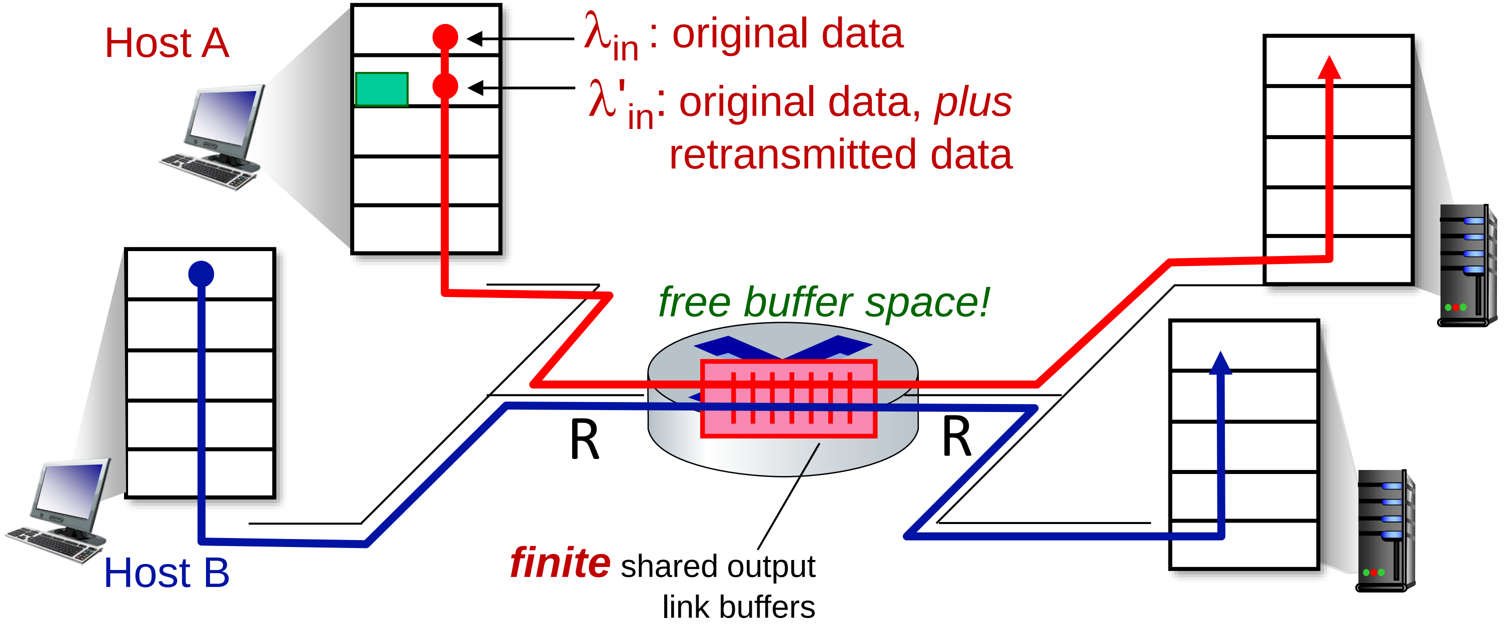
Idealization: *some* perfect knowledge

- packets can be lost (dropped at router) due to full buffers
- sender knows when packet has been dropped: only resends if packet *known* to be lost



"wasted" capacity due to retransmissions

when sending at R/2, some packets are needed retransmissions



λ_{in} : original data

λ'_{in}: original data, *plus* retransmitted data

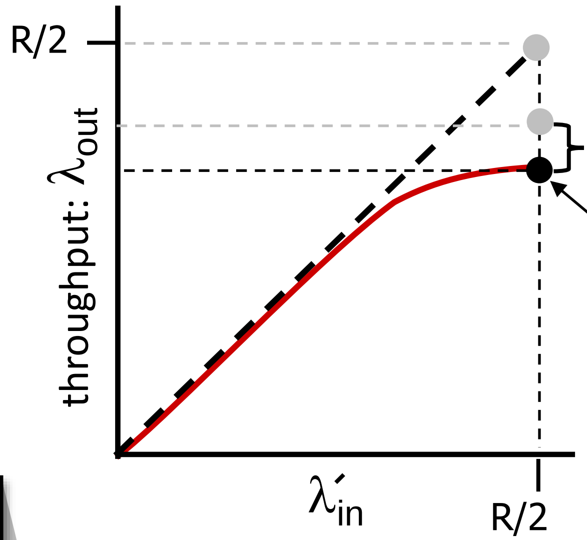
free buffer space!

finite shared output link buffers

Causes/costs of congestion: scenario 2

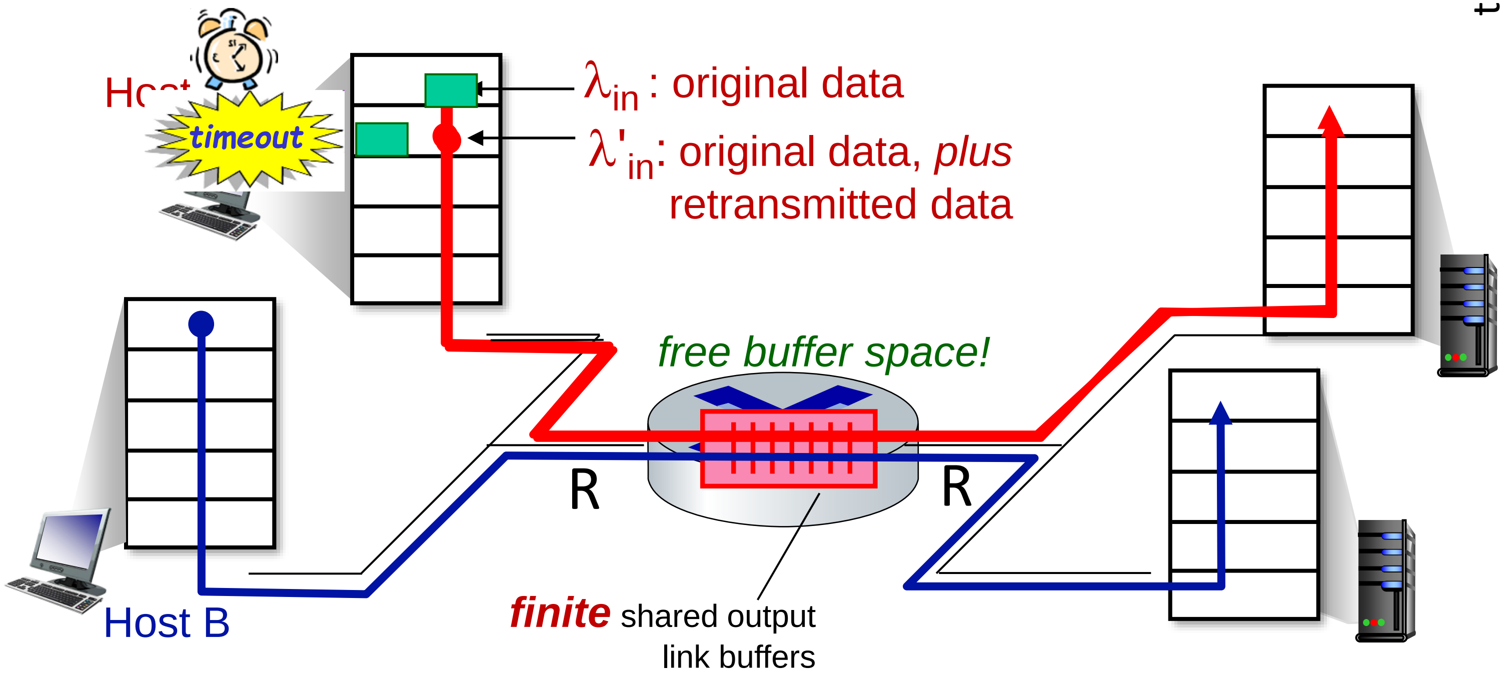
Realistic scenario: *un-needed duplicates*

- packets can be lost, dropped at router due to full buffers – requiring retransmissions
- but sender times can time out prematurely, sending *two* copies, *both* of which are delivered



"wasted" capacity due to un-needed retransmissions

when sending at R/2, some packets are retransmissions, including needed and *un-needed* duplicates, that are delivered!



λ_{in} : original data

λ'_{in}: original data, *plus* retransmitted data

free buffer space!

finite shared output link buffers

Causes/costs of congestion: scenario 2

Realistic scenario: *un-needed duplicates*

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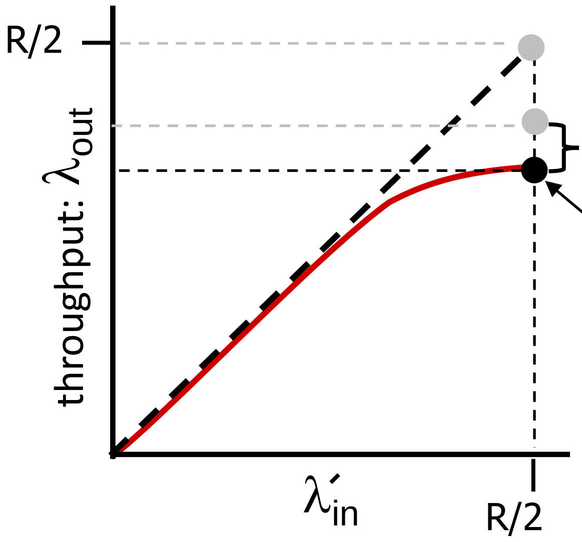


"wasted" capacity due to un-needed retransmissions

when sending at R/2, some packets are retransmissions, including needed and *un-needed* duplicates, that are delivered!

"costs" of congestion:

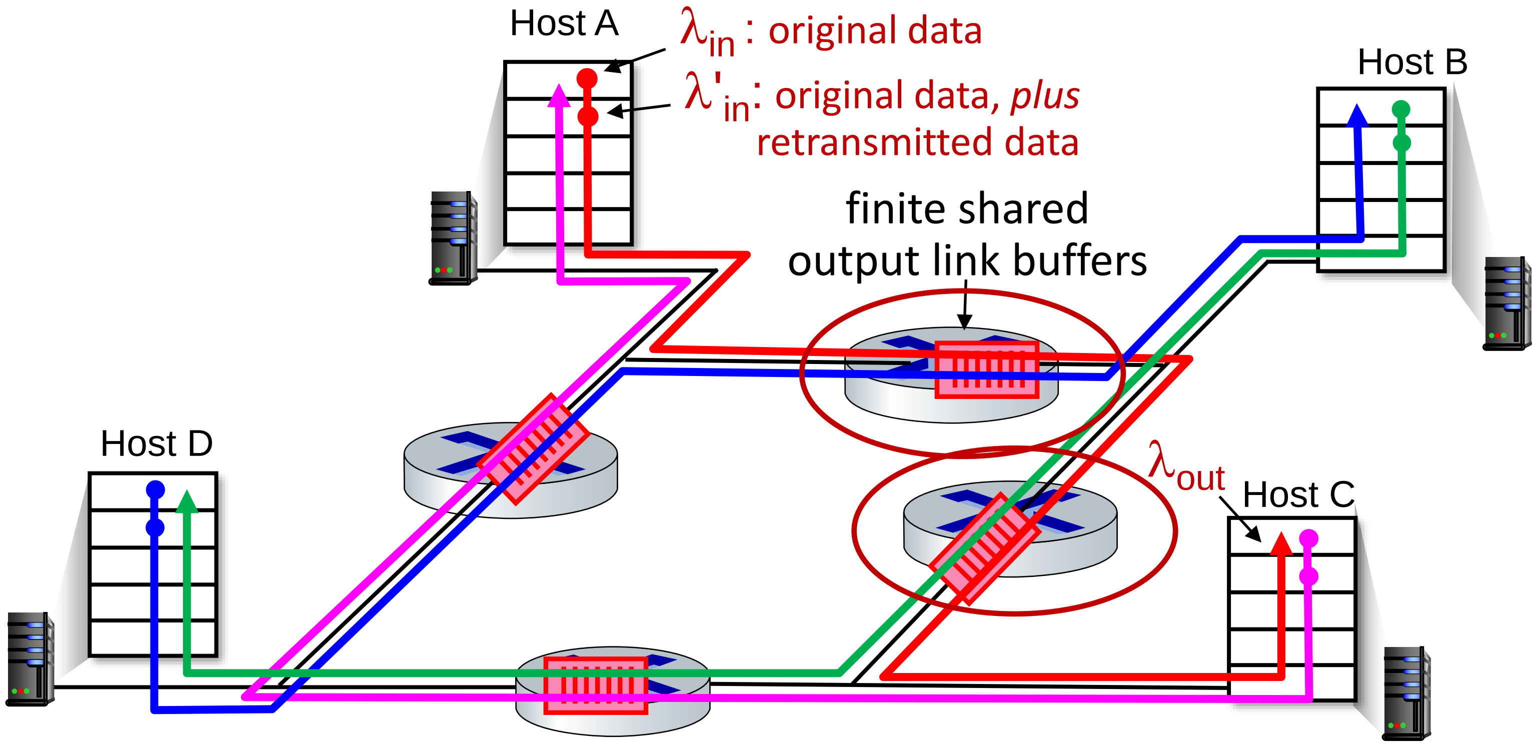
- more work (retransmission) for given receiver throughput
- unneeded retransmissions: link carries multiple copies of a packet
  - decreasing maximum achievable throughput

Causes/costs of congestion: scenario 3

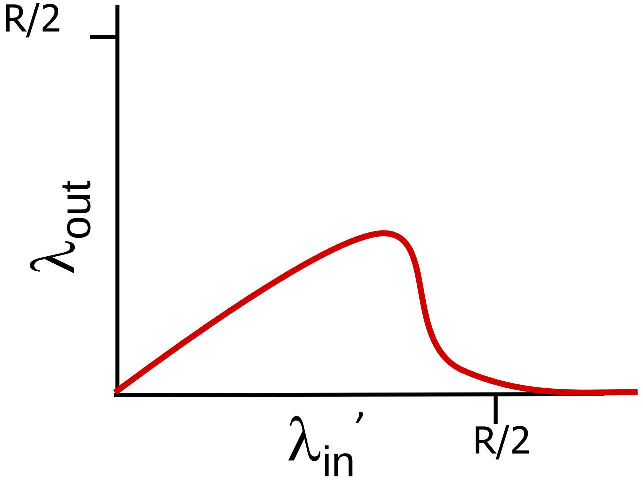
- *four* senders
- *multi-hop* paths
- timeout/retransmit

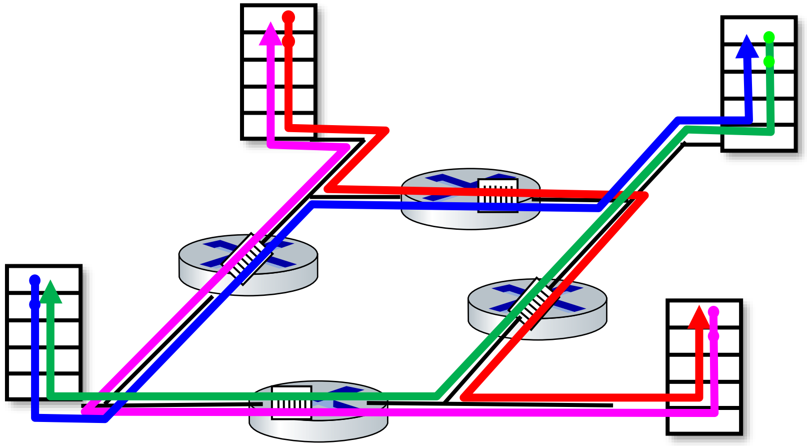
Q: what happens as λ_{in} and λ_{in}' increase ?

A: as red λ_{in}' increases, all arriving blue pkts at upper queue are dropped, blue throughput → 0


Host A
λ_{in} : original data
λ'_{in}: original data, *plus* retransmitted data
Host B
finite shared output link buffers
Host D
λ_{out}
Host C

Causes/costs of congestion: scenario 3



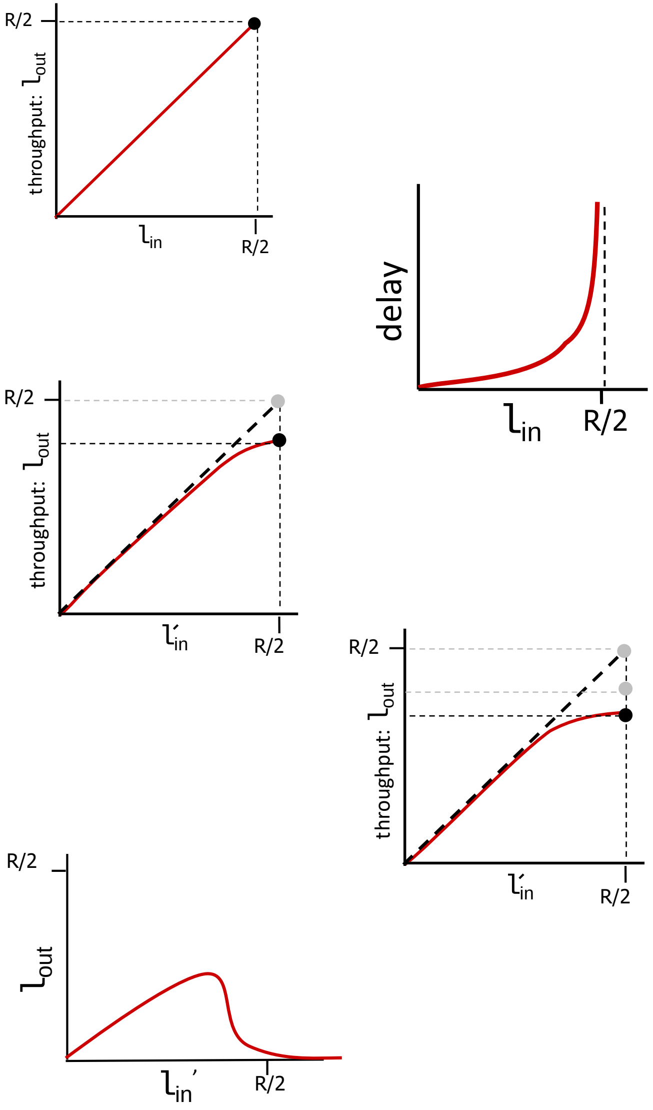


another "cost" of congestion:

- when packet dropped, any upstream transmission capacity and buffering used for that packet was wasted!

Causes/costs of congestion: insights

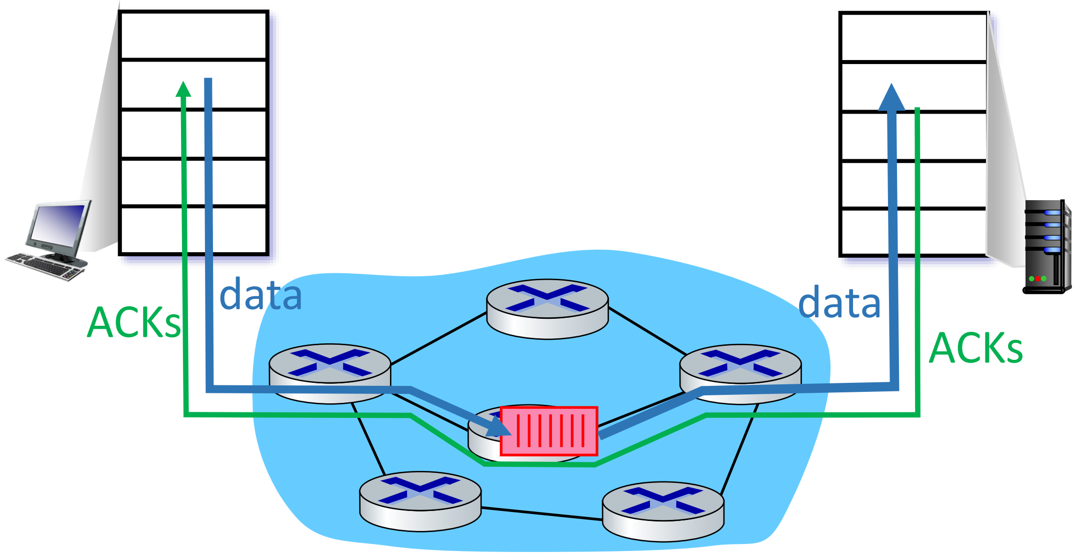
- throughput can never exceed capacity
- delay increases as capacity approached
- loss/retransmission decreases effective throughput
- un-needed duplicates further decreases effective throughput
- upstream transmission capacity / buffering wasted for packets lost downstream



Approaches towards congestion control

End-end congestion control:

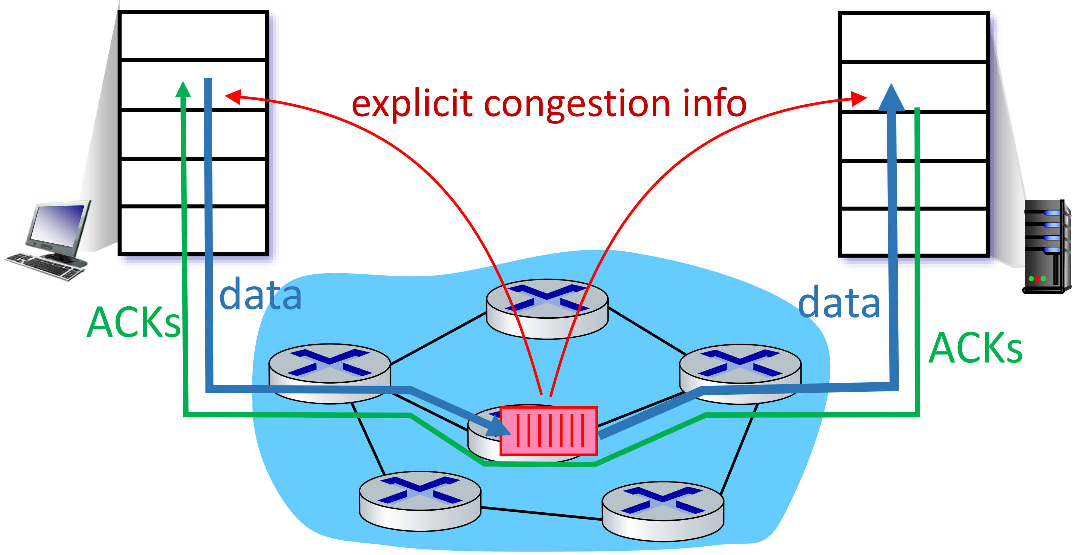
- no explicit feedback from network
- congestion *inferred* from observed loss, delay
- approach taken by TCP



Approaches towards congestion control

Network-assisted congestion control:

- routers provide *direct* feedback to sending/receiving hosts with flows passing through congested router
- may indicate congestion level or explicitly set sending rate
- TCP ECN, ATM, DECbit protocols


explicit congestion info
ACKs
data
data
ACKs

Chapter 3: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- Principles of reliable data transfer
- Connection-oriented transport: TCP
- Principles of congestion control
- TCP congestion control
- Evolution of transport-layer functionality

TCP congestion control: AIMD

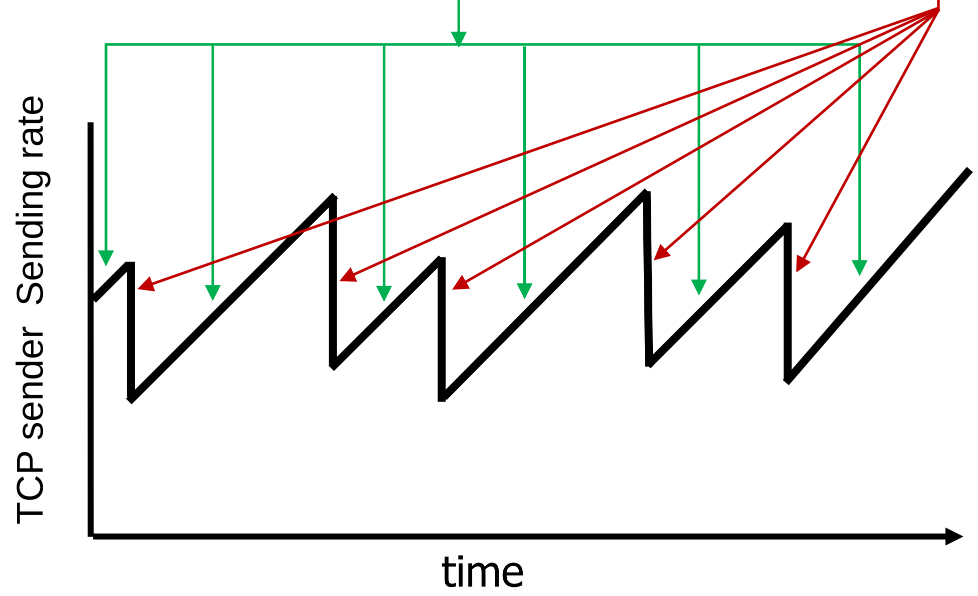
- *approach:* senders can increase sending rate until packet loss (congestion) occurs, then decrease sending rate on loss event

Additive Increase

increase sending rate by 1 maximum segment size every RTT until loss detected

Multiplicative Decrease

cut sending rate in half at each loss event



AIMD sawtooth behavior: *probing* for bandwidth

TCP AIMD: more

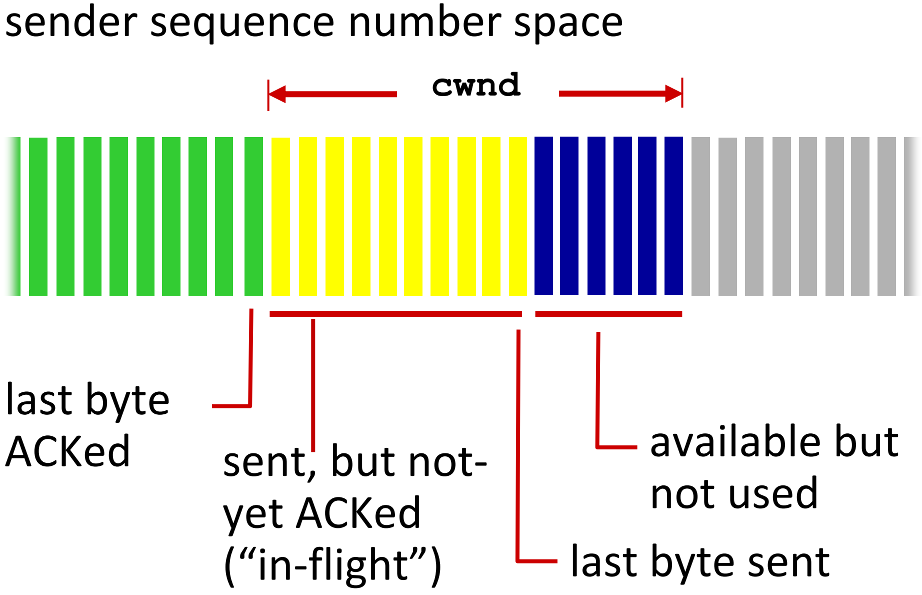
Multiplicative decrease detail: sending rate is

- Cut in half on loss detected by triple duplicate ACK (TCP Reno)
- Cut to 1 MSS (maximum segment size) when loss detected by timeout (TCP Tahoe)

Why AIMD?

- AIMD – a distributed, asynchronous algorithm – has been shown to:
    - optimize congested flow rates network wide!
    - have desirable stability properties

TCP congestion control: details


sender sequence number space
cwnd
last byte ACKed
sent, but not-yet ACKed ("in-flight")
available but not used
last byte sent

TCP sending behavior:

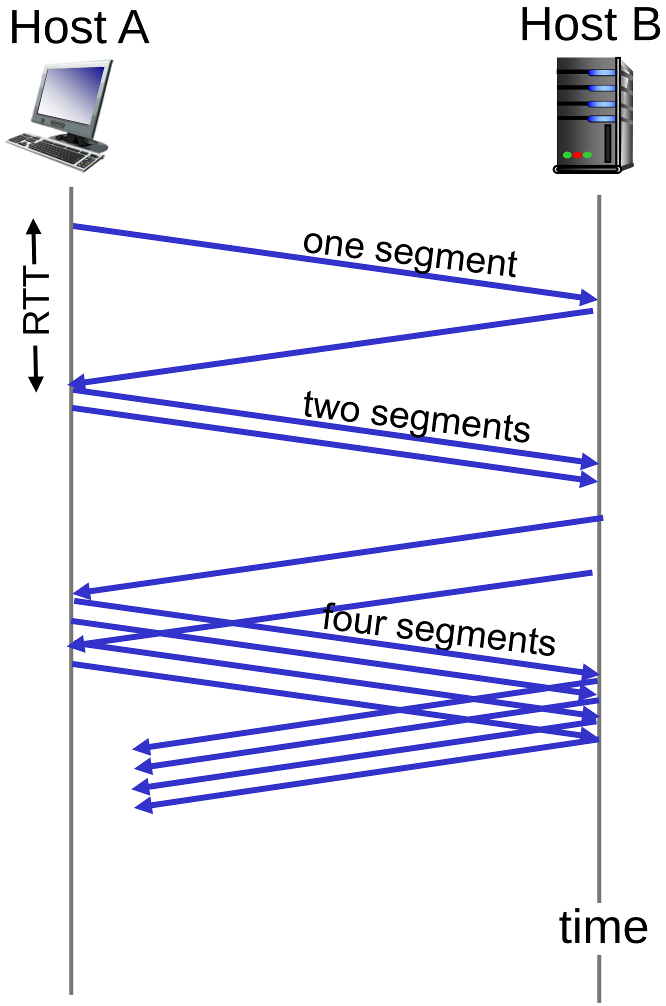
- *roughly:* send `cwnd` bytes, wait RTT for ACKS, then send more bytes

$$\text{TCP rate} \approx \frac{\texttt{cwnd}}{\text{RTT}} \text{ bytes/sec}$$

- TCP sender limits transmission: `LastByteSent- LastByteAcked ≤ cwnd`
- `cwnd` is dynamically adjusted in response to observed network congestion (implementing TCP congestion control)

TCP slow start

- when connection begins, increase rate exponentially until first loss event:
  - initially **`cwnd`** = 1 MSS
  - double **`cwnd`** every RTT
  - done by incrementing **`cwnd`** for every ACK received
- *summary:* initial rate is slow, but ramps up exponentially fast


Host A
Host B
RTT
one segment
two segments
four segments
time

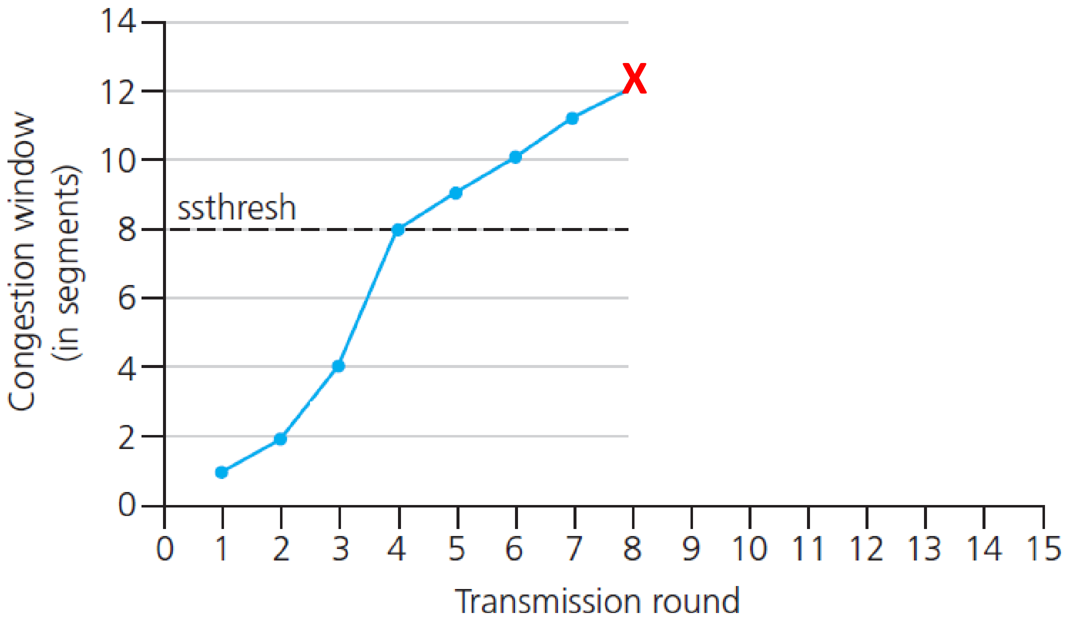
TCP: from slow start to congestion avoidance

Q: when should the exponential increase switch to linear?

A: when **`cwnd`** gets to 1/2 of its value before timeout.

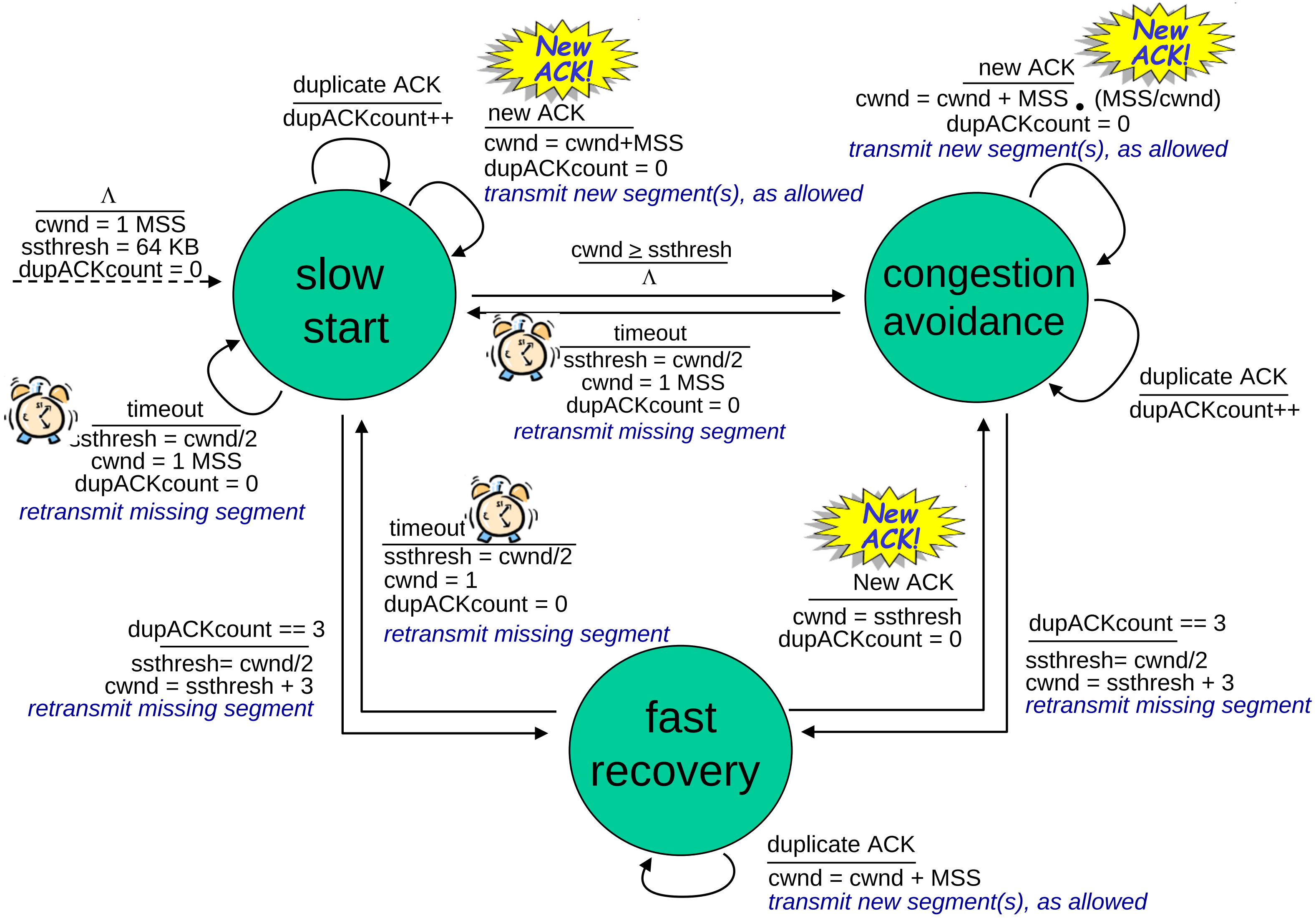
Implementation:

- variable **`ssthresh`**
- on loss event, **`ssthresh`** is set to 1/2 of **`cwnd`** just before loss event



* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

Summary: TCP congestion control



Λ

cwnd = 1 MSS
ssthresh = 64 KB
dupACKcount = 0

duplicate ACK

dupACKcount++

new ACK

cwnd = cwnd+MSS
dupACKcount = 0
transmit new segment(s), as allowed

timeout

ssthresh = cwnd/2
cwnd = 1 MSS
dupACKcount = 0
retransmit missing segment

cwnd ≥ ssthresh

Λ

timeout

ssthresh = cwnd/2
cwnd = 1 MSS
dupACKcount = 0
retransmit missing segment

new ACK

cwnd = cwnd + MSS • (MSS/cwnd)
dupACKcount = 0
transmit new segment(s), as allowed

duplicate ACK

dupACKcount++

timeout

ssthresh = cwnd/2
cwnd = 1
dupACKcount = 0
retransmit missing segment

dupACKcount == 3

ssthresh= cwnd/2
cwnd = ssthresh + 3
retransmit missing segment

New ACK

cwnd = ssthresh
dupACKcount = 0

dupACKcount == 3

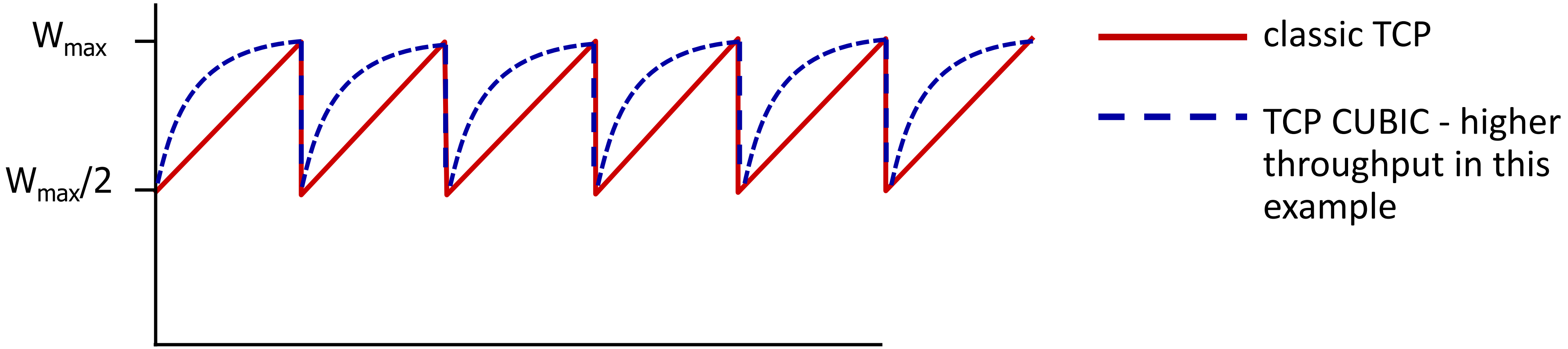
ssthresh= cwnd/2
cwnd = ssthresh + 3
retransmit missing segment

duplicate ACK

cwnd = cwnd + MSS
transmit new segment(s), as allowed

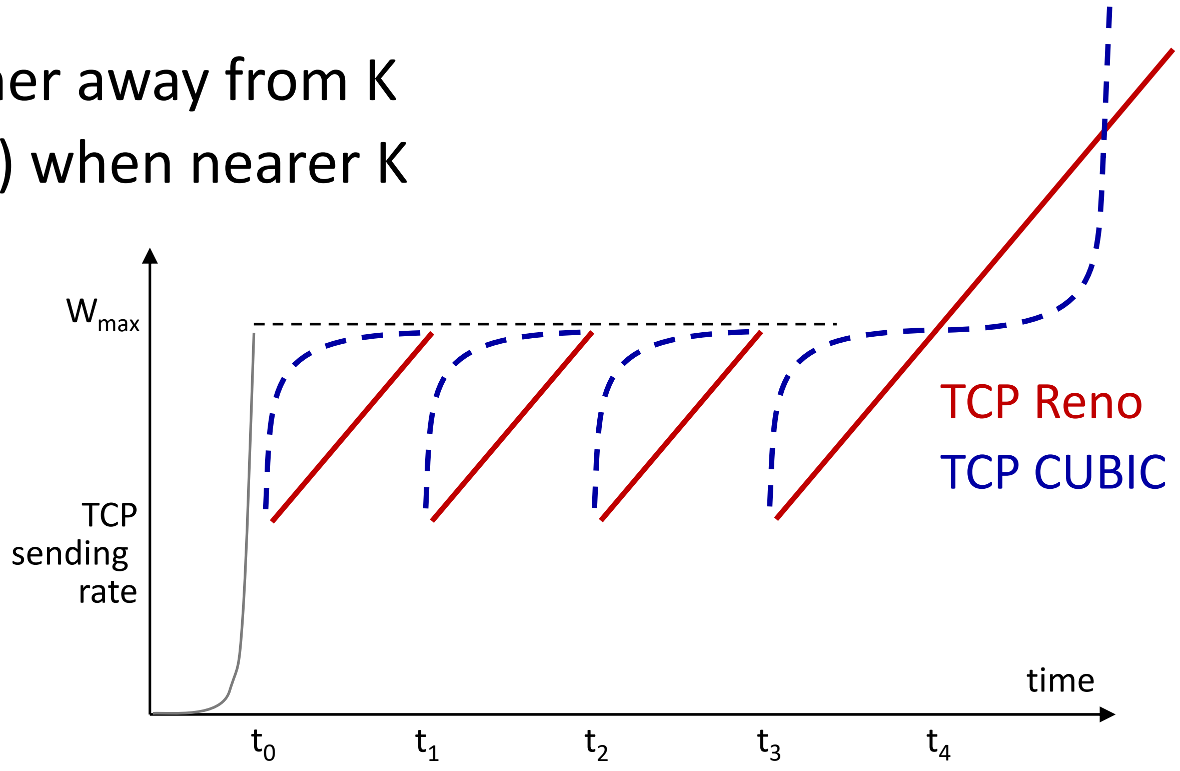
TCP CUBIC

- Is there a better way than AIMD to "probe" for usable bandwidth?
- Insight/intuition:
  - W_{max}: sending rate at which congestion loss was detected
  - congestion state of bottleneck link probably (?) hasn't changed much
  - after cutting rate/window in half on loss, initially ramp to to W_{max} *faster*, but then approach W_{max} more *slowly*



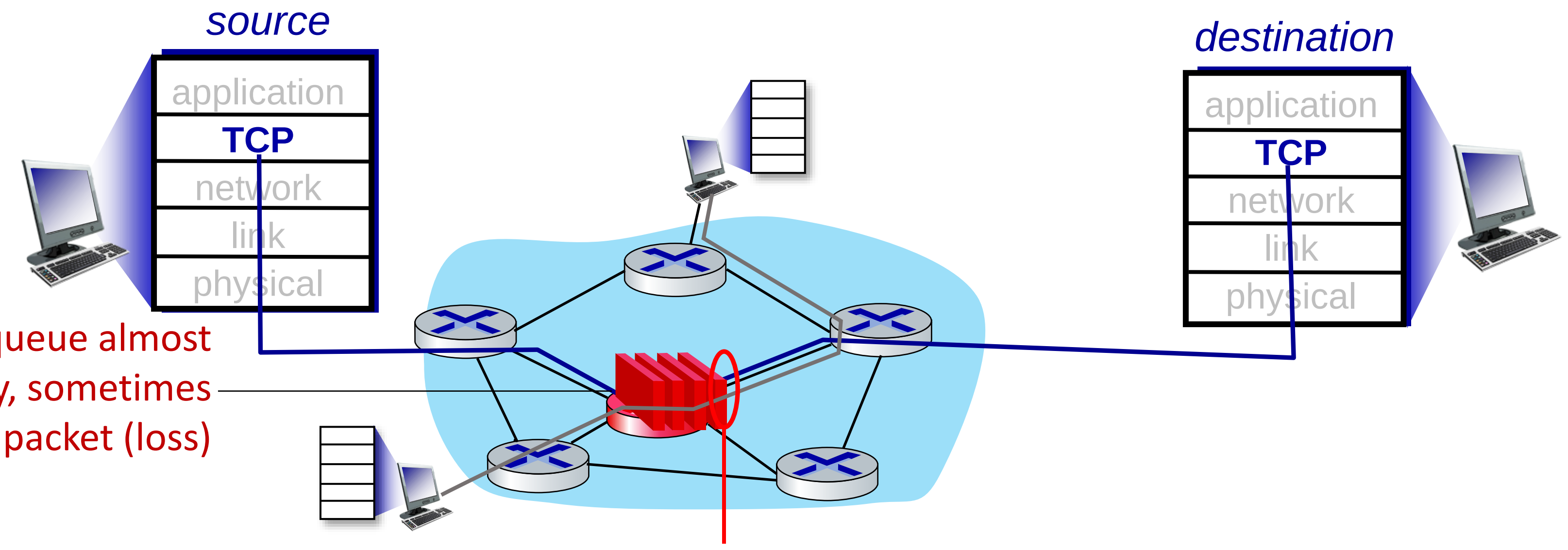
TCP CUBIC

- K: point in time when TCP window size will reach W_{max}
  - K itself is tuneable
- increase W as a function of the *cube* of the distance between current time and K
  - larger increases when further away from K
  - smaller increases (cautious) when nearer K
- TCP CUBIC default in Linux, most popular TCP for popular Web servers



TCP and the congested "bottleneck link"

- TCP (classic, CUBIC) increase TCP's sending rate until packet loss occurs at some router's output: the *bottleneck link*

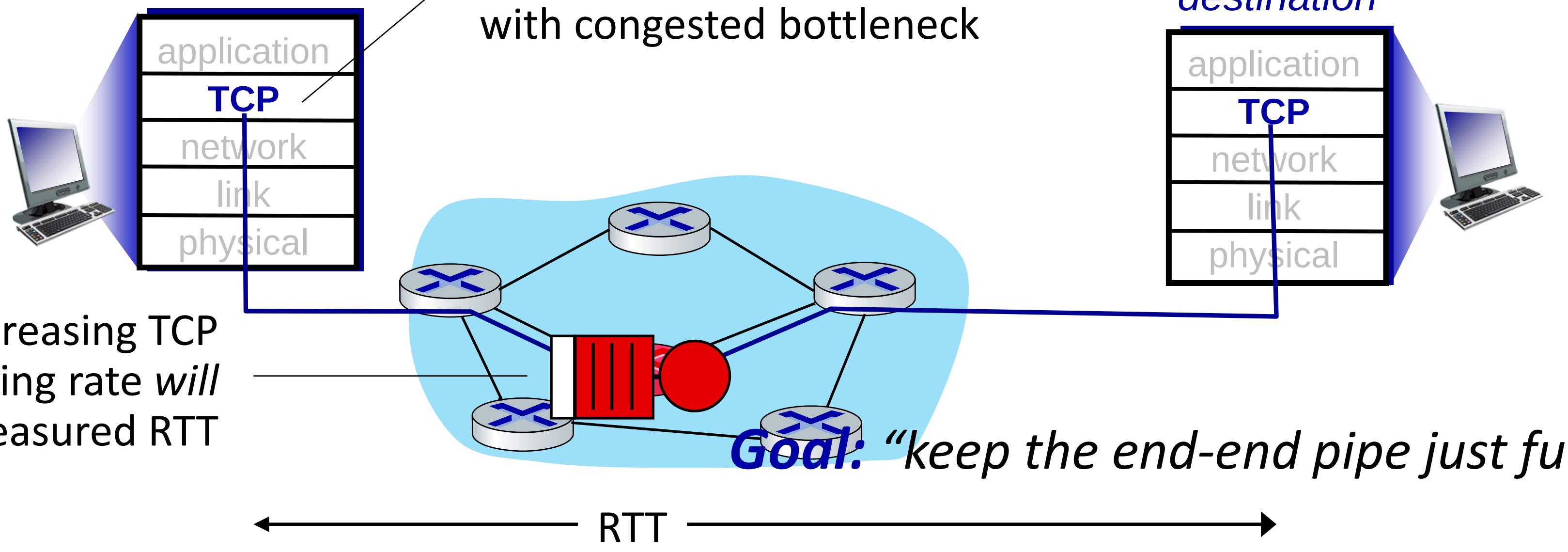

source
application
TCP
network
link
physical
destination
application
TCP
network
link
physical

packet queue almost never empty, sometimes overflows packet (loss)

bottleneck link (almost always busy)

TCP and the congested "bottleneck link"

- TCP (classic, CUBIC) increase TCP's sending rate until packet loss occurs at some router's output: the *bottleneck link*
- understanding congestion: useful to focus on congested bottleneck link


source
application
TCP
network
link
physical
destination
application
TCP
network
link
physical
RTT

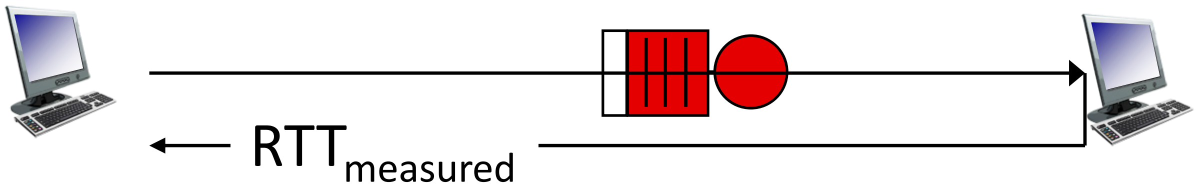
insight: increasing TCP sending rate will *not* increase end-end throughout with congested bottleneck

insight: increasing TCP sending rate *will* increase measured RTT

Goal: *"keep the end-end pipe just full, but not fuller"*

Delay-based TCP congestion control

Keeping sender-to-receiver pipe “just full enough, but no fuller”: keep bottleneck link busy transmitting, but avoid high delays/buffering



$$\text{measured throughput} = \frac{\text{\# bytes sent in last RTT interval}}{RTT_{measured}}$$

Delay-based approach:

- RTT_{min} - minimum observed RTT (uncongested path)
- uncongested throughput with congestion window `cwnd` is cwnd/RTT_{min}

```
if measured throughput “very close” to  uncongested throughput
    increase cwnd linearly            /* since path not congested */
else if measured throughput “far below” uncongested throughout
    decrease cwnd  linearly           /* since path is congested */
```

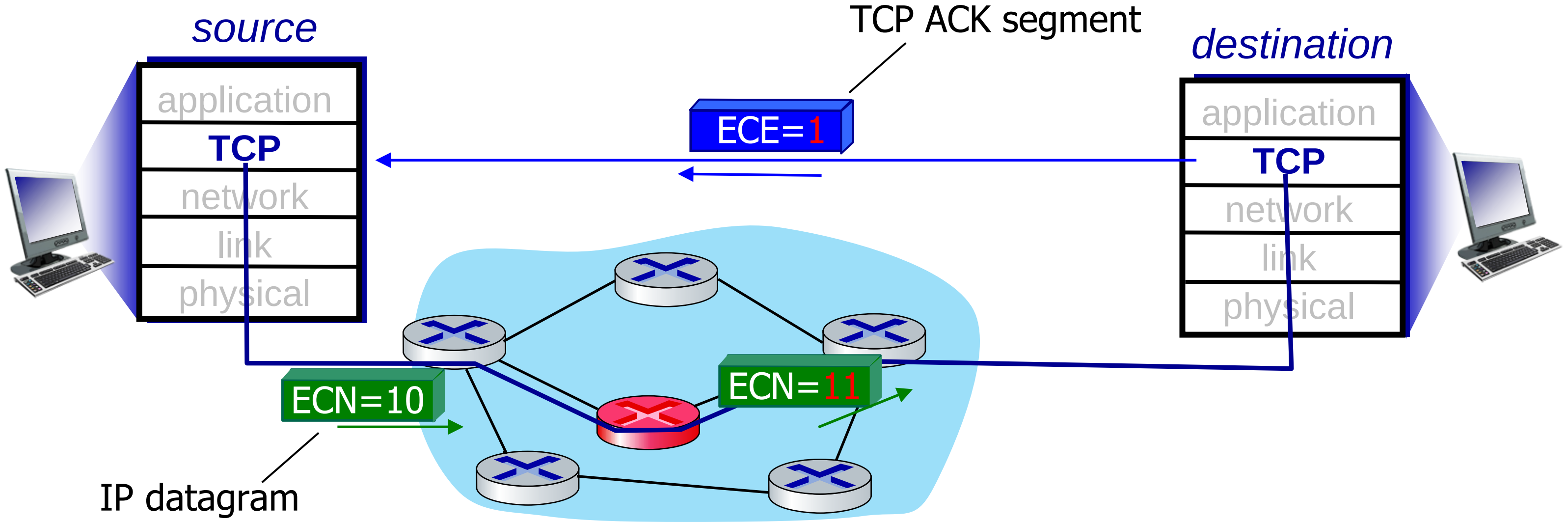
Delay-based TCP congestion control

- congestion control without inducing/forcing loss
- maximizing throughout ("keeping the just pipe full... ") while keeping delay low ("…but not fuller")
- a number of deployed TCPs take a delay-based approach
  - BBR deployed on Google's (internal) backbone network

Explicit congestion notification (ECN)

TCP deployments often implement *network-assisted* congestion control:

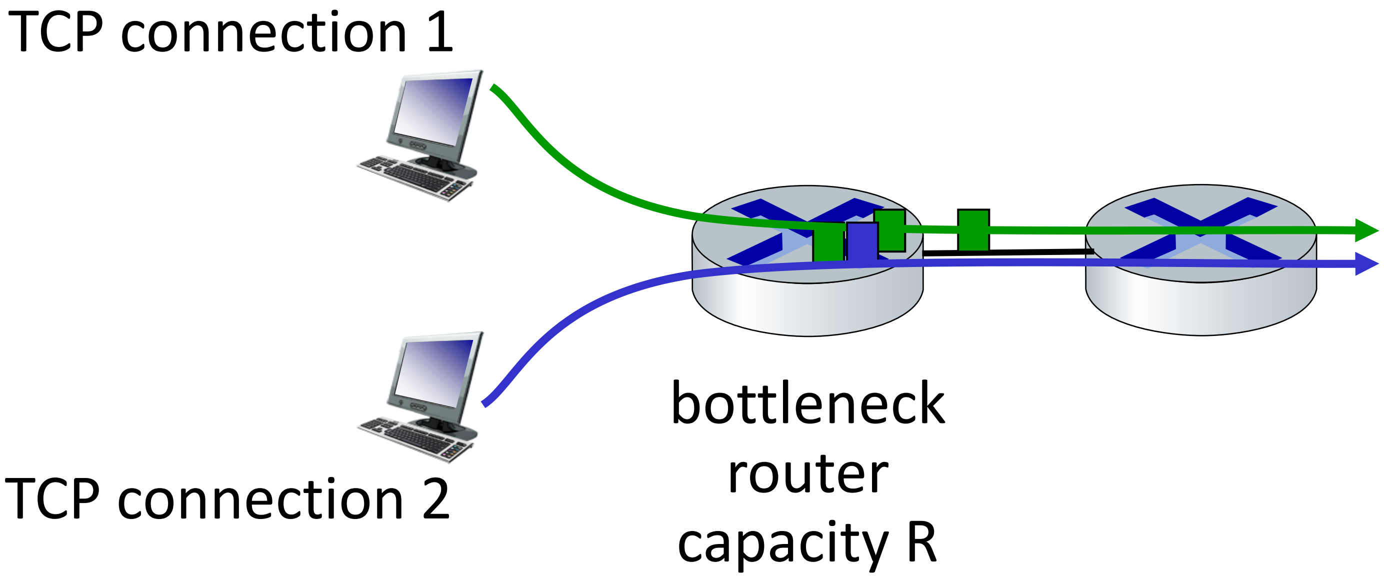
- two bits in IP header (ToS field) marked *by network router* to indicate congestion
  - *policy* to determine marking chosen by network operator
- congestion indication carried to destination
- destination sets ECE bit on ACK segment to notify sender of congestion
- involves both IP (IP header ECN bit marking) and TCP (TCP header C,E bit marking)



source
application
TCP
network
link
physical
TCP ACK segment
ECE=1
destination
application
TCP
network
link
physical
ECN=10
ECN=11
IP datagram

TCP fairness

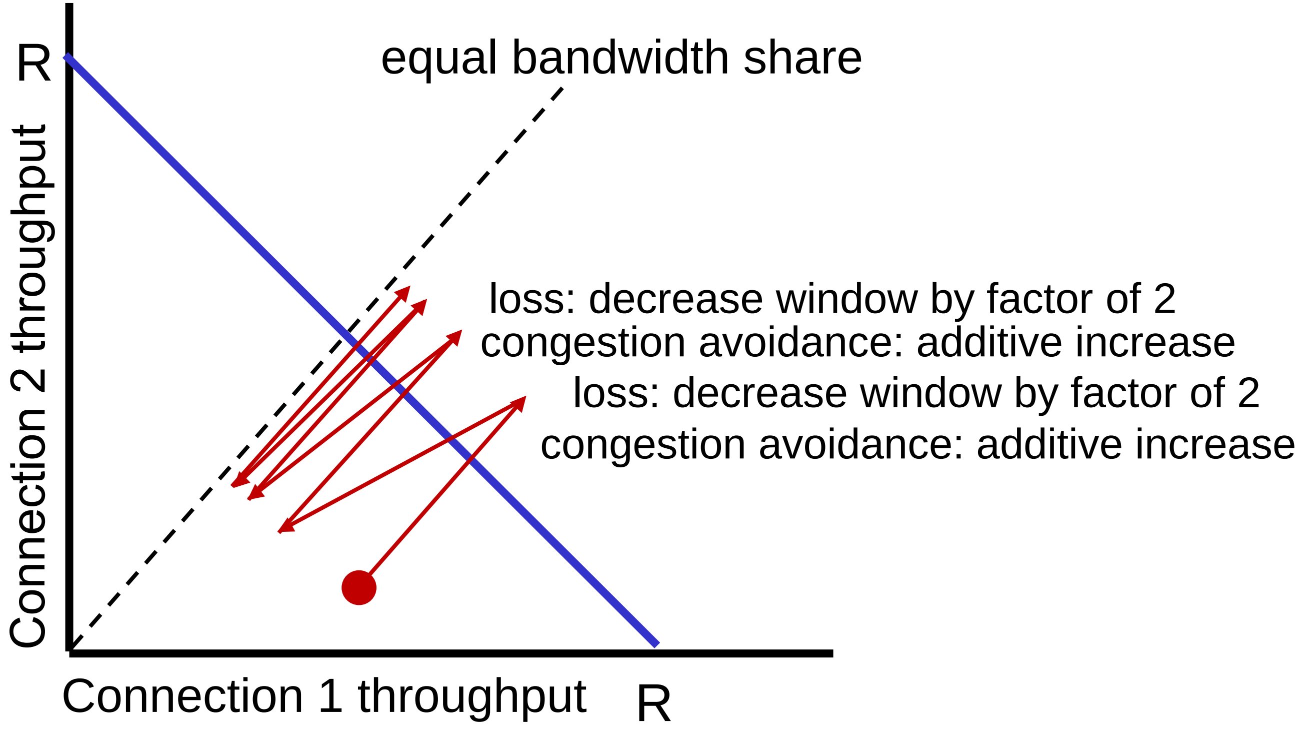
Fairness goal: if *K* TCP sessions share same bottleneck link of bandwidth *R*, each should have average rate of *R/K*


TCP connection 1
TCP connection 2
bottleneck router capacity R

Q: is TCP Fair?

Example: two competing TCP sessions:

- additive increase gives slope of 1, as throughout increases
- multiplicative decrease decreases throughput proportionally



Is TCP fair?

A: Yes, under idealized assumptions:

- same RTT
- fixed number of sessions only in congestion avoidance

Fairness: must all network apps be "fair"?

Fairness and UDP

- multimedia apps often do not use TCP
  - do not want rate throttled by congestion control
- instead use UDP:
  - send audio/video at constant rate, tolerate packet loss
- there is no "Internet police" policing use of congestion control

Fairness, parallel TCP connections

- application can open *multiple* parallel connections between two hosts
- web browsers do this , e.g., link of rate R with 9 existing connections:
  - new app asks for 1 TCP, gets rate R/10
  - new app asks for 11 TCPs, gets R/2

Transport layer: roadmap

- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- Principles of reliable data transfer
- Connection-oriented transport: TCP
- Principles of congestion control
- TCP congestion control
- Evolution of transport-layer functionality

Evolving transport-layer functionality

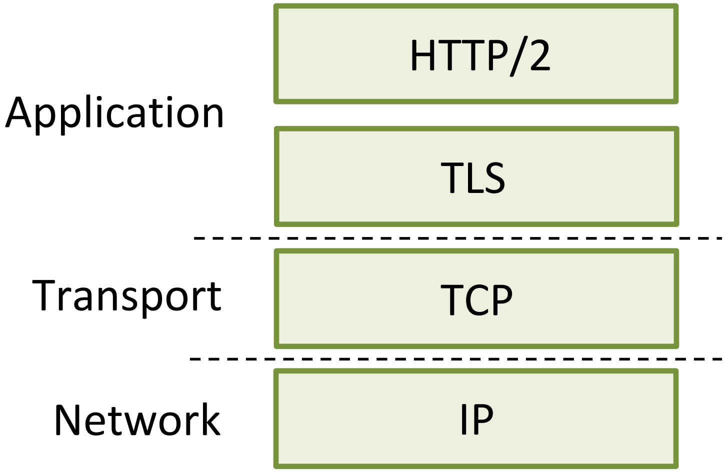
- TCP, UDP: principal transport protocols for 40 years
- different “flavors” of TCP developed, for specific scenarios:

Scenario	Challenges
Long, fat pipes (large data transfers)	Many packets “in flight”; loss shuts down pipeline
Wireless networks	Loss due to noisy wireless links, mobility; TCP treat this as congestion loss
Long-delay links	Extremely long RTTs
Data center networks	Latency sensitive
Background traffic flows	Low priority, “background” TCP flows

- moving transport–layer functions to application layer, on top of UDP
  - HTTP/3: QUIC

QUIC: Quick UDP Internet Connections

- application-layer protocol, on top of UDP
  - increase performance of HTTP
  - deployed on many Google servers, apps (Chrome, mobile YouTube app)


Application
HTTP/2
TLS
Transport
TCP
Network
IP

HTTP/2 over TCP

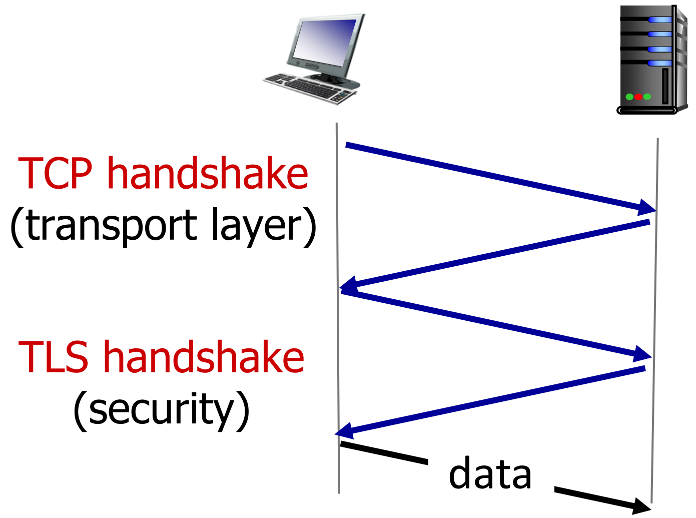
QUIC: Quick UDP Internet Connections

adopts approaches we've studied in this chapter for connection establishment, error control, congestion control

- **error and congestion control:** "Readers familiar with TCP's loss detection and congestion control will find algorithms here that parallel well-known TCP ones." [from QUIC specification]
- **connection establishment:** reliability, congestion control, authentication, encryption, state established in one RTT

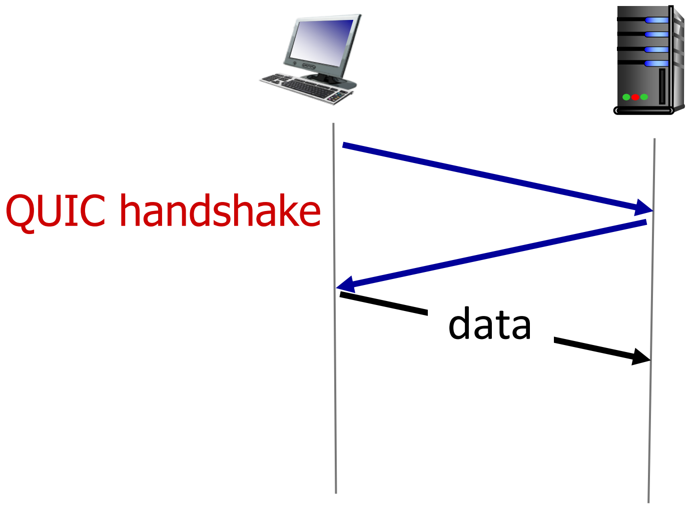
- multiple application-level "streams" multiplexed over single QUIC connection
  - separate reliable data transfer, security
  - common congestion control

QUIC: Connection establishment


TCP handshake
(transport layer)
TLS handshake
(security)
data

TCP (reliability, congestion control state) + TLS (authentication, crypto state)

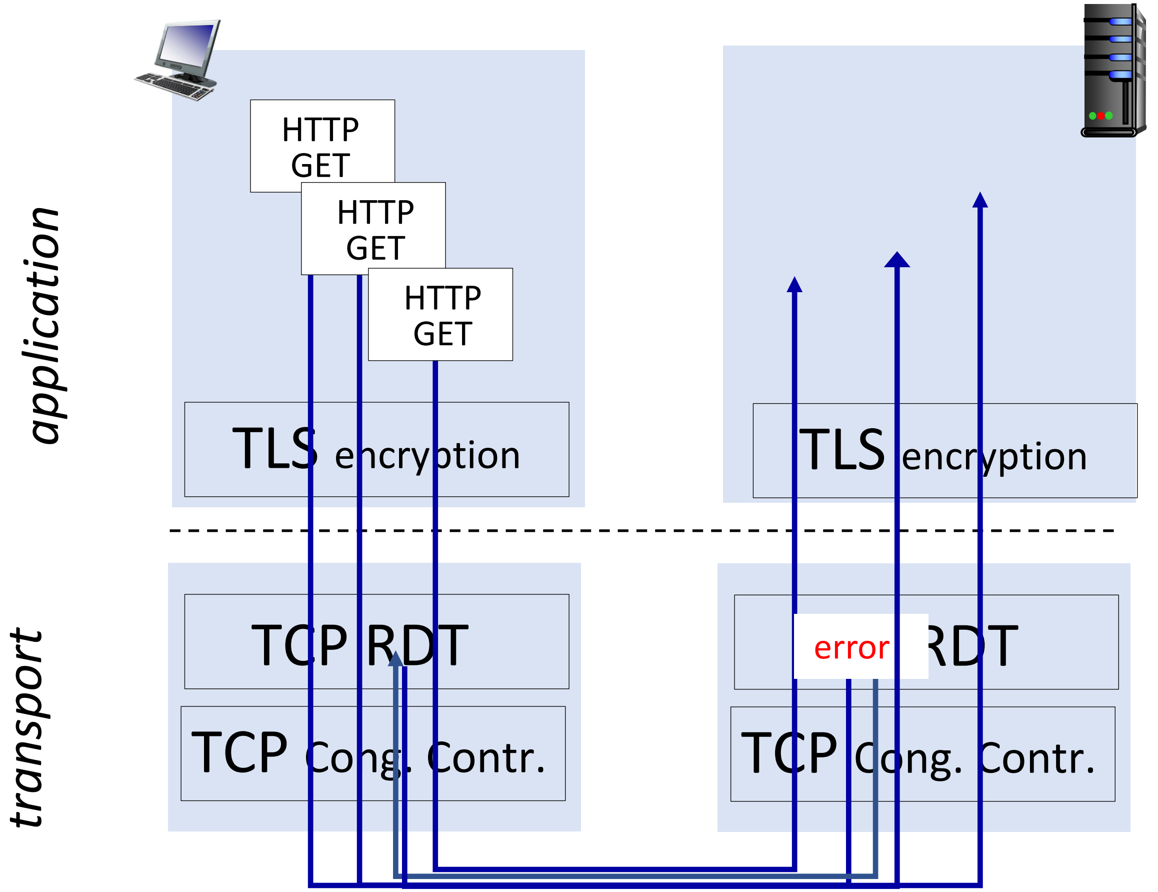
- 2 serial handshakes


QUIC handshake
data

QUIC: reliability, congestion control, authentication, crypto state

- 1 handshake

QUIC: streams: parallelism, no HOL blocking


application
transport
HTTP GET
HTTP GET
HTTP GET
TLS encryption
TLS encryption
TCP RDT
error
RDT
TCP Cong. Contr.
TCP Cong. Contr.

(a) HTTP 1.1

Chapter 3: summary

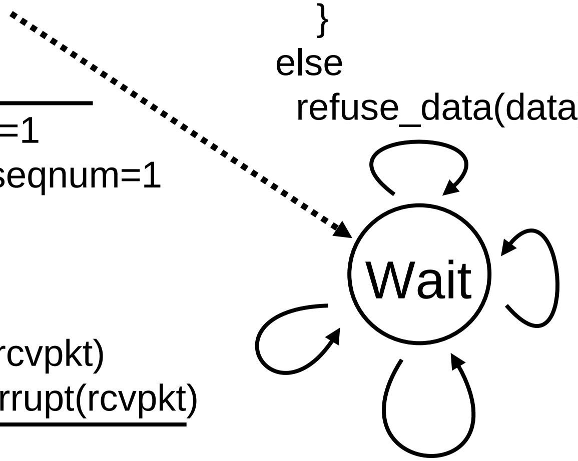
- principles behind transport layer services:
  - multiplexing, demultiplexing
  - reliable data transfer
  - flow control
  - congestion control
- instantiation, implementation in the Internet
  - UDP
  - TCP

Up next:

- leaving the network "edge" (application, transport layers)
- into the network "core"
- two network-layer chapters:
  - data plane
  - control plane

Additional Chapter 3 slides

Go-Back-N: sender extended FSM



```
rdt_send(data)
______________
if (nextseqnum < base+N) {
    sndpkt[nextseqnum] = make_pkt(nextseqnum,data,chksum)
    udt_send(sndpkt[nextseqnum])
    if (base == nextseqnum)
       start_timer
    nextseqnum++
    }
else
  refuse_data(data)
```

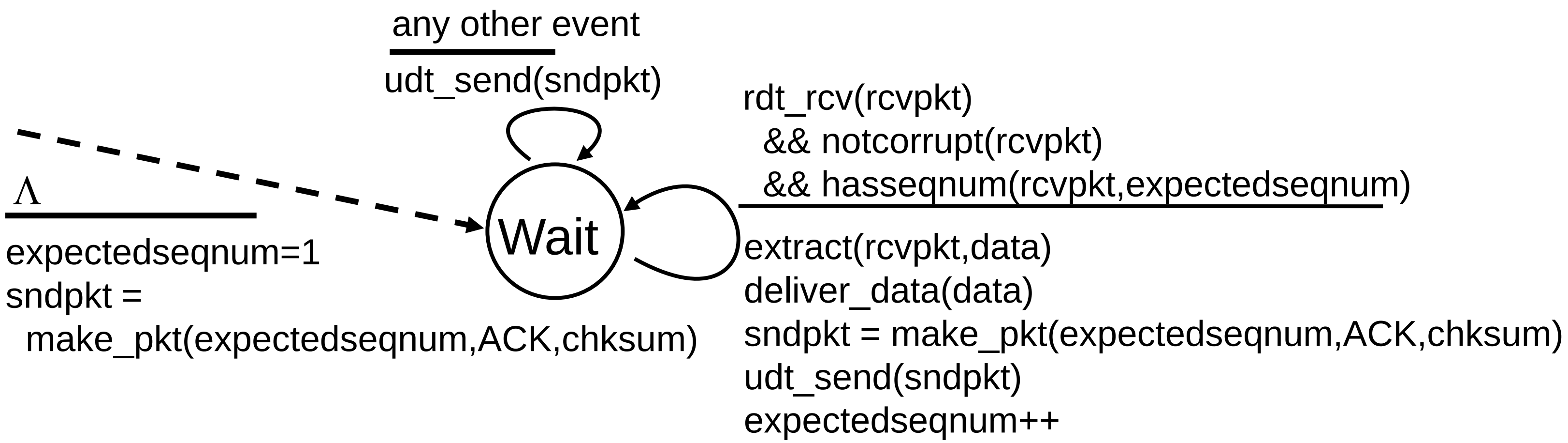
```
Λ
______________
base=1
nextseqnum=1
```

```
timeout
______________
start_timer
udt_send(sndpkt[base])
udt_send(sndpkt[base+1])
…
udt_send(sndpkt[nextseqnum-1])
```

```
rdt_rcv(rcvpkt)
   && corrupt(rcvpkt)
______________
```

```
rdt_rcv(rcvpkt) &&
   notcorrupt(rcvpkt)
______________
base = getacknum(rcvpkt)+1
If (base == nextseqnum)
    stop_timer
  else
    start_timer
```

Go-Back-N: receiver extended FSM



any other event

udt_send(sndpkt)

Λ

expectedseqnum=1
sndpkt =
  make_pkt(expectedseqnum,ACK,chksum)

Wait

rdt_rcv(rcvpkt)
  && notcorrupt(rcvpkt)
  && hasseqnum(rcvpkt,expectedseqnum)

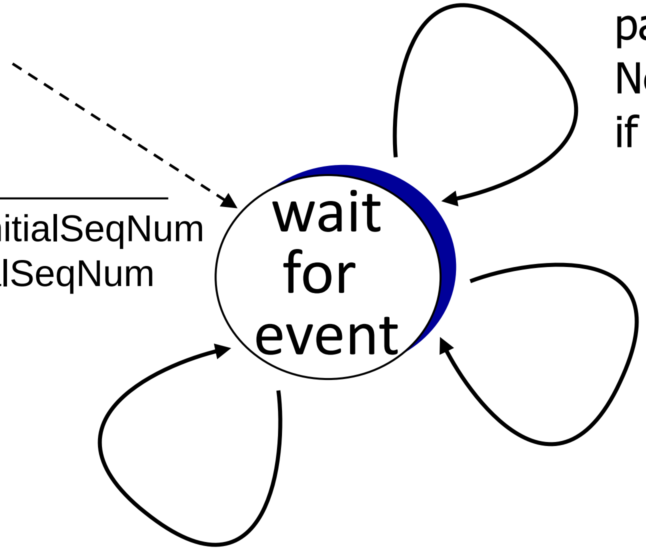
extract(rcvpkt,data)
deliver_data(data)
sndpkt = make_pkt(expectedseqnum,ACK,chksum)
udt_send(sndpkt)
expectedseqnum++

ACK-only: always send ACK for correctly-received packet with highest *in-order* seq #

  - may generate duplicate ACKs
  - need only remember **`expectedseqnum`**

- out-of-order packet:
  - discard (don’t buffer): *no receiver buffering!*
  - re-ACK pkt with highest in-order seq #

TCP sender (simplified)



Λ

```
NextSeqNum = InitialSeqNum
SendBase = InitialSeqNum
```

data received from application above

```
create segment, seq. #: NextSeqNum
pass segment to IP (i.e., “send”)
NextSeqNum = NextSeqNum + length(data)
if (timer currently not running)
    start timer
```

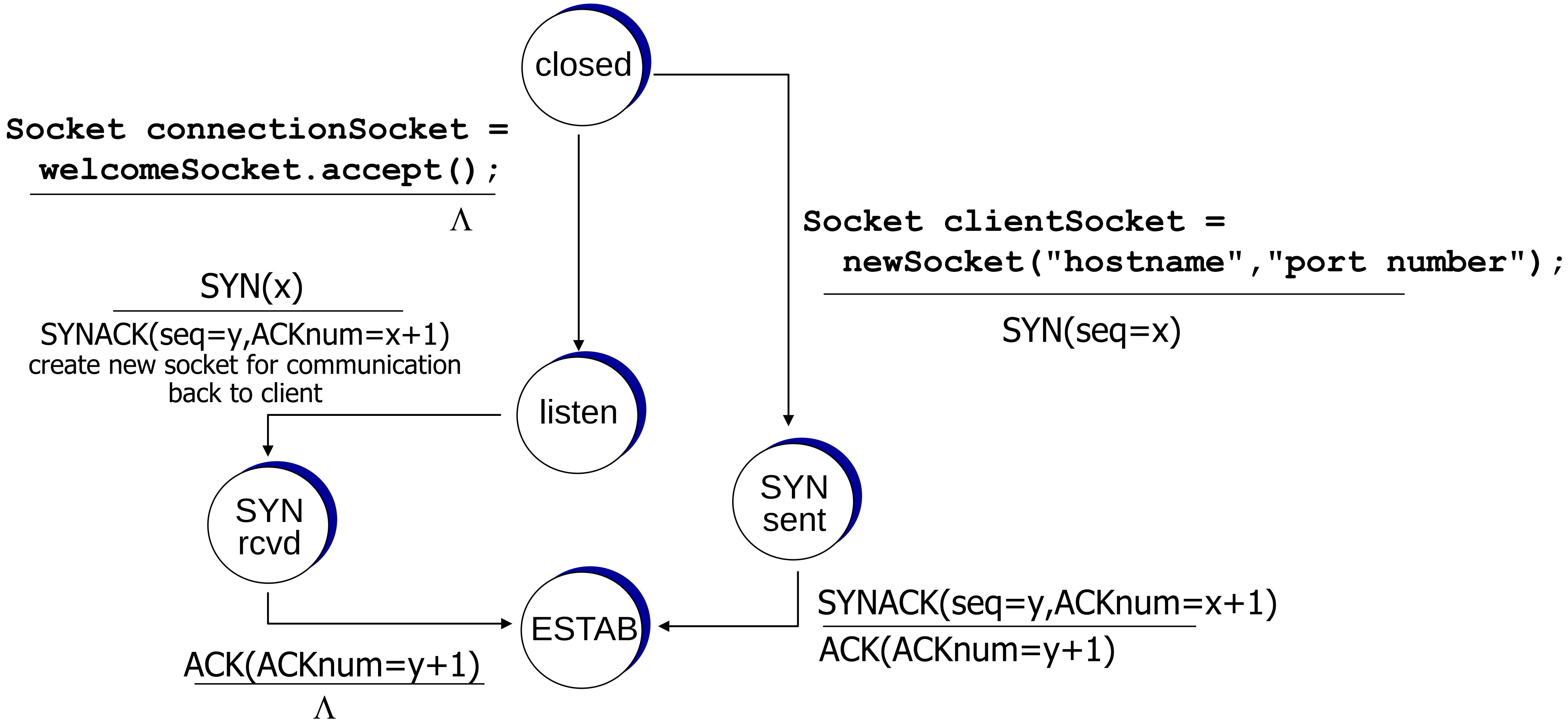
timeout

```
retransmit not-yet-acked segment
            with smallest seq. #
start timer
```

ACK received, with ACK field value y

```
if (y > SendBase) {
   SendBase = y
   /* SendBase–1: last cumulatively ACKed byte */
   if (there are currently not-yet-acked segments)
       start timer
     else stop timer
 }
```

TCP 3-way handshake FSM



closed

`Socket connectionSocket = welcomeSocket.accept();`

Λ

`Socket clientSocket = newSocket("hostname","port number");`

SYN(seq=x)

listen

SYN(x)

SYNACK(seq=y,ACKnum=x+1)
create new socket for communication
back to client

SYN rcvd

SYN sent

ESTAB

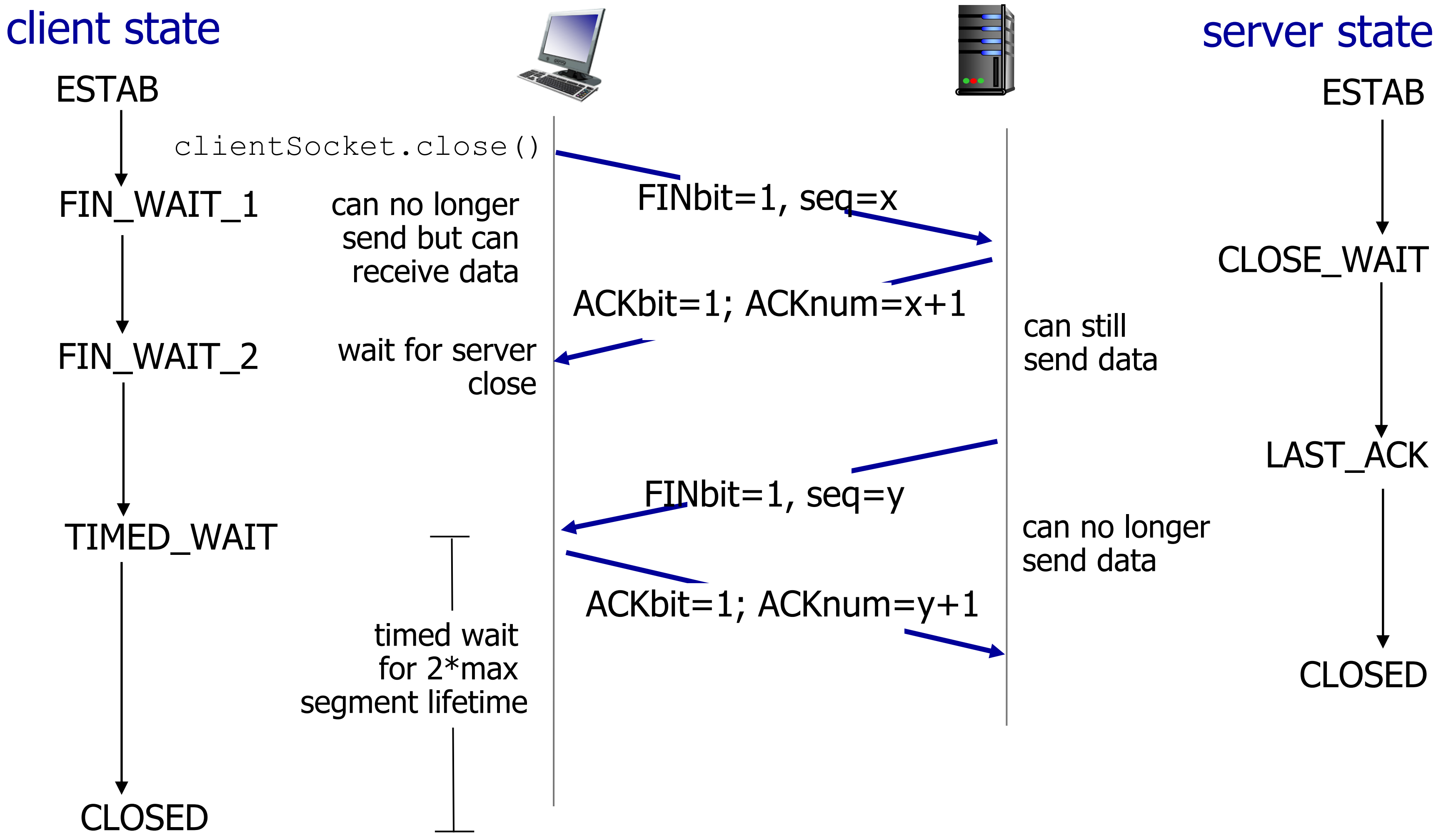
SYNACK(seq=y,ACKnum=x+1)

ACK(ACKnum=y+1)

ACK(ACKnum=y+1)

Λ

Closing a TCP connection



client state

ESTAB

`clientSocket.close()`

FIN_WAIT_1

can no longer send but can receive data

FINbit=1, seq=x

server state

ESTAB

CLOSE_WAIT

ACKbit=1; ACKnum=x+1

FIN_WAIT_2

wait for server close

can still send data

LAST_ACK

FINbit=1, seq=y

TIMED_WAIT

can no longer send data

ACKbit=1; ACKnum=y+1

timed wait for 2*max segment lifetime

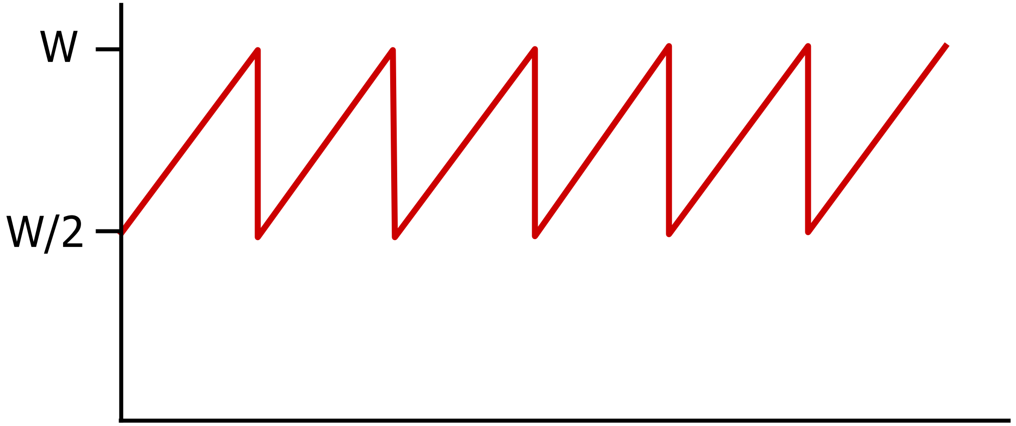
CLOSED

CLOSED

TCP throughput

- avg. TCP thruput as function of window size, RTT?
    - ignore slow start, assume there is always data to send
- W: window size (measured in bytes) where loss occurs
    - avg. window size (# in-flight bytes) is ¾ W
    - avg. thruput is 3/4W per RTT

$$\text{avg TCP thruput} = \frac{3}{4}\frac{W}{RTT} \text{ bytes/sec}$$



TCP over "long, fat pipes"

- example: 1500 byte segments, 100ms RTT, want 10 Gbps throughput
- requires W = 83,333 in-flight segments
- throughput in terms of segment loss probability, L [Mathis 1997]:

$$\text{TCP throughput} = \frac{1.22 \cdot \text{MSS}}{\text{RTT}\sqrt{L}}$$

➜ to achieve 10 Gbps throughput, need a loss rate of L = $2 \cdot 10^{-10}$ *– a very small loss rate!*

- versions of TCP for long, high-speed scenarios